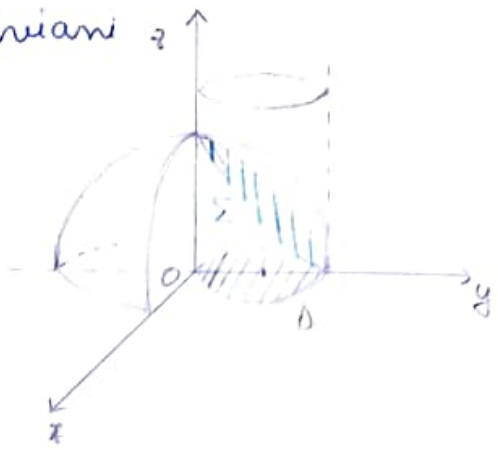
Aria Lui Viviani



$\Sigma: \begin{cases} x^2+y^2+z^2 = R^2 \\ z \geq 0 \\ x^2+y^2 = Ry \end{cases}$

$R>0$, fixed

$x^2+y^2+z^2 = R^2 \Leftrightarrow z^2 = R^2 - x^2 - y^2 \Leftrightarrow$

$\Leftrightarrow z = \sqrt{R^2-x^2-y^2} = f(x,y)$

$x^2+y^2 = Ry \Leftrightarrow x^2+y^2 - \frac{1}{2}\cdot 2 \cdot Ry + \left(\frac{R}{2}\right)^2 = \left(\frac{R}{2}\right)^2 \Leftrightarrow$

$\Leftrightarrow x^2 + \left(y-\frac{R}{2}\right)^2 = \left(\frac{R}{2}\right)^2$

$D: x^2 + \left(y-\frac{R}{2}\right)^2 \leq \frac{R^2}{4}$

$$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx\,dy =$$

$$= \iint_D \sqrt{1+\left(\frac{-x}{\sqrt{R^2-x^2-y^2}}\right)^2 + \left(\frac{-y}{\sqrt{R^2-x^2-y^2}}\right)^2}\, dx\,dy =$$

$$= \iint_D \sqrt{1+\frac{x^2+y^2}{R^2-x^2-y^2}}\, dx\,dy = \iint_D \sqrt{\frac{R^2-x^2-y^2+x^2+y^2}{R^2-x^2-y^2}}\, dx\,dy =$$

$$= R \iint_D \frac{1}{\sqrt{R^2-(x^2+y^2)}}\, dx\,dy$$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$

$x^2+y^2 \leq Ry \Leftrightarrow \rho^2 \leq R\rho\sin\theta \Leftrightarrow \begin{cases} \rho \leq R\sin\theta \\ \rho \geq 0 \end{cases} \Rightarrow \rho \in [0;\ R\sin\theta]$

$\rho \geq 0,\ R>0 \Rightarrow \sin\theta \geq 0 \Rightarrow \theta \in [0;\pi]$

$$\text{Aria}(\Sigma) = R \cdot \int_0^\pi \int_0^{R\sin\theta} \frac{1}{\sqrt{R^2-\rho^2}} \cdot \rho\, d\rho\, d\theta = -\frac{R}{2}\int_0^\pi \sqrt{R^2-\rho^2}\cdot 2 \Big|_0^{R\sin\theta} d\theta =$$

$$= -R\int_0^\pi \left(\sqrt{R^2-R^2\sin^2\theta} - \sqrt{R^2}\right)d\theta = -R\int_0^\pi \left(R\sqrt{1-\sin^2\theta} - R\right)d\theta =$$

$$= -R^2\int_0^\pi (|\cos\theta| - 1)\, d\theta = \cancel{-R^2\left(\int_0^\pi \cos\theta\, d\theta - \int_0^\pi d\theta\right)} =$$

$$\cancel{= -R^2(\sin\theta - \theta)\Big|_0^\pi = -R^2(0-\pi) + R^2(0-0) = \pi R^2}$$

$$= -R^2\left(\int_0^{\frac{\pi}{2}} \cos\theta\, d\theta - \int_{\frac{\pi}{2}}^\pi \cos\theta\, d\theta - \int_0^\pi d\theta\right) = -R^2\left(\sin\theta\Big|_0^{\frac{\pi}{2}} - \sin\theta\Big|_{\frac{\pi}{2}}^\pi - \theta\Big|_0^\pi\right) =$$

$$= -R^2(1-0-0+1-\pi+0) = -R^2(2-\pi) = R^2(\pi-2)$$

Exerciții date - sesiune

1) Să se calculeze următoarele integrale cu ajutorul funcțiilor lui Euler:

a) $\int_0^{\infty} \frac{dx}{\sqrt[3]{x^2(1+x^3)}}$; b) $\int_0^{1} \sqrt{\ln^3\left(\frac{1}{x}\right)}\,dx$; c) $\int_0^{\infty} x^4 e^{-x^2}\,dx$; d) $\int_0^{1} \frac{x^4}{\sqrt{1-x^2}}\,dx$

e) $\int_0^{\frac{\pi}{2}} \sin^{1/2}x \cdot \cos^{3/2}x\,dx$; f) $\int_{-\infty}^{\infty} e^{-2x^2+x+1}\,dx$; g) $\int_{-1}^{2} \frac{dx}{\sqrt[6]{(x+1)^5(2-x)}}$;

h) $\int_0^{\infty} \frac{x\sqrt{x}}{(x^2-2x+4)^2}\,dx$

2) $\int_\gamma x \cdot \frac{\sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}}\,dx + y \cdot \frac{\sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}}\,dy$

γ este o curbă cu capetele în $A(\pi, \pi\sqrt{3})$ și $B(2\pi, 2\pi\sqrt{3})$, situată în primul cadran, ce nu trece prin origine

3) $\int_\gamma (\cos x y - xy \sin xy + e^{2x})\,dx + (y^2 - x^2 \cdot \sin xy)\,dy$

γ este o curbă cu capetele în $A(0, -1)$, $B(0, 2)$

4) $\iint_D (x^2+y^2)\,dx\,dy$; D mărginit de $y^2 = 4x$, $x+y=3$, $y=0$

5) $\iint_D \sqrt{x^2+y^2}\,dx\,dy$, $D: x^2+y^2 \le y$, $x>0$

6) $\iint_D \sqrt{3-x^2-y^2}\,dx\,dy$; $D: x^2+y^2 \le 2$, $y \ge 0$

7) $\iint_D (1+x^2+y^2)\,dx\,dy$; $D: x^2+y^2 \le 2x$, $y \ge 0$

8) $\iint_D \sqrt{1+x^2+y^2}\,dx\,dy$; $D: 1 \le x^2+y^2 \le 9$, $y \le x \le y\sqrt{3}$, $y \ge 0$

9) $\iint_D \sqrt{x^2+y}\,dx\,dy$; $D: x \le x^2+y^2 \le 1$, $0 \le y \le x$

10) $Vol(\Omega) = ?$, Ω mărg. de $\begin{cases} z^2 = \frac{x^2+y^2}{2} \\ x^2+y^2+z^2 = 3 \end{cases}$, $z>0$

11) $Vol(\Omega) = ?$, Ω mărg. de $\begin{cases} z = 6 - x^2 - y^2 \\ z = 3 + 2x^2 + 2y^2 \end{cases}$

12) $Vol(\Omega) = ?$; Ω mărg. de $\begin{cases} z = 3 - 2x^2 - 2y^2 \\ x^2 + y^2 = z^2 \end{cases}$, $z>0$

13) Calculați $\int_\gamma \bar{v}\, d\bar{r}$, γ drum închis arbitrar, $O \notin \gamma$

14) Să se calculeze $\int_\gamma \bar{v}\, d\bar{r}$ (circulația), $\bar{v} = \frac{-y}{x^2+y^2}\bar{i} + \frac{x}{x^2+y^2}\bar{j}$

a) $\mathcal{C}(O,R)$; b) γ curbă arbitrar închisă, $O \notin \gamma$;

c) este $\bar{v}$ câmp de gradienți?

15) Aria $(\Sigma) = ?$; $\Sigma : z = 5 - x^2 - y^2$, $z \in [2,4]$

16) Aria $(\Sigma) = ?$; $\Sigma : z^2 = 3x^2 + 3y^2$, $z \in [1,3]$

17) Să se determine fluxul lui $\bar{v}$ prin Σ

$\bar{v} = (x^2 + y^2 + z^2)\bar{i} + (z^2 - 2yx)\bar{j} + (3z - x - y)\bar{k}$

$\Sigma : x^2 + y^2 + z^2 = 4$, $z > 0$

18) $\bar{v} = (2xy - z^2)\bar{i} + (z^2 - y^2)\bar{j} + (x^2 + y^3 - 3z)\bar{k}$

$\Sigma : z^2 = x^2 + y^2$, $z \in [0,2]$

19) $\bar{v} = (x + y^2)\bar{i} + (z^2 - y)\bar{j} + (z^2 + x)\bar{k}$

$\Sigma : z - 1 = x^2 + y^2$, $z \in [1,2]$

20) Să se calculeze circulația câmpului vectorial $\bar{v}$ prin curba γ

$\bar{v} = (y - z)\bar{i} + (z - x)\bar{j} + (x - y)\bar{k}$

$\gamma : \begin{cases} x^2 + y^2 = a^2 \\ \frac{x}{a} + \frac{z}{h} = 1 \end{cases}$, $a, h > 0$

24) Să se calculeze fluxul vectorului lui $\bar{v}$ prin suprafața Σ, orientată după normala exterioară, unde Σ este semisfera superioară a sferei unitate. $\bar{v} = z\bar{i} + z\bar{j} + xyz\bar{k}$

22) $x^3 + 9x - 3 = 0$, $\varepsilon < 10^{-2}$

Calculația cu o precizie mai mică decât 10^{-2} unica rădăcină reală a ecuației.

23) $x^3 + 4x - 1 = 0$, $\varepsilon < 10^{-3}$

24) $\bar{v} = \left(-\frac{y}{x^2+y^2} , \frac{x}{x^2+y^2} \right)$

a), b), c)

25) $\bar{v} = x^3 y^2 \bar{i} + x^2 y^3 \bar{j} + 3z\bar{k}$; $\Sigma : z = x^2 + y^2$, $z \le 6 - x^2 - y^2$, $0 \le z \le 6$

Să se calculeze fluxul lui $\bar{v}$ prin Σ

Rezolvare exerciții date - sesiune

1) a) $I = \int_0^{\infty} \frac{dx}{\sqrt[3]{x^3}(1+x^3)}$

$x^3 = y \Rightarrow x = \sqrt[3]{y} \Rightarrow dx = \frac{1}{3\sqrt[3]{y^2}} dy \Rightarrow dx = \frac{1}{3} \cdot y^{-\frac{2}{3}} dy \Rightarrow$

$\Rightarrow I = \int_0^{\infty} \frac{\frac{1}{3} \cdot y^{-\frac{2}{3}} dy}{y^{\frac{1}{3}}(1+y)} = \frac{1}{3} \int_0^{\infty} \frac{y^{-\frac{2}{3}} \cdot y^{-\frac{2}{3}}}{1+y} dy = \frac{1}{3} \int_0^{\infty} \frac{y^{-\frac{4}{3}}}{1+y} dy$

$$B(p,q) = \int_0^{\infty} \frac{y^{p-1}}{(1+y)^{p+q}} dy$$

$\begin{cases} p-1 = -\frac{4}{9} \mid +1 \Rightarrow p = -\frac{4}{9} + \frac{9}{9} = \frac{1}{9} \\ p+q = 1 \Rightarrow q = 1-p = 1-\frac{1}{9} = \frac{9}{9} - \frac{1}{9} = \frac{8}{9} \end{cases} \Rightarrow$

$\Rightarrow I = \frac{1}{3} B\left(\frac{1}{9}; \frac{8}{9}\right)$

$$B(p,q) = \frac{\Gamma(p) \cdot \Gamma(q)}{\Gamma(p+q)}$$

$\Rightarrow I = \frac{1}{3} \frac{\Gamma\left(\frac{1}{9}\right) \cdot \Gamma\left(\frac{8}{9}\right)}{\Gamma\left(\frac{1}{9}+\frac{8}{9}\right)} = \frac{1}{3} \frac{\Gamma\left(\frac{1}{9}\right) \cdot \Gamma\left(\frac{8}{9}\right)}{\Gamma(1)}$

$\frac{8}{9} = 1 - \frac{1}{9} \Rightarrow \Gamma\left(\frac{1}{9}\right) \Gamma\left(\frac{8}{9}\right) = \frac{\pi}{\sin\left(\frac{\pi}{9}\right)}$

$\Rightarrow I = \frac{1}{3} \frac{\pi}{\sin\left(\frac{\pi}{9}\right)} = \frac{\pi}{3\sin\left(\frac{\pi}{9}\right)}$

$1 \in \mathbb{N} \Rightarrow \Gamma(1) = (1-1)! = 0! = 1$

b) $I = \int_0^1 \sqrt{\ln^3\left(\frac{1}{x}\right)} dx$

$\ln\left(\frac{1}{x}\right) = y \Leftrightarrow \frac{1}{x} = e^y \Leftrightarrow x = \frac{1}{e^y} \Leftrightarrow x = e^{-y} \Rightarrow dx = -e^{-y} dy \Rightarrow$

$\Rightarrow I = \int_{\infty}^0 \sqrt{y^3} \cdot (-e^{-y}) dy = \int_0^{\infty} \sqrt{y^3} \cdot e^{-y} dy = \int_0^{\infty} y^{\frac{3}{2}} e^{-y} dy$

$$\Gamma(p) = \int_0^{\infty} y^{p-1} \cdot e^{-y} dy \qquad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$p-1 = \frac{3}{2} \Rightarrow p = \frac{5}{2} \Rightarrow I = \Gamma\left(\frac{5}{2}\right) = \Gamma\left(1+\frac{3}{2}\right) = \frac{3}{2}\Gamma\left(1+\frac{1}{2}\right) =$

$= \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{3}{4} \cdot \sqrt{\pi} = \frac{3\sqrt{\pi}}{4}$

c) $J = \int_0^{\infty} x^4 \cdot e^{-x^2} dx$

$x^2 = y \Rightarrow x = \sqrt{y} \Rightarrow dx = \frac{1}{2\sqrt{y}} dy \Leftrightarrow dx = \frac{1}{2} y^{-\frac{1}{2}} dy \Rightarrow$

$\Rightarrow J = \int_0^{\infty} y^2 \cdot e^{-y} \cdot \frac{1}{2} y^{-\frac{1}{2}} dy = \frac{1}{2} \int_0^{\infty} y^{\frac{3}{2}} \cdot e^{-y} dy$

$\Gamma(p) = \int_0^{\infty} y^{p-1} \cdot e^{-y} dy$

$p-1 = \frac{3}{2} \Rightarrow p = \frac{5}{2} \Rightarrow J = \frac{1}{2} \cdot \Gamma\left(\frac{5}{2}\right) = \frac{1}{2} \cdot \Gamma\left(1+\frac{3}{2}\right) = \frac{1}{2} \cdot \frac{3}{2} \cdot \Gamma\left(1+\frac{1}{2}\right) =$

$= \frac{3}{4} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{3}{8} \cdot \sqrt{\pi} = \frac{3\sqrt{\pi}}{8}$

d) $\int_0^1 \frac{x^4}{\sqrt{1-x^2}} dx$

$x^2 = y \Leftrightarrow x = \sqrt{y} \Rightarrow dx = \frac{1}{2\sqrt{y}} dy \Leftrightarrow dx = \frac{1}{2} y^{-\frac{1}{2}} dy \Rightarrow$

$\Rightarrow J = \int_0^1 \frac{y^2}{\sqrt{1-y}} \cdot \frac{1}{2} \cdot y^{-\frac{1}{2}} dy = \frac{1}{2} \int_0^1 y^{\frac{3}{2}} \cdot (1-y)^{-\frac{1}{2}}$

~~$B(p;q) = \int_0^{\infty} \frac{y^{p-1}}{(1+y)^{p+q}} dy$~~

$p-1 = \frac{3}{2} \Rightarrow p = \frac{5}{2}$

~~$p+q = -\frac{1}{2}$~~ $\Rightarrow q = -\frac{1}{2} - \frac{5}{2} = -\frac{4}{2} = -\frac{1}{2}$

$$B(p;q) = \int_0^1 y^{p-1} (1-y)^{q-1} dy$$

$p-1 = \frac{3}{2} \Rightarrow p = \frac{5}{2}$

$q-1 = -\frac{1}{2} \Rightarrow q = \frac{1}{2}$

$\Rightarrow J = \frac{1}{2} B\left(\frac{5}{2}; \frac{1}{2}\right)$

$B(p;q) = \frac{\Gamma(p) \cdot \Gamma(q)}{\Gamma(p+q)} \Rightarrow J = \frac{1}{2} \cdot \frac{\Gamma\left(\frac{5}{2}\right) \cdot \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{6}{2}\right)} =$

$= \frac{1}{2} \cdot \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot \Gamma\left(1+\frac{3}{2}\right) \cdot \sqrt{\pi}}{2!} = \frac{15}{8} \cdot \frac{3}{2} \cdot \Gamma\left(1+\frac{1}{2}\right) = \frac{3\sqrt{\pi}}{8} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right)$

$\Gamma(3) = (3-1)! = 2! = 2$

$J = \frac{3\pi}{16}$

3

e) $J = \int_0^{\frac{\pi}{2}} \sin^{1/2} x \cdot \cos^{3/2} x \, dx$

$\sin^2 x = y \Rightarrow dy = 2 \sin x \cos x \, dx$

~~$J = \int_0^{\frac{\pi}{2}} 2 \sin x \cos x \, dx$~~

$2 \sin x \cos x \cdot \alpha = \sin^{1/2} x \cdot \cos^{3/2} x \Rightarrow \alpha = \frac{1}{2} \left(\sin^{-\frac{1}{2}} x \cdot \cos^{\frac{1}{2}} x \right)$

$\Rightarrow J = \int_0^{\frac{\pi}{2}} \frac{1}{2} \sin^{-\frac{1}{2}} x \cdot \cos^{\frac{1}{2}} x \cdot 2 \sin x \cos x \, dx =$

$= \int_0^{\frac{\pi}{2}} \frac{1}{2} \left(\sin^2 x \right)^{-\frac{1}{4}} \cdot (1 - \sin^2 x)^{\frac{1}{4}} \cdot 2 \sin x \cos x \, dx =$

$= \frac{1}{2} \int_0^1 y^{-\frac{1}{4}} \cdot (1-y)^{\frac{1}{4}} \cdot dy$

$\sin^2 0 = 0$

$\sin^2 \frac{\pi}{2} = 1^2 = 1$

$\beta(p, q) = \int_0^1 y^{p-1} (1-y)^{q-1} \, dy$

$p - 1 = -\frac{1}{4} \Rightarrow p = \frac{3}{4}$

$q - 1 = \frac{1}{4} \Rightarrow q = \frac{5}{4}$

$\Rightarrow J = \frac{1}{2} \beta(p, q)$

$\beta(p, q) = \frac{\Gamma(p) \cdot \Gamma(q)}{\Gamma(p+q)}$

$J = \frac{1}{2} \cdot \frac{\Gamma\left(\frac{3}{4}\right) \cdot \Gamma\left(\frac{5}{4}\right)}{\Gamma(2)} = \frac{1}{2} \cdot \frac{\Gamma\left(1 + \frac{1}{4}\right) \cdot \Gamma\left(\frac{3}{4}\right)}{1!} =$

$= \frac{1}{2} \cdot \frac{1}{4} \cdot \Gamma\left(\frac{1}{4}\right) \cdot \Gamma\left(1 - \frac{1}{4}\right) = \frac{1}{8} \cdot \frac{\pi}{\sin \frac{\pi}{4}} = \frac{\pi}{8} \cdot \frac{1}{\frac{\sqrt{2}}{2}} =$

$= \frac{\pi}{8} \cdot \frac{2}{\sqrt{2}} = \frac{\pi \cdot 2\sqrt{2}}{8 \cdot 2} = \frac{\pi \sqrt{2}}{8}$

g) $J = \int_{-\infty}^{\infty} e^{-2x^2 + x + 1} \, dx = \int_{-\infty}^{\infty} e^{-2\left(x^2 - \frac{1}{2}x - \frac{1}{2}\right)} \, dx =$

$= \int_{-\infty}^{\infty} e^{-2\left(x^2 - 2 \cdot \frac{1}{4} \cdot x + \frac{1}{16} - \frac{1}{16} - \frac{1}{2}\right)} \, dx = \int_{-\infty}^{\infty} e^{-2\left[\left(x - \frac{1}{4}\right)^2 - \frac{9}{16}\right]} \, dx =$

$= \int_{-\infty}^{\infty} e^{-2\left(x - \frac{1}{4}\right)^2} \cdot e^{\frac{9}{8}} \, dx = e^{\frac{9}{8}} \int_{-\infty}^{\infty} e^{-2\left(x - \frac{1}{4}\right)^2} \, dx$

$\sqrt{2}\left(x+\frac{1}{4}\right)^2 = y \Rightarrow x+\frac{1}{4} = \frac{y}{\sqrt{2}} \Rightarrow x = \frac{1}{\sqrt{2}}y + \frac{1}{4} \Rightarrow$

$\Rightarrow dx = \frac{1}{\sqrt{2}}\,dy$

$I = \int_{-\infty}^{\infty} e^{\frac{9}{8}} e^{-y^2} \cdot \frac{1}{\sqrt{2}}\,dy = \frac{e^{\frac{9}{8}}}{\sqrt{2}} \int_{-\infty}^{\infty} e^{-y^2}\,dy = \frac{e^{\frac{9}{8}}\sqrt{\pi}}{2}$

g) $I = \int_{-1}^{2} \frac{dx}{\sqrt{(x+1)^5(2-x)}}$

2) $\int_\gamma \underbrace{x \cdot \frac{\sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}}}_{P}\,dx + \underbrace{y \cdot \frac{\sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}}}_{Q}\,dy$ — ω

Verific dacă $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

$\frac{\partial P}{\partial y} = x \cdot \frac{\cos\left(\sqrt{x^2+y^2}\right)\cdot \frac{2y}{2\sqrt{x^2+y^2}} \cdot \sqrt{x^2+y^2} - \sin\left(\sqrt{x^2+y^2}\right)\cdot \frac{2y}{2\sqrt{x^2+y^2}}}{x^2+y^2} =$

$= x \cdot \frac{y\cos\left(\sqrt{x^2+y^2}\right) - y\sin\left(\sqrt{x^2+y^2}\right)\cdot \frac{1}{\sqrt{x^2+y^2}}}{x^2+y^2}$

$\frac{\partial Q}{\partial x} = y \cdot \frac{\cos\left(\sqrt{x^2+y^2}\right)\cdot \frac{2x}{2\sqrt{x^2+y^2}} \cdot \sqrt{x^2+y^2} - \sin\left(\sqrt{x^2+y^2}\right)\cdot \frac{2x}{2\sqrt{x^2+y^2}}}{x^2+y^2} =$

$= x \cdot \frac{y\cos\left(\sqrt{x^2+y^2}\right) - y\sin\left(\sqrt{x^2+y^2}\right)\cdot \frac{1}{\sqrt{x^2+y^2}}}{x^2+y^2}$

$\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow \omega$ este închisă $\Rightarrow \omega$ este exactă $\Rightarrow$

$\{(x,y) \in \mathbb{R}^2 \mid x>0,\ y>0\}$ stelată

$\Rightarrow$ integrala independentă de drum $\Rightarrow \int_\gamma \omega = \int_{[AB]} \omega$

$[AB]: \begin{cases} x = t \\ y = t\sqrt{3} \end{cases}, t \in [\pi, 2\pi]$

$\int_\gamma \omega = \int_{\pi}^{2\pi} t \cdot \frac{\sin\sqrt{t^2+3t^2}}{\sqrt{4t^2}} + t\sqrt{3} \cdot \frac{\sin(2t)}{2t}\,dt =$

5

$= \frac{1}{2}\int_{\pi}^{2\pi} \left(\sin(2t) + \sqrt{3}\cdot \sin 2t\right) dt = \frac{1+\sqrt{3}}{2}\int_{\pi}^{2\pi} \sin 2t\, dt =$

$= \frac{1+\sqrt{3}}{2}\cdot \frac{\cos(2t)}{-2}\cdot(-1)\Big|_{\pi}^{2\pi} = (1+\sqrt{3})(\cos 4\pi - \cos 2\pi) = 0$

4) $\iint (x^2+y^2)\,dx\,dy$; D mărginit de $y^2 = 4x$, $x+y=3$, $y=0$

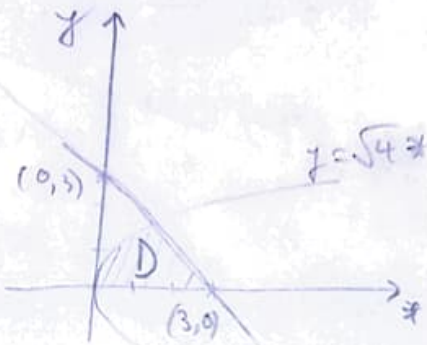
$\begin{cases} y^2 = 4x \Rightarrow y^2 - 12 + 4y = 0 \\ x+y=3 \Rightarrow x = 3-y \end{cases}$

$\Delta = 16 + 48 = 64 = 8^2$

$y_{1,2} = \frac{-4 \pm 8}{2} = -2 \pm 4$

$y_1 = -6$

$y_2 = 2$



$x, y > 0 \Rightarrow y = 2 \Rightarrow x = 3-2 = 1 \Rightarrow (x,y) = (1,2)$

$\iint_D (x^2+y^2)\,dx\,dy = \int_0^2 \left(\int_{\frac{y^2}{4}}^{3-y} (x^2+y^2)\,dx\right) dy =$

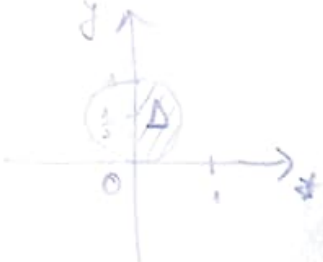
$= \int_0^2 \left(\frac{x^3}{3} + y^2 x\right)\Big|_{\frac{y^2}{4}}^{3-y} dy = \int_0^2 \frac{(3-y)^3}{3} + (3-y)y^2 - \frac{y^4}{12} - \frac{y^4}{4}\, dy$

5) $\iint_D \sqrt{x^2+y^2}\,dx\,dy$, $D: x^2+y^2 \le y,\ x>0$

$x^2+y^2 \le y \Leftrightarrow x^2+y^2-y \le 0 \Leftrightarrow$

$\Leftrightarrow x^2 + y^2 - 2\cdot\frac{1}{2}\cdot y + \left(\frac{1}{2}\right)^2 - \frac{1}{4} \le 0 \Leftrightarrow$

$\Leftrightarrow x^2 + \left(y - \frac{1}{2}\right)^2 \le \left(\frac{1}{2}\right)^2$



$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$

$\rho^2(\cos^2\theta + \sin^2\theta) \le \rho\sin\theta \Leftrightarrow \rho^2 \le \rho\sin\theta \Leftrightarrow \rho\sin\theta \le \sin\theta \Rightarrow$

$\Rightarrow \sin\theta \ge 0 \Rightarrow \theta \in [0,\pi]$; $x>0$ $\Big\} \Rightarrow \theta \in [0; \frac{\pi}{2}]$; $\rho \in [0, \sin\theta]$

$$J = \iint_D \sqrt{x^2+y^2}\,dx\,dy = \int_0^{\frac{\pi}{2}} \int_0^{\sin\theta} \sqrt{\rho^2} \cdot \rho\,d\rho\,d\theta =$$

$$= \int_0^{\frac{\pi}{2}} \frac{\rho^3}{3}\Big|_0^{\sin\theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{\sin^3\theta}{3}\,d\theta = \frac{\sin^3(\frac{\pi}{2})}{3} - \frac{\sin^3(0)}{3}$$

$$= \frac{1}{3}\int_0^{\frac{\pi}{2}} \sin\theta \cdot \sin^2\theta\,d\theta = \frac{1}{3}\int_0^{\frac{\pi}{2}} \sin\theta(1-\cos^2\theta)\,d\theta =$$

$$= \frac{1}{3}\int_0^{\frac{\pi}{2}} \sin\theta\,d\theta + \frac{1}{3}\int_0^{\frac{\pi}{2}} (\cos\theta)' \cdot \cos^2\theta\,d\theta =$$

$$= \frac{1}{3}(\cos\tfrac{\pi}{2} - \cos 0) + \frac{1}{3}\left(\cos^3\theta\Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos\theta \cdot 2\cos^4\theta \cdot \sin\theta\,d\theta\right)$$

$\cos\theta = t \Leftrightarrow -\sin\theta\,d\theta = dt$

$t' = -\sin\theta$

$$J = -\frac{1}{3}\int_0^{\frac{\pi}{2}} (1-t^2)\,dt =$$

$$= \frac{1}{3}\int_0^{\frac{\pi}{2}} \sin\theta\,d\theta + \frac{1}{3}\int_0^{\frac{\pi}{2}} (\cos\theta)' \cdot \cos^2\theta\,d\theta =$$

$$= -\frac{1}{3}\cos\theta\Big|_0^{\frac{\pi}{2}} + \frac{1}{3}\cdot 2\cos\theta\Big|_0^{\frac{\pi}{2}} = \frac{1}{3}\cos\theta\Big|_0^{\frac{\pi}{2}} = 0 - \frac{1}{3} = -\frac{1}{3}$$

6) $\iint_D \sqrt{3-x^2-y^2}\,dx\,dy$; $D: x^2+y^2 \le 2,\ y \ge 0$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \rho \in [0, \sqrt{2}],\ \theta \in [0, 2\pi]$$

$y \ge 0 \Leftrightarrow \rho\sin\theta \ge 0 \Big\} \Rightarrow \sin\theta \ge 0 \Rightarrow \theta \in [0, \pi]$
$\rho > 0$

$$\iint_D \sqrt{3-x^2-y^2}\,dx\,dy = \iint_D \sqrt{3-(x^2+y^2)}\,dx\,dy =$$

$$= \int_0^{\pi}\left(\int_0^{\sqrt{2}} \sqrt{3-\rho^2} \cdot \rho\,d\rho\right)d\theta = \frac{-1}{2}\int_0^{\pi}\left(\int_0^{\sqrt{2}} (3-\rho^2)' \cdot \sqrt{3-\rho^2}\,d\rho\right)d\theta =$$

$$= -\frac{1}{2}\int_0^{\pi}\left[\frac{2}{3}\cdot(3-\rho^2)^{\frac{3}{2}}\Big|_0^{\sqrt{2}}\right]d\theta = \frac{1}{2}\cdot\frac{2}{3}\int_0^{\pi}(1-3^{\frac{3}{2}})\,d\theta =$$

$$= -\frac{3}{4}\left(\theta - \sqrt[3]{3}\,\theta\right)\Big|_0^{\pi} = -\frac{3\pi}{4} + \frac{3\pi\sqrt[3]{3}}{4}$$

$$= -\frac{1}{2}\cdot\frac{2}{3}\int_0^{\pi}(3-\rho^2)^{\frac{1}{2}}\Big|_0^{\sqrt{2}}d\theta = -\frac{1}{3}\int_0^{\pi}(1-\sqrt{27})\,d\theta = -\frac{1}{3}(\theta - 3\sqrt{3}\theta)\Big|_0^{\pi} =$$

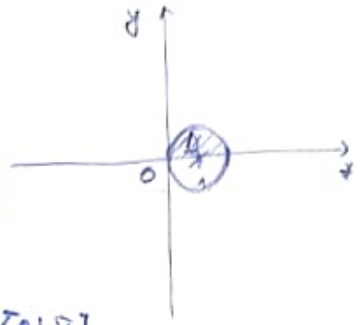
$$= -\frac{\pi}{3} - 3\sqrt{3}\pi = -\pi\left(\frac{1}{3} - 3\sqrt{3}\right)$$

7

7) $\iint_D (1+x^2+y^2)\,dx\,dy$; $D: x^2+y^2 \le 2x,\ y \ge 0$

$x^2+y^2 \le 2x \Leftrightarrow x^2-2x+1-1+y^2 \le 0 \Leftrightarrow$

$\Leftrightarrow (x-1)^2+y^2 \le 1$



$\iint_D$ ~~(1+x)~~ $\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho \in [0;1]$, $\theta \in [0;2\pi]$

$y \ge 0 \Rightarrow \rho\sin\theta \ge 0 \\ \rho \ge 0 \Rightarrow \sin\theta \ge 0 \Rightarrow \theta \in [0;\pi]$

$\rho^2\cos^2\theta + \rho^2\sin^2\theta \le 2\rho\cos\theta \Leftrightarrow$

$\Leftrightarrow \rho^2 \le 2\rho\cos\theta \Leftrightarrow \rho \le 2\cos\theta \Leftrightarrow$

$\Leftrightarrow 2\cos\theta \ge \rho \\ 0 \le \rho \le 1 \Rightarrow 2\cos\theta \ge 0 \Rightarrow \cos\theta \ge 0 \Rightarrow \theta \in [0, \frac{\pi}{2}]$

$\rho \le 2\cos\theta \Rightarrow \rho \in [0; 2\cos\theta]$

$$\iint_D (1+x^2+y^2)\,dx\,dy = \int_0^{\frac{\pi}{2}} \int_0^{2\cos\theta} (1+\rho^2)\cdot\rho\,d\rho\,d\theta =$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{\rho^2}{2} + \frac{\rho^4}{4}\right)\Big|_0^{2\cos\theta} d\theta = \int_0^{\frac{\pi}{2}} (2\cos^2\theta + 4\cos^4\theta)\,d\theta =$$

$$= 2\underbrace{\int_0^{\frac{\pi}{2}} \cos^2\theta\,d\theta}_{J_1} + 4\underbrace{\int_0^{\frac{\pi}{2}} \cos^4\theta\,d\theta}_{J_2}$$

$$J_1 = \int_0^{\frac{\pi}{2}} \frac{1+\cos 2\theta}{2}\,d\theta = \frac{1}{2}\int_0^{\frac{\pi}{2}} d\theta + \frac{1}{4}\int_0^{\frac{\pi}{2}} (2\theta)'\cdot\cos(2\theta)\,d\theta =$$

$$= \frac{\theta}{2}\Big|_0^{\frac{\pi}{2}} + \frac{\sin(2\theta)}{4}\Big|_0^{\frac{\pi}{2}} = \frac{\pi}{4} - 0 + 0 - 0 = \frac{\pi}{4}$$

$$J_2 = \int_0^{\frac{\pi}{2}} \cos^2\theta\cdot\cos^2\theta\,d\theta = \int_0^{\frac{\pi}{2}} \cos^2\theta(1-\sin^2\theta)\,d\theta =$$

$$= \int_0^{\frac{\pi}{2}} \cos^2\theta\,d\theta - \int_0^{\frac{\pi}{2}} \sin^2\theta\cdot\cos^2\theta\,d\theta =$$

$$= \frac{\pi}{4} - \frac{1}{4}\int_0^{\frac{\pi}{2}} (2\sin\theta\cos\theta)\cdot(2\sin\theta\cos\theta)\,d\theta =$$

$$= \frac{\pi}{4} - \frac{1}{4}\int_0^{\frac{\pi}{2}} (\sin(2\theta))^2\,d\theta = \frac{\pi}{4} - \frac{1}{4}\int_0^{\frac{\pi}{2}} \frac{1-\cos 4\theta}{2}\,d\theta =$$

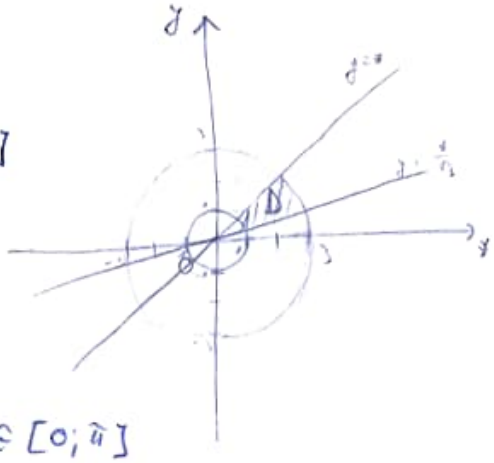
$$= \frac{\pi}{4} - \frac{1}{8}\int_0^{\frac{\pi}{2}} (1-\cos(4\theta))\,d\theta = \frac{\pi}{4} - \frac{1}{8}\theta\Big|_0^{\frac{\pi}{2}} + \frac{1}{8}\int_0^{\frac{\pi}{2}} \cos(4\theta)\,d\theta =$$

$$= \frac{\pi}{4} - \frac{\pi}{16} + \frac{1}{8}\cdot\frac{1}{4}\cdot\int_0^{\frac{\pi}{2}} (4\theta)'\cdot\cos(4\theta)\,d\theta =$$

$$= \frac{3\pi}{16} - \frac{1}{32}\cdot\sin(4\theta)\Big|_0^{\frac{\pi}{2}} = \frac{3\pi}{16} - 0 + 0 = \frac{3\pi}{16}$$

8) $\iint_D \sqrt{1+x^2+y^2}\,dx\,dy$; D: $1 \le x^2+y^2 \le 9$, $y \le x \le y\sqrt{3}$, $y \ge 0$

$x \le y\sqrt{3} \Rightarrow \frac{x}{\sqrt{3}} \le y$



$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \rho \in [1;3];\ \theta \in [0;2\pi]$$

$1 \le \rho^2 \le 9$

$\rho\sin\theta \le \rho\cos\theta \le \rho\sin\theta\cdot\sqrt{3}$

$\left.\begin{array}{l}\rho\sin\theta \ge 0 \\ \rho > 0\end{array}\right\} \Rightarrow \sin\theta \ge 0 \Rightarrow \theta \in [0;\pi]$

$\sin\theta \le \cos\theta \le \sqrt{3}\sin\theta \Rightarrow 1 \le \operatorname{tg}\theta \le \sqrt{3} \Rightarrow$

$\Rightarrow \operatorname{tg}\frac{\pi}{4} \le \operatorname{tg}\theta \le \operatorname{tg}\frac{\pi}{6} \Rightarrow \frac{\pi}{4} \le \theta \le \frac{\pi}{6} \Rightarrow$

$\Rightarrow \theta \in [\frac{\pi}{4};\frac{\pi}{6}]$

$\Rightarrow \theta \in [\frac{\pi}{6};\frac{5\pi}{4}]$, $\rho \in [1;3]$

$$\iint_D \sqrt{1+x^2+y^2}\,dx\,dy = \int_{\frac{\pi}{6}}^{\frac{5\pi}{4}}\int_1^3 \left(\sqrt{1+\rho^2}\right)\cdot\rho\,d\rho\,d\theta =$$

$$= \frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{4}} \frac{(1+\rho^2)^{\frac{3}{2}}}{\frac{3}{2}}\Big|_1^3 d\theta =$$

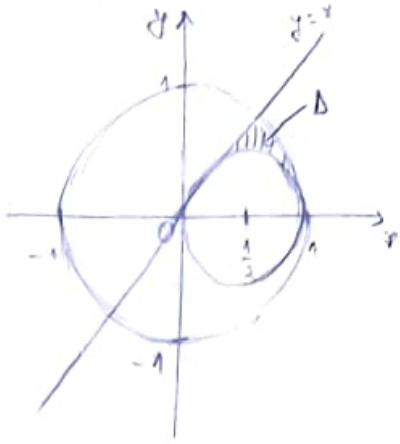
$$= 3\int_{\frac{\pi}{6}}^{\frac{5\pi}{4}} \left(10^{\frac{3}{2}} - 2^{\frac{3}{2}}\right)d\theta = \cancel{3\sqrt{10^3}\,\theta - 3\sqrt{2^3}}$$

$$= \left(3\cdot 10\sqrt{10}\,\theta - 3\cdot 2\sqrt{2}\,\theta\right)\Big|_{\frac{\pi}{6}}^{\frac{5\pi}{4}} = 30\sqrt{10}\cdot\frac{5\pi}{4} - 6\sqrt{2}\cdot\frac{5\pi}{4} - 30\sqrt{10}\cdot\frac{\pi}{6} + 6\sqrt{2}\cdot\frac{\pi}{6} =$$

$$= \frac{6\pi}{4}\left(5\sqrt{10} - \sqrt{2}\right) - 5\sqrt{10}\cdot\pi + \sqrt{2}\cdot\pi = \frac{3\pi}{2}\cdot\sqrt{2}\left(5\sqrt{5} - 1\right) - \pi\sqrt{2}\left(5\sqrt{5} - 1\right) =$$

$$= \pi\sqrt{2}\left(5\sqrt{5} - 1\right)\left(\frac{3}{2} - 1\right) = \frac{\pi\sqrt{2}}{2}\left(5\sqrt{5} - 1\right)$$

9) $\iint_D \sqrt{x^2+y^2}\,dx\,dy$; $D: x \le x^2+y^2 \le 1$, $0 \le y \le x$



$*$ $\le x^2+y^2 \Leftrightarrow x^2+y^2 \ge x$ (*)

a) $x^2+y^2-x \ge 0$ (*)

b) $x^2 - 2\cdot\frac{1}{2}\cdot x + \frac{1}{4} + y^2 - \frac{1}{4} \ge 0$ (*)

c) $\left(x-\frac{1}{2}\right)^2 + y^2 \ge \left(\frac{1}{2}\right)^2$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$

$\rho\cos\theta \le \rho^2 \le 1$

$\rho^2 \le 1 \Rightarrow \rho \in [0;1]$

$\rho\cos\theta \le 0 \Rightarrow \cos\theta \le 0 \Rightarrow \theta \in \left[\frac{\pi}{2}; \frac{3\pi}{2}\right]$ ~~$\left[\frac{\pi}{2}; \frac{4\pi}{2}\right]$~~ $[0; \frac{\pi}{2}]$

$0 \le \rho\sin\theta \le \rho\cos\theta$ (*) $0 \le \sin\theta \le \cos\theta$ ~~$\Rightarrow \theta \in \left[\frac{3\pi}{2}; 2\pi\right]$~~

$\Rightarrow \theta \in [0; \frac{\pi}{2}]$

$0 \le \sin\theta \le \cos\theta$ (:) $0 \le \operatorname{tg}\theta \le 1 \Rightarrow \theta \in [0; \frac{\pi}{4}]$

$\rho\cos\theta \le \rho^2$ (*) $\rho^2 - \rho\cos\theta \ge 0$ (*) $\rho(\rho - \cos\theta) \ge 0$ (*)

(*) $\rho - \cos\theta \ge 0$ ($\Rightarrow \rho \ge \cos\theta$; $\rho \le 1$) $\Rightarrow \rho \in [\cos\theta; 1]$

$$\iint_D \sqrt{x^2+y^2}\,dx\,dy = \int_0^{\frac{\pi}{4}} \int_{\cos\theta}^{1} \sqrt{\rho^2}\cdot\rho\,d\rho\,d\theta =$$

$$= \frac{1}{3}\int_0^{\frac{\pi}{4}} \rho^3\Big|_{\cos\theta}^{1} d\theta = \frac{1}{3}\int_0^{\frac{\pi}{4}} (1-\cos^3\theta)\,d\theta =$$

$$= \frac{1}{3}\theta\Big|_0^{\frac{\pi}{4}} - \frac{1}{3}\int_0^{\frac{\pi}{4}} \cos\theta(1-\sin^2\theta)\,d\theta =$$

$$= \frac{\pi}{12} - \frac{1}{3}\int_0^{\frac{\pi}{4}} \cos\theta\,d\theta + \frac{1}{3}\int_0^{\frac{\pi}{4}} (\sin\theta)'\cdot\sin^2\theta\,d\theta =$$

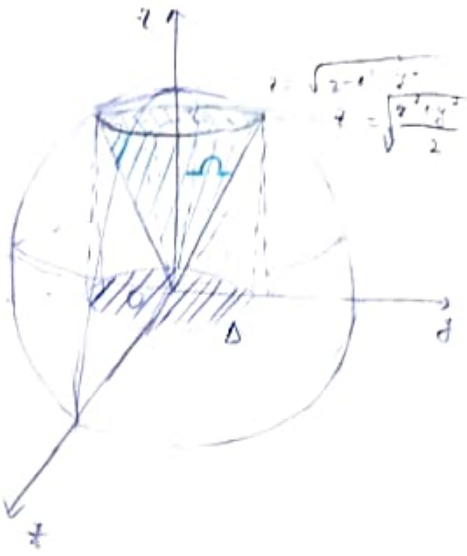
$$= \frac{\pi}{12} - \frac{\sin\theta}{3}\Big|_0^{\frac{\pi}{4}} + \frac{1}{3}\cdot\frac{\sin^3\theta}{3}\Big|_0^{\frac{\pi}{4}} =$$

$$= \frac{\pi}{12} - \frac{\sqrt{2}}{6} + \frac{1}{9}\cdot\left(\frac{\sqrt{2}}{2}\right)^3 = \frac{\pi}{12} - \frac{\sqrt{2}}{6} + \frac{1}{9}\cdot\frac{2\sqrt{2}}{8} =$$

$$= \frac{3\pi - 2\sqrt{2}}{12} + \frac{1}{9}\cdot\frac{\sqrt{2}}{4} = \frac{3\pi - 6\sqrt{2} + \sqrt{2}}{36} = \frac{3\pi - 5\sqrt{2}}{36}$$

10) $Vol(\Omega) = ?$

Ω mărg. de $\begin{cases} z^2 = \dfrac{x^2+y^2}{2} \\ x^2+y^2+z^2 = 3 \end{cases}$, $z>0$



$$Vol(\Omega) = \iiint_{\Omega} dx\,dy\,dz =$$

$$= \iint_D \int_{\sqrt{\frac{x^2+y^2}{2}}}^{\sqrt{3-x^2-y^2}} dz\,dx\,dy = \iint_D \left(\sqrt{3-x^2-y^2} - \sqrt{\frac{x^2+y^2}{2}} \right) dx\,dy =$$

$$= \iint_D \sqrt{3-x^2-y^2} - \sqrt{\frac{x^2+y^2}{2}}\; dx\,dy$$

~~$D: x^2+y^2 \le$~~ $\begin{cases} z^2 = \dfrac{x^2+y^2}{2} \\ x^2+y^2+z^2 = 3 \end{cases}$ $\Rightarrow x^2+y^2+\frac{x^2+y^2}{2} = 3 \Rightarrow$

$\Rightarrow 2x^2+2y^2+x^2+y^2 = 6 \Rightarrow 3x^2+3y^2 = 6 \Rightarrow x^2+y^2 = 2 \Rightarrow$

$\Rightarrow z^2 = \frac{2}{2} = 1 \Rightarrow z = 1 \Rightarrow$

$\Rightarrow D: x^2+y^2 \le 2$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho \in [0;\sqrt{2}]$, $\theta \in [0;2\pi]$

$$Vol(\Omega) = \int_0^{2\pi} \int_0^{\sqrt{2}} \left(\sqrt{3-\rho^2} - \sqrt{\frac{\rho^2}{2}} \right) \cdot \rho\, d\rho\, d\theta =$$

$$= \int_0^{2\pi} \left(\int_0^{\sqrt{2}} \sqrt{3-\rho^2} \cdot \rho\, d\rho - \frac{1}{\sqrt{2}} \int_0^{\sqrt{2}} \rho^3\, d\rho \right) d\theta =$$

$$= \int_0^{2\pi} \left(-\frac{1}{2} \cdot \frac{2}{3} \cdot (3-\rho^2)^{\frac{3}{2}} \Big|_0^{\sqrt{2}} - \frac{1}{\sqrt{2}} \cdot \frac{\rho^4}{4} \Big|_0^{\sqrt{2}} \right) d\theta =$$

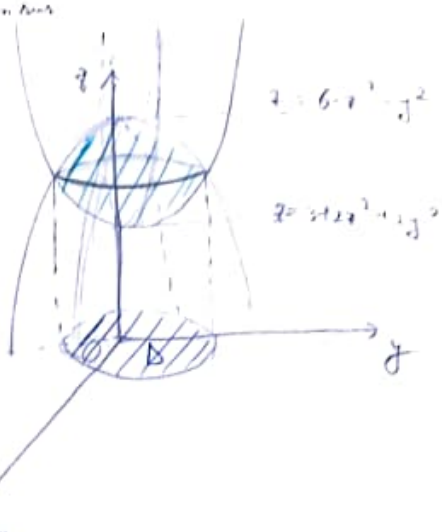
$$= \int_0^{2\pi} \left(-\frac{1}{3}(1-3\sqrt{3}) - \frac{1}{\sqrt{2}} \right) d\theta =$$

$$= \int_0^{2\pi} \left(-\frac{1}{3} + \sqrt{3} - \frac{1}{\sqrt{2}} \right) d\theta = \left(-\frac{\theta}{3} + \sqrt{3}\theta - \frac{\theta}{\sqrt{2}} \right) \Big|_0^{2\pi} =$$

$$= -\frac{2\pi}{3} + 2\pi\sqrt{3} - \frac{2\pi}{\sqrt{2}} = \pi\left(2\sqrt{3} - \frac{4}{3} - \sqrt{2} \right) = \pi \cdot \frac{6\sqrt{3}-2-\sqrt{2}}{3}$$

11) $Vol(\Omega) = ?$

Ω mărg. de $\begin{cases} z = 6 - x^2 - y^2 \\ z = 3 + 2x^2 + 2y^2 \end{cases}$ paraboloid eliptic



$$Vol(\Omega) = \iiint_{\Omega} dx\,dy\,dz = \iint_{D} \int_{3+2x^2+2y^2}^{6-x^2-y^2} dz\,dx\,dy =$$

$$= \iint_{D} (6 - x^2 - y^2 - 3 - 2x^2 - 2y^2)\,dx\,dy =$$

$$= \iint_{D} (3 - 3x^2 - 3y^2)\,dx\,dy =$$

$$= 3\iint_{D} (1 - x^2 - y^2)\,dx\,dy = 3\iint_{D} 1 - (x^2 + y^2)\,dx\,dy$$

$$\begin{cases} z = 6 - x^2 - y^2 \,/\cdot 2 \\ z = 3 + 2x^2 + 2y^2 \end{cases} \Leftrightarrow \begin{cases} 2z = 12 - 2x^2 - 2y^2 \\ z = 3 + 2x^2 + 2y^2 \end{cases}$$

$$3z = 15 \Rightarrow z = 5 \Rightarrow$$

$$\Rightarrow 5 = 6 - x^2 - y^2 \Rightarrow x^2 + y^2 = 1 \Rightarrow$$

$$\Rightarrow D: x^2 + y^2 \le 1$$

~~$Vol(\Omega) =$~~ $\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho \in [0;1]$, $\theta \in [0; 2\pi]$

$$Vol(\Omega) = 3\int_0^{2\pi}\int_0^1 (1 - \rho^2)\cdot\rho\,d\rho\,d\theta = 3\int_0^{2\pi} \left(\frac{\rho^2}{2} - \frac{\rho^4}{4}\right)\Big|_0^1 d\theta =$$

$$= 3\int_0^{2\pi} \frac{1}{2} - \frac{1}{4}\,d\theta = \frac{3}{2}\theta\,|_0^{2\pi} - \frac{3}{4}\theta\,|_0^{2\pi} =$$

$$= 3\pi - \frac{3\pi}{2} = \frac{3\pi}{2}$$

12) $Vol(\Omega) = ?$; Ω mărg. de $\begin{cases} z = 3 - 2x^2 - 2y^2 \\ x^2 + y^2 = z^2 \end{cases}$, $z > 0$

$$\begin{cases} z = 3 - 2x^2 - 2y^2 \\ z^2 = x^2 + y^2 \,/\cdot(+2) \end{cases} \Leftrightarrow \begin{cases} z = 3 - 2x^2 - 2y^2 \\ +2z^2 = +2x^2 + 2y^2 \end{cases}$$

$$z + 2z^2 = 3 \Rightarrow$$

$$\Rightarrow 2z^2 + z - 3 = 0$$

$$\Delta = 1 + 24 = 25 \Rightarrow z_{1,2} = \frac{-1 \pm 5}{4}$$

$$\left.\begin{array}{l} z_1 = -\frac{6}{4} = -\frac{3}{2} \\ z_2 = 1 \\ z > 0 \end{array}\right\} \Rightarrow z = 1 \Rightarrow$$

$$\Rightarrow x^2 + y^2 = 1 \Rightarrow D: x^2 + y^2 \le 1$$

$$Vol(\Omega) = \iiint_{\Omega} dx\,dy\,dz = \iint_{D} \int_{\sqrt{x^2+y^2}}^{3-2x^2-2y^2} dz\,dx\,dy =$$

$$= \iint_{D} \left(3 - 2x^2 - 2y^2 - \sqrt{x^2+y^2}\right) dx\,dy = \iint_{D} \left[3 - 2(x^2+y^2) - \sqrt{x^2+y^2}\right] dx\,dy.$$

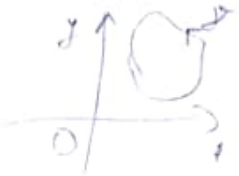
$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases} \quad \rho \in [0;1],\ \theta \in [0; 2\pi]$$

$$Vol(\Omega) = \int_0^{2\pi}\int_0^1 \left(3 - 2\rho^2 - \sqrt{\rho^2}\right)\rho\,d\rho\,d\theta = \int_0^{2\pi}\int_0^1 (3\rho - 2\rho^3 - \rho^2)\,d\rho\,d\theta =$$

$$= \int_0^{2\pi} \left(\frac{3}{2} - \frac{1}{2} - \frac{1}{3}\right) d\theta = \int_0^{2\pi} \left(1 - \frac{1}{3}\right) d\theta = \left(\theta - \frac{\theta}{3}\right)\Big|_0^{2\pi} =$$

$$= 2\pi - \frac{2\pi}{3} = 2\pi\left(1 - \frac{1}{3}\right)$$

13) $\int_\gamma \bar{v}\,d\bar{r} = ?$



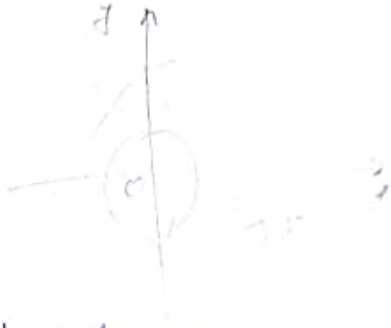
γ – drum închis arbitrar

$O \notin \gamma$

Cazul I) $O \in Ext(\gamma)$

Aplicând Green – Riemann $\Rightarrow \int_\gamma \bar{v}\,d\bar{r} = 0$

Cazul II) $O \in Int(\gamma)$



Nu se poate aplica Green-Riemann

Fie $C(O,R) \subset Int(\gamma)$

Fie $K = Ext\,C(O,R) \cap Int(\gamma)$

$\partial K = \overline{C(O,R)} \cup \gamma$

$$\int_{\partial K} P\,dx + Q\,dy = 0 \Rightarrow \int_{C(O,R)} P\,dx + Q\,dy + \int_{\gamma} P\,dx + Q\,dy = 0 \Rightarrow$$

$$\Rightarrow \int_\gamma P\,dx + Q\,dy = \int_{C(O,R)} P\,dx + Q\,dy = 2\pi$$

14) Să se calculeze circulația lui $\bar{v}$, $\bar{v} = \underbrace{\frac{-y}{x^2+y^2}}_{P}\bar{i} + \underbrace{\frac{x}{x^2+y^2}}_{Q}\bar{j}$

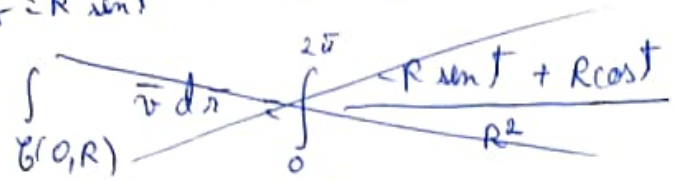
a) $$\int_{C(O,R)} \bar{v}\,d\bar{r} = \int_{C(O,R)} \frac{-y}{x^2+y^2}\,dx + \frac{x}{x^2+y^2}\,dy$$

$$\frac{\partial Q}{\partial x} = \frac{x^2+y^2 - 2x^2}{(x^2+y^2)^2}$$

$$\frac{\partial P}{\partial y} = \frac{-x^2 - y^2 + 2y^2}{(x^2+y^2)^2}$$

$$\Rightarrow \int_{C(O,R)} \bar{v}\,d\bar{r} = \iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,dx\,dy =$$

$\begin{cases} x = R\cos t \\ y = R\sin t \end{cases} \quad t \in [0; 2\pi]$



$$\int_{\mathcal{C}(0,R)} \bar{v}\, d\bar{r} = \int_0^{2\pi} \frac{-R\sin t}{R^2} \cdot (-R\sin t) + \frac{R\cos t}{R^2} \cdot R\cos t \, dt =$$

$$= \int_0^{2\pi} \frac{R^2\sin t + R^2\cos t}{R^2} dt = \int_0^{2\pi} \frac{R^2}{R^2} dt = \int_0^{2\pi} dt = 2\pi$$

b) γ curbă arbitrară închisă, $0 \notin \gamma$



Cazul 1) $0 \in \mathrm{Ext}(\gamma)$

$$\int_\gamma \bar{v}\, d\bar{r} = \int_\gamma P dx + Q dy = \iint_K \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \, dx dy$$

$$\frac{\partial Q}{\partial x} = \frac{x^2 + y^2 - 2x^2}{(x^2+y^2)^2}$$

$$\frac{\partial P}{\partial y} = \frac{-x^2 - y^2 + 2y^2}{(x^2+y^2)^2}$$

$$\Rightarrow \int_\gamma \bar{v}\, d\bar{r} = \iint_K \frac{x^2 + y^2 - 2x^2 + x^2 + y^2 - 2y^2}{(x^2+y^2)^2} dx dy = 0$$

Cazul 2) $0 \in \mathrm{Int}(\gamma)$

$\exists R > 0$ a.î. $D(0,R) \subset K$

$A = K \setminus D(0,R)$

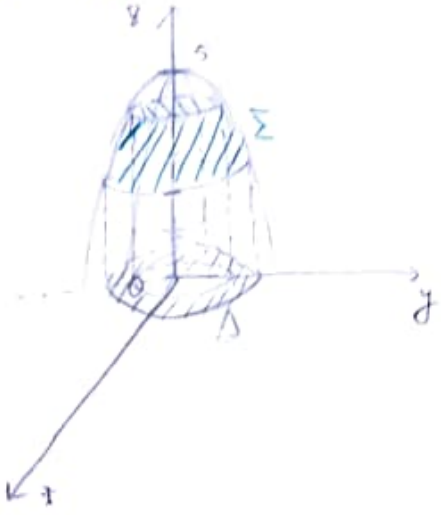
$\mathrm{Fr}\, A = \gamma \cup \mathcal{C}(0,R)$

$$\left.\begin{aligned} &\int_{\mathrm{Fr}\, A} P dx + Q dy \overset{G-R}{=} \iint_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx dy = 0 \\ &\int_{\mathrm{Fr}\, A} P dx + Q dy = \int_\gamma P dx + Q dy + \int_{\mathcal{C}(0,R)} P dx + Q dy \end{aligned}\right\} \Rightarrow$$

$$\Rightarrow \int_\gamma P dx + Q dy = \int_{\mathcal{C}(0,R)} P dx + Q dy \overset{\text{conf. lui a)}}{=} 2\pi$$

c) $\bar{v}$ câmp de gradienți $\Leftrightarrow \int_\gamma \bar{v}\, d\bar{r} = 0, \forall \gamma$ curbă închisă

15) Aria (Σ) = ? ; $\Sigma : z = 5 - x^2 - y^2$, $z \in [2,4]$



$z = 5 - x^2 - y^2 = f(x,y)$

$$\int_\Sigma d\sigma = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx\,dy =$$

$$= \iint_D \sqrt{1 + 4x^2 + 4y^2}\, dx\,dy =$$

$$= \iint_D \sqrt{1 + 4(x^2+y^2)}\, dx\,dy$$

~~$z = f(x,y)$~~ $\left.\begin{array}{l} z = 5 - x^2 - y^2 \\ 2 \le z \le 4 \end{array}\right\} \Rightarrow$

$\Rightarrow 2 \le 5 - x^2 - y^2 \le 4 \Leftrightarrow -3 \le -x^2 - y^2 \le -1 \Leftrightarrow$

$\Leftrightarrow 3 \ge x^2 + y^2 \ge 1$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases} \quad \rho \in [1; \sqrt{3}] \ , \ \theta \in [0; 2\pi]$$

$$\int_\Sigma d\sigma = \int_0^{2\pi}\int_1^{\sqrt{3}} \sqrt{1+4\rho^2} \cdot \rho\, d\rho\, d\theta = \frac{1}{8}\int_0^{2\pi}\int_1^{\sqrt{3}} (1+4\rho^2)^1 \cdot \sqrt{1+4\rho^2}\, d\rho\, d\theta =$$

$$= \frac{1}{8}\int_0^{2\pi} \frac{2}{5}\cdot(1+4\rho^2)^{\frac{5}{2}} \Big|_1^{\sqrt{3}} d\theta = \frac{1}{12}\int_0^{2\pi} (13\sqrt{13} + 5\sqrt{5})\, d\theta =$$

$$= \frac{1}{12}(13\sqrt{13} + 5\sqrt{5})\,\theta \Big|_0^{2\pi} = \frac{\pi}{6}(13\sqrt{13} + 5\sqrt{5})$$

16) Aria (Σ) = ? ; $\Sigma : z^2 = 3x^2 + 3y^2$, $z \in [1,3]$

~~z^2~~

~~$\int_\Sigma d\sigma = \iint_D \sqrt{1+}$~~



$z^2 = 3x^2 + 3y^2 \Leftrightarrow z = \sqrt{3x^2+3y^2} = f(x,y)$

$$\int_\Sigma d\sigma = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx\,dy =$$

$$= \iint_D \sqrt{1 + \left(\frac{3x}{\sqrt{3x^2+3y^2}}\right)^2 + \left(\frac{3y}{\sqrt{3x^2+3y^2}}\right)^2}\, dx\,dy =$$

$$= \iint_D \sqrt{1 + \frac{9x^2+9y^2}{3x^2+3y^2}}\, dx\,dy = \iint_D \sqrt{1 + \frac{9(x^2+y^2)}{3(x^2+y^2)}}\, dx\,dy =$$

$$= \iint_D \sqrt{1+3}\, dx\,dy = \iint_D 2\, dx\,dy = 2\iint_D dx\,dy = 2\,\text{Aria}(D)$$

$\Rightarrow \int_\Sigma d\sigma = \frac{16\pi}{3}$

Aria $(D) = \pi \cdot (\sqrt{3})^2 - \pi \cdot \left(\frac{1}{\sqrt{3}}\right)^2 = 3\pi - \frac{\pi}{3} = \frac{8\pi}{3}$

$3x^2 + 3y^2 = 1 \Leftrightarrow x^2 + y^2 = \frac{1}{3}$

$3x^2 + 3y^2 = 9 \Leftrightarrow x^2 + y^2 = 3$

15

(7) $\bar{v} = \underbrace{(x^2+y^2+z^2)}_{P}\bar{i} + \underbrace{(z^2-2yx)}_{Q}\bar{j} + \underbrace{(3z-x-y)}_{R}\bar{k}$

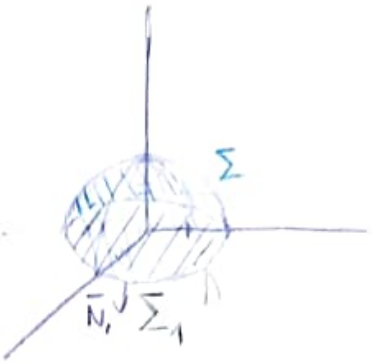
$\Sigma : x^2+y^2+z^2 = 4 \ , \ z>0$

$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \iiint_{\Omega} \operatorname{div}(\bar{v})\,dx\,dy\,dz$

$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} =$

$= 2x + 0 - 2x + 3 = 3 \Rightarrow \mathcal{F}_{\Sigma\cup\Sigma_1}\bar{v} =$

$= 3\iiint_{\Omega} dx\,dy\,dz = 3 \cdot \frac{2}{3}\pi R^3 \cdot \frac{1}{2} = 2\pi R^3$ (volumul semisferei, $R=2$) $\Rightarrow \mathcal{F}_{\Sigma\cup\Sigma_1}\bar{v} = 16\pi$



$\mathcal{F}_{\Sigma}(\bar{v}) + \mathcal{F}_{\Sigma_1}(\bar{v}) = 16\pi \Rightarrow \mathcal{F}_{\Sigma}(\bar{v}) = 16\pi - \mathcal{F}_{\Sigma_1}(\bar{v})$

$\mathcal{F}_{\Sigma_1}(\bar{v}) = \iint_{\Sigma_1} \bar{v}\,\bar{N}_1\,d\sigma$

$\bar{N}_1 = -\bar{k}$

$\Sigma_1 : z = 0$

$D : x^2+y^2 \le 4$

$\mathcal{F}_{\Sigma_1}(\bar{v}) = -\iint_D (0-x-y)\,dx\,dy = \iint_D (x+y)\,dx\,dy$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \ \rho\in[0;2], \ \theta\in[0;2\pi]$

$\mathcal{F}_{\Sigma_1}(\bar{v}) = \int_0^{2\pi}\int_0^2 (\rho\cos\theta + \rho\sin\theta)\cdot\rho\,d\rho\,d\theta = \int_0^{2\pi}\int_0^2 \rho^2(\sin\theta+\cos\theta)\,d\rho\,d\theta =$

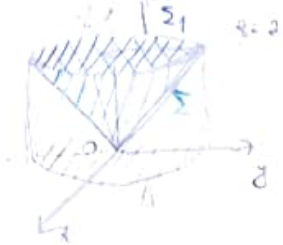
$= \frac{1}{3}\int_0^{2\pi}\int_0^2 \rho^3(\sin\theta+\cos\theta)\Big|_0^2\,d\theta = \frac{8}{3}\int_0^{2\pi}(\sin\theta+\cos\theta)\,d\theta =$

$= \frac{8}{3}(-\cos\theta+\sin\theta)\Big|_0^{2\pi} = -\frac{8}{3}+\frac{8}{3} = 0$

$\Rightarrow \mathcal{F}_{\Sigma}(\bar{v}) = 16\pi - 0 = 16\pi$

(18) $\bar{v} = \underbrace{(2xy - z^2)}_{P}\bar{i} + \underbrace{(z^2-y^2)}_{Q}\bar{j} + \underbrace{(x^2+y^2-3z)}_{R}\bar{k}$

$\Sigma : z^2 = x^2+y^2 \ , \ z\in[0,2]$



$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \iiint_{\Omega} \operatorname{div}(\bar{v})\,dx\,dy\,dz$

$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 2y - 2y - 3 \Rightarrow$

$\Rightarrow \mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \iiint_{\Omega} (2z-2y-3)\,dx\,dy\,dz = \iint_D \int_0^{\cdots} -3\iiint_{\Omega} dx\,dy\,dz =$

$= -3 \cdot \bar{i} \cdot 2^2 \cdot \frac{1}{3_1} \cdot 2 = -8\bar{i}$

$\mathcal{F}_{\Sigma \cup \Sigma_1}(\bar{v}) = \mathcal{F}_{\Sigma}(\bar{v}) + \mathcal{F}_{\Sigma_1}(\bar{v}) \Rightarrow \mathcal{F}_{\Sigma}(\bar{v}) = \mathcal{F}_{\Sigma \cup \Sigma_1}(\bar{v}) - \mathcal{F}_{\Sigma_1}(\bar{v}) =$

$= -8\bar{i} - \mathcal{F}_{\Sigma_1}(\bar{v})$

$\mathcal{F}_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v} \cdot \bar{N}_1 \, d\sigma =$

$\bar{N}_1 = \bar{k}$

$z = 2 \Rightarrow x^2 + y^2 = 4 \Rightarrow D: x^2 + y^2 \le 4$

$\mathcal{F}_{\Sigma_1}(\bar{v}) = \iint_D (x^2 + y^2 - 6) \, dx \, dy$

$\begin{cases} x = \rho \cos\theta \\ y = \rho \sin\theta \end{cases} \quad \rho \in [0, 2], \ \theta \in [0, 2\pi] \Rightarrow \mathcal{F}_{\Sigma_1}(\bar{v}) = \int_0^{2\pi} \int_0^2 (\rho^2 - 6) \rho \, d\rho \, d\theta =$

$= \int_0^{2\pi} \left(\frac{\rho^4}{4} - \frac{6\rho^2}{2} \right) \Big|_0^2 d\theta = \int_0^{2\pi} (4 - 12) d\theta = -8 \int_0^{2\pi} d\theta =$

$= -8\theta \Big|_0^{2\pi} = -16\pi$

$\mathcal{F}_{\Sigma}(\bar{v}) = -8\bar{i} + 16\bar{i} = 8\bar{i}$

19) $\bar{v} = \underbrace{(x + y^2)}_{P} \bar{i} + \underbrace{(z^2 - y)}_{Q} \bar{j} + \underbrace{(z^2 + z)}_{R} \bar{k}$

$\Sigma: z - 1 = x^2 + y^2, \ z \in [1, 2]$



$\mathcal{F}_{\Sigma \cup \Sigma_1}(\bar{v}) = \iiint_{\Omega} \operatorname{div}(\bar{v}) \, dx \, dy \, dz$

$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 1 - 1 + 2z = 2z \Rightarrow$

$\Rightarrow \mathcal{F}_{\Sigma \cup \Sigma_1}(\bar{v}) = \iiint_{\Omega} 2z \, dz \, dx \, dy =$

$= 2 \iint_D \int_{x^2+y^2+1}^{2} z \, dz \, dx \, dy = \cancel{2 \iint_D 2 - x^2 - y^2 - 1 \, dx \, dy =}$

$= \iint_D 4 - (x^2 + y^2 + 1)^2 \, dx \, dy$

$0 \le x^2 + y^2 \le 1 \Rightarrow D: x^2 + y^2 \le 1$

$\begin{cases} x = \rho \cos\theta \\ y = \rho \sin\theta \end{cases} \quad ; \rho \in [0; 1], \ \theta \in [0, 2\pi]$

$\mathcal{F}_{\Sigma \cup \Sigma_1}(\bar{v}) = \int_0^{2\pi} \int_0^1 [4 - (\rho^2 + 1)^2] \cdot \rho \, d\rho \, d\theta = \int_0^{2\pi} \int_0^1 (4 - \rho^4 - 2\rho^2 - 1) \rho \, d\rho \, d\theta =$

$= \int_0^{2\pi} \int_0^1 (3\rho - \rho^5 - 2\rho^3) \, d\rho \, d\theta = \int_0^{2\pi} \left(3\rho^2 \cdot \frac{1}{2} - \rho^6 \cdot \frac{1}{6} - 2 \cdot \rho^4 \cdot \frac{1}{4} \right) \Big|_0^1 d\theta =$

$= \int_0^{2\pi} \left(\frac{3}{2} - \frac{1}{6} - \frac{1}{2} \right) d\theta = \frac{5}{2} \theta \Big|_0^{2\pi} = 5\pi$

$\Phi_{\Sigma\cup\Sigma_1}(\bar v) = \Phi_\Sigma(\bar v) + \Phi_{\Sigma_1}(\bar v) \Rightarrow \Phi_\Sigma(\bar v) = \Phi_{\Sigma\cup\Sigma_1}(\bar v) - \Phi_{\Sigma_1}(\bar v) =$

$= 5\bar u - \Phi_{\Sigma_1}(\bar v)$

$\Phi_{\Sigma_1}(\bar v) = \int_{\Sigma_1} \bar v \cdot \bar N_1 \, d\sigma$

$\bar N_1 = \bar k$

$\Sigma_1 : z = 2$

$D: x^2 + y^2 \le 1$

$\Phi_{\Sigma_1}(\bar v) = \iint_D (\bar b \cdot \bar k)\, dx\,dy = \int_0^{2\pi}\int_0^1 \rho\cos\theta\,\rho\,d\rho\,d\theta = \int_0^{2\pi}\cos\theta\,d\theta = \sin\theta\Big|_0^{2\pi} =$

$= 0 - 0 = 0 \Rightarrow \Phi_\Sigma(\bar v) = 5\bar u - 0 = 5\bar u$

$= \int_0^{2\pi}\int_0^1 \rho\cos\theta\cdot\rho\,d\rho\,d\theta = \int_0^{2\pi}\int_0^1 \rho^2\cos\theta\,d\rho\,d\theta = \int_0^{2\pi} \frac{1}{3}\rho^3\cos\theta\Big|_0^1 d\theta =$

$= \frac{1}{3}\int_0^{2\pi}\cos\theta\,d\theta = \frac{1}{3}\sin\theta\Big|_0^{2\pi} = 0 \Rightarrow \Phi_\Sigma(\bar v) = 5\bar u - 0 = 5\bar u$

$= \iint_D (4 + x)\,dx\,dy = \int_0^{2\pi}\int_0^1 (4 + \rho\cos\theta)\,\rho\,d\rho\,d\theta = \int_0^{2\pi}\int_0^1 (4\rho + \rho^2\cos\theta)\,d\rho\,d\theta =$

$= \int_0^{2\pi}\left(4\cdot\frac{\rho^2}{2} + \frac{\rho^3}{3}\cos\theta\right)\Big|_0^1 d\theta = \int_0^{2\pi}\left(2 + \frac{1}{3}\cos\theta\right)d\theta =$

$= \left(2\theta + \frac{1}{3}\sin\theta\right)\Big|_0^{2\pi} = 4\bar u \Rightarrow \Phi_\Sigma(\bar v) = 5\bar u - 4\bar u = \bar u$

20) $\bar v = (y - z)\bar i + (z - x)\bar j + (x - y)\bar k$

$\gamma: \begin{cases} x^2 + y^2 = a^2 \\ \frac{x}{a} + \frac{z}{h} = 1, \quad a, h > 0 \end{cases}$

$C = \int_\gamma \bar v\, d\bar r \overset{\text{Stokes}}{=\!=\!=} \int_\Sigma \operatorname{rot}(\bar v)\cdot\bar N\, d\sigma = \iint_D \operatorname{rot}(\bar v)\cdot\frac{\bar n_x \times \bar n_y}{\|\bar n_x \times \bar n_y\|}\cdot$

$\cdot\|\bar n_x \times \bar n_y\|\, dx\,dy = \iint_D \operatorname{rot}(\bar v)(\bar n_x \times \bar n_y)\, dx\,dy$

$\operatorname{rot}(\bar v) = \begin{vmatrix} \bar i & \bar j & \bar k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y - z & z - x & x - y \end{vmatrix} = -\bar i - \bar j - \bar k - \bar k - \bar i - \bar j =$

$= -2(\bar i + \bar j + \bar k)$

~~$\bar r = x\bar i + y\bar j + z$~~

$\frac{z}{h} = 1 - \frac{x}{a} \Rightarrow z = h - x\cdot\frac{h}{a} = h\left(1 - \frac{x}{a}\right)$, $\Sigma: z = h\left(1 - \frac{x}{a}\right)$

$D: x^2 + y^2 \le a^2$

$r = x\bar i + y\bar j + h\left(1 - \frac{x}{a}\right)\bar k$

$\bar n_x = \bar i - \frac{h}{a}\bar k$

$\bar n_y = \bar j$

$$\text{rot}\,\bar{v}\cdot(\bar{n}_x \times \bar{n}_y) = \begin{vmatrix} -2 & -2 & -2 \\ a & 0 & -\frac{h}{a} \\ 0 & 1 & 0 \end{vmatrix} = -2 + \frac{2h}{a} = 2\left(\frac{h}{a}-1\right) \Rightarrow$$

$$\Rightarrow C = \iint_D 2\left(\frac{h}{a}-1\right)dxdy = 2\left(\frac{h}{a}-1\right)\iint_D dxdy = 2\left(\frac{h}{a}-1\right)\cdot\tilde{u}\cdot a^2$$

21) $\bar{v} = \underbrace{z}_{P}\bar{i} + \underbrace{xy}_{Q}\bar{j} + \underbrace{xyz}_{R}\bar{k}$

$\Sigma: x^2+y^2+z^2 = 1,\ z \geq 0$

$\Rightarrow z = \sqrt{1-(x^2+y^2)}$

$D: x^2+y^2 \leq 1$

$\gamma: \begin{cases} x^2+y^2 = 1 \\ z = 0 \end{cases}$

$$\mathcal{T}_\Sigma(\text{rot}(\bar{v})) = \int_\Sigma \text{rot}(\bar{v})\cdot\bar{N}\,d\sigma \overset{\text{Stokes}}{=\!=} \int_\gamma \bar{v}\,d\bar{r} = \int_\gamma P dx + Q dy + R dz =$$

$$= \int_\gamma z dx + x dy + xyz\,dz$$

$$\begin{cases} x = \cos t \\ y = \sin t \\ z = 0 \end{cases}, t \in [0, 2\tilde{u}]$$

$$\Rightarrow \mathcal{T}_\Sigma(\text{rot}\,\bar{v}) = \int_0^{2\tilde{u}} (0 + \cos t\cdot\cos t + \cancel{\sin t\cdot 0}\cdot 0)\,dt =$$

$$= \int_0^{2\tilde{u}} \cos^2 t\,dt = \int_0^{2\tilde{u}} \frac{1+\cos 2t}{2}\,dt = \frac{1}{2}\left(\int_0^{2\tilde{u}} 1\,dt + \cancel{\int_0^{2\tilde{u}} \cos 2t\,dt}\right) =$$

$$= \frac{1}{2}t\Big|_0^{2\tilde{u}} + \frac{1}{4}\sin(2t)\Big|_0^{2\tilde{u}} = \tilde{u} + 0 - 0 = \tilde{u}$$

22) $x^3+9x-3=0,\ \varepsilon < 10^{-2}$

$f(x) = x^3+9x-3$

$f(0) = -3$

$f(1) = 7$

$f(0)\cdot f(1) < 0$, f continuă $\Rightarrow f(x)=0$ are cel puțin o soluție reală în $(0,1)$

$f'(x) = 3x^2+9 > 0$ pe $\mathbb{R}$ $\Rightarrow$

$\Rightarrow f$ are o unică soluție în $(0,1)$

$x^3+9x-3=0$

$x(x^2+9) = 3 \Rightarrow x = \frac{3}{x^2+9}$

fie $T(x) = \frac{3}{x^2+9}$, $T: [0;1] \to [0;1]$

19

$[0,1]$ – spațiu metric complet

Conform teoremei Lagrange $\Rightarrow \exists c \in [0,1]$ a.î. $|T(x) - T(y)| = |T'(c)| \cdot |x-y| \leq \sup |T'(c)| \cdot |x-y|$

$T'(x) = -\frac{3}{(x^2+9)^2} \cdot 2x \Rightarrow |T'(x)| = \frac{6x}{(x^2+9)^2} = g(x)$

$g'(x) = \frac{6(x^2+9)' - 2(x^2+9)\cdot 2x \cdot 6x}{(x^2+9)^3} = \frac{6x^2+54-24x^2}{(x^2+9)^3} > 0, (\forall) x \in [0,1]$

g crescătoare $\Rightarrow \sup g(x) = \sup |T'(x)| = g(1) = \frac{6}{100} = \frac{3}{50}$

$x \in [0,1]$

Din (1) $\Rightarrow T$ contracție de factor $K = \frac{3}{50}$ $\Rightarrow$ Conform Th. Contracției

$[0,1]$ spațiu metric complet

că $\exists! \xi \in [0,1]$ a.î. $T(\xi) = \xi$

$x_0 = 0$ și $x_{n+1} = T(x_n), (\forall) n \in \mathbb{N}$

$|x_n - \xi| \leq \frac{K^n}{1-K} \cdot |x_1 - x_0| < 10^{-2}$

$x_1 = T(x_0) = \frac{3}{9} = \frac{1}{3}$

$\frac{\left(\frac{3}{50}\right)^n}{1-\frac{3}{50}} \cdot \frac{1}{3} < 10^{-2} \Leftrightarrow \left(\frac{3}{50}\right)^n \cdot \frac{50}{47} \cdot \frac{1}{3} < 10^{-2} \Rightarrow$

$\Rightarrow \left(\frac{3}{50}\right)^{n-1} \cdot \frac{1}{47} < \frac{1}{100} \Rightarrow n = 2 \Rightarrow$

$\Rightarrow \xi \simeq x_2 = T(x_1) = \frac{3}{\frac{1}{9}+9} = \frac{3}{28} \cdot 3 = \frac{9}{28}$

23) $x^3 + 4x - 1 = 0$, $\varepsilon < 10^{-3}$

$f(x) = x^3 + 4x - 1$

$f(0) = -1 < 0$

$f(1) = 4 > 0$

$f(0) \cdot f(1) < 0$
f continuă
$f'(x) = 3x^2 + 4$
$x \in [0,1]$

$\Rightarrow f$ are o unică soluție în $[0,1]$

aprox $\xi \in [0,1]$ cu $\varepsilon < 10^{-3}$

$T: X \to X$ contractie de factor k

$x^3 + 4x - 1 = 0 \Rightarrow x^3 + 4x = 1 \Rightarrow x(x^2+4) = 1 \Rightarrow$

$\Rightarrow x = \frac{1}{x^2+4}$

fie $T(x) = \frac{1}{x^2+4}$

$T: [0,1] \to [0,1]$

$T'(x) = \frac{-2x}{(x^2+4)^2}$

$|T'(x)| = \frac{2x}{(x^2+4)^2} = g(x)$

$\max |g(x)| < 1$

$x \in [0,1]$

$g'(x) = \ldots = \frac{8-6x^2}{(x^2+4)^3}$, $x \in [0,1]$

$\max |T'(x)| = \frac{2}{25} < 1$

$x_0 = 0$

$x_1 = T(0) = \frac{1}{4}$

$|x_m - \xi| \leq \frac{\left(\frac{2}{25}\right)^m}{1 - \frac{2}{25}} \cdot \frac{1}{4}$

$|x_m - \xi| \leq \left(\frac{2}{25}\right)^m \cdot \frac{25}{23} \cdot \frac{1}{4} < \varepsilon$

$\left(\frac{2}{25}\right)^m \cdot \frac{1}{3} < \frac{1}{1000}$

$\xi \approx x_3$

$x_2 = T\left(\frac{1}{4}\right) = \frac{1}{\frac{1}{16}+4} = \frac{16}{65}$

$x_3 = T\left(\frac{16}{65}\right) = \ldots$

24) $\bar{v} = \left(\underbrace{\frac{-y}{x^2+y^2}}_{P}, \underbrace{\frac{x}{x^2+y^2}}_{Q} \right)$

a) $\int_{\mathcal{C}(O,R)} \bar{v}\, d\bar{r} = ?$

$O \in \text{Int}\, \mathcal{C}(O,R) \Rightarrow$ nu se poate aplica G-R

$\begin{cases} x = \cancel{\cos t}\ R\cos t \\ y = R \sin t \end{cases}, t \in [0, 2\pi] \Rightarrow \int_{\mathcal{C}(O,R)} \bar{v}\, d\bar{r} = \int_0^{2\pi} \left(\frac{-R\sin t}{R^2} \cdot (-R\sin t) + \frac{R\cos t}{R^2} \cdot R\cos t \right) dt = \int_0^{2\pi} \frac{R^2}{R^2}\, dt = \int_0^{2\pi} dt = t\Big|_0^{2\pi} = 2\pi - 0 = 2\pi$

b) Fie γ o curbă arbitrară închisă

Cazul I: $O \in \text{Ext}(\gamma) \Rightarrow$ se ~~poate~~ poate aplica G-R $\Rightarrow$

$\Rightarrow \int_\gamma \bar{v}\, d\bar{r} = \iint_K \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dx dy$

$\frac{\partial Q}{\partial x} = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2}$

$\frac{\partial P}{\partial y} = \frac{-x^2-y^2+2y^2}{(x^2+y^2)^2}$

$\Rightarrow \int_\gamma \bar{v}\, d\bar{r} = \iint_K \frac{0}{(x^2+y^2)^2}\, dx dy = 0$

Cazul II: $O \in \text{Int}(\gamma) \Rightarrow$ nu se poate aplica G-R

Fie $R > 0$ a.î. $D(O,R) \subset K$

$A = K \setminus D(O,R)$

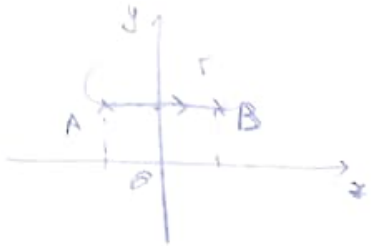
$\text{Fr}\, A = \gamma \cup \mathcal{C}(O,R)$

$\left. \begin{aligned} \int_{\text{Fr}\, A} P dx + Q dy \overset{G-R}{=} \iint_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = 0 \end{aligned} \right\} \Rightarrow$

$\int_{\text{Fr}\, A} P dx + Q dy = \int_\gamma P dx + Q dy - \int_{\mathcal{C}(O,R)} P dx + Q dy$

$\Rightarrow \int_\gamma P dx + Q dy = \int_{\mathcal{C}(O,R)} P dx + Q dy \overset{\text{conf. lui a}}{=\!=} 2\pi$

c) $\int_{\gamma \cup [AB]} \bar{v}\, d\bar{r} \overset{G-R}{=\!=} \iint_K \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dx dy = 0 \Rightarrow \int_\gamma = -\int_{[AB]} \bar{v}\, d\bar{r}$



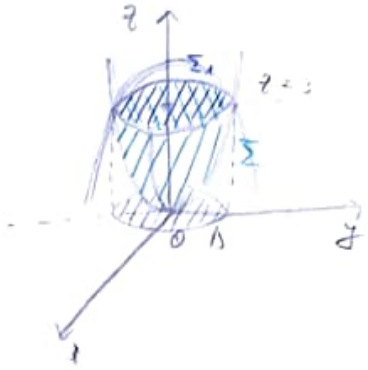
$$\begin{cases} x = t \\ y = 1 \end{cases}, t \in [-1;1]$$

$$\int_\gamma \bar{v}\,d\bar{r} = -\int_{-1}^{1} \left(\frac{t}{t^2+1} - \frac{1}{t^2+1}\cdot t + \frac{t}{t^2+1}\cdot 0 \right) dt =$$

$$= \int_{-1}^{1} \frac{1}{t^2+1}\,dt = \operatorname{arctg} t \Big|_{-1}^{1} = \operatorname{arctg} 1 - \operatorname{arctg}(-1) =$$

$$= \operatorname{arctg} 1 + \operatorname{arctg} 1 = \frac{\pi}{4} + \frac{\pi}{4} = \frac{2\pi}{4} = \frac{\pi}{2}$$

25) $\bar{v} = \underbrace{x^3 y^2}_{P}\,\bar{i} + \underbrace{x^2 y^3}_{Q}\,\bar{j} + \underbrace{3z}_{R}\,\bar{k}$



$\Sigma : z = x^2+y^2, \; z \le 6 - x^2 - y^2, \; 0 \le z \le 9$

$\Phi_\Sigma(\bar{v}) = ?$

$$\Phi_{\Sigma \cup \Sigma_1} \bar{v} \overset{G-O}{=\!=} \iiint_\Omega \operatorname{div}(\bar{v})\,dx\,dy\,dz$$

$$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 3x^2y^2 + 3x^2y^2 + 3 = 6x^2y^2 + 3$$

$$\Rightarrow \Phi_{\Sigma \cup \Sigma_1} \bar{v} = \iiint_\Omega (6x^2y^2 + 3)\,dx\,dy\,dz = 3\iiint_\Omega (2x^2y^2 + 1)\,dx\,dy\,dz$$

$$\begin{cases} z = x^2+y^2 \\ z = 6 - x^2 - y^2 \\ 0 \le z \le 6 \end{cases} \Rightarrow z = 6 - z \Rightarrow 2z = 6 \Rightarrow z = 3 \in [0;6] \Rightarrow$$

$$\Rightarrow \Phi_{\Sigma \cup \Sigma_1} \bar{v} = 3 \iint_D \left(\int_{x^2+y^2}^{3} (2x^2y^2 + 1)\,dz \right) dx\,dy = \cancel{\text{fff}}$$

$$= 3\iint_D \left(2x^2y^2 z + z \right) \Big|_{x^2+y^2}^{3} dx\,dy = 3\iint_D \left(6x^2y^2 + 3 - 2x^4y^4 - x^2y^2 - 2x^2y^2(x^2+y^2) - (x^2+y^2) \right) dx\,dy = 3\iint_D (2x^2y^2 + 1)\left[3 - (x^2+y^2) \right] dx\,dy$$

$$\begin{cases} z \le 6 - x^2 - y^2 \\ z = 3 \end{cases} \Rightarrow 3 \le 6 - x^2 - y^2 \Rightarrow -3 \le -x^2 - y^2 \Rightarrow x^2 + y^2 \le 3 \Rightarrow$$

$$\Rightarrow D : x^2 + y^2 \le 3 \Rightarrow \Phi_{\Sigma \cup \Sigma_1} \bar{v} = 3\int_0^{2\pi} \int_0^{\sqrt{3}} (2\rho^4 \cos^2\theta \cdot \sin^2\theta + 1)(3 - \rho^2) \cdot \rho\,d\rho\,d\theta =$$

$$= \cancel{3\int_0^{2\pi} \frac{1}{3}\rho^6 \cos^2\theta \cdot \sin^2\theta} = \alpha$$

$$\Phi_\Sigma(\bar{v}) = \alpha - \Phi_{\Sigma_1}(\bar{v})$$

$$\Phi_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v} \cdot \bar{N}_1\,d\sigma = \int_{\Sigma_1} 3z\,dx\,dy = \iint_D 3\rho\,d\rho\,d\varphi = 3 \cdot \pi \cdot (\sqrt{3})^2 = 2\pi$$

$$\Rightarrow \Phi_\Sigma(\bar{v}) = \alpha - 2\pi$$

Rezolvare exercitii date seminar 13

1) a) $J=\int_0^\infty \frac{1}{\sqrt[3]{x^2}(1+x^3)}dx = \int_0^\infty \frac{x^{-\frac{2}{3}}}{(1+x^3)}dx$

$x^3=y \Leftrightarrow x=\sqrt[3]{y} \Rightarrow dx=\frac{1}{3\sqrt[3]{y^2}}dy$

$J=\int_0^\infty \frac{y^{-\frac{2}{3}}}{(1+y)^1}\cdot\frac{1}{3\sqrt[3]{y^2}}dy=\frac{1}{3}\int_0^\infty \frac{y^{-\frac{2}{3}}\cdot y^{-\frac{2}{3}}}{(1+y)^1}dy=$

$=\frac{1}{3}\int_0^\infty \frac{y^{-\frac{8}{9}}}{(1+y)^1}dy$

$\beta(p,q)=\int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}dy$

$p-1=-\frac{8}{9} \Rightarrow p=-\frac{8}{9}+\frac{9}{9}=\frac{1}{9}$

$p+q=1 \Rightarrow q=\frac{9}{9}-\frac{1}{9}=\frac{8}{9}$

$\Rightarrow J=\frac{1}{3}\beta\left(\frac{1}{9};\frac{8}{9}\right)$

$\beta(p,q)=\frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$\Rightarrow J=\frac{1}{3}\cdot\frac{\Gamma(\frac{1}{9})\cdot\Gamma(\frac{8}{9})}{\Gamma(1)}=\frac{1}{3}\Gamma\left(\frac{1}{9}\right)\cdot\Gamma\left(1-\frac{1}{9}\right)=$

$=\frac{1}{3}\cdot\frac{\pi}{\sin\frac{\pi}{9}}=\frac{\pi}{3}\cdot\frac{1}{\sin\frac{\pi}{9}}$

d) $J=\int_0^1 \frac{x^4}{\sqrt{1-x^2}}dx$

$x^2=y \Rightarrow x=\sqrt{y} \Rightarrow dx=\frac{1}{2\sqrt{y}}dy$

$J=\int_0^1 y^2\cdot(1-y)^{-\frac{1}{2}}\cdot\frac{1}{2}\cdot y^{-\frac{1}{2}}dy=\frac{1}{2}\int_0^1 y^{\frac{3}{2}}\cdot(1-y)^{-\frac{1}{2}}dy$

$\beta(p,q)=\int_0^1 y^{p-1}\cdot(1-y)^{q-1}dy$

$p-1=\frac{3}{2} \Rightarrow p=\frac{3}{2}+\frac{2}{2}=\frac{5}{2}$

$q-1=-\frac{1}{2} \Rightarrow q=-\frac{1}{2}+\frac{2}{2}=\frac{1}{2}$

$J=\frac{1}{2}\beta\left(\frac{5}{2};\frac{1}{2}\right)$

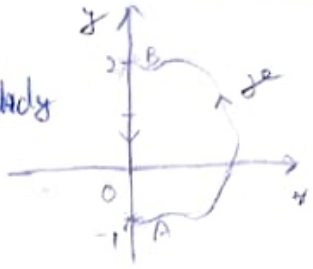
$$B(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$$

$$\Rightarrow I = \frac{1}{2}\cdot\frac{\Gamma\left(\frac{5}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(3)} = \frac{1}{4}\Gamma\left(1+\frac{3}{2}\right)\cdot\Gamma\left(\frac{1}{2}\right) =$$

$$= \frac{1}{4}\cdot\frac{3}{2}\cdot\Gamma\left(\frac{3}{2}\right)\cdot\sqrt{\pi} = \frac{3\sqrt{\pi}}{8}\Gamma\left(1+\frac{1}{2}\right) = \frac{3\sqrt{\pi}}{16}\cdot\Gamma\left(\frac{1}{2}\right) = \frac{3\pi}{16}$$

3) $\int_\gamma \underbrace{(\cos xy - xy\sin xy + e^{2x})}_{P}\,dx + \underbrace{(y^2 - x^2\sin xy)}_{Q}\,dy$

$A(0,-1)$, $B(0,2)$



~~$C = \int_\gamma P\,dx + Q\,dy \overset{G-R}{=} \int_\gamma \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,dx\,dy$~~

$$\frac{\partial Q}{\partial x} = -2x\sin(xy) + (-x^2)\cdot\cos(xy)\cdot y =$$

$$= -2x\sin(xy) - x^2 y\cos(xy)$$

$$\frac{\partial P}{\partial y} = -\sin(xy)\cdot x - y\sin xy - xy\cos(xy)\cdot x$$

~~$C = \int_\gamma$~~ $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} \Rightarrow \omega$ închisă

$\mathbb{R}^2$ mulțime stelată $\Rightarrow \omega$ exactă $\Rightarrow$

$\Rightarrow$ integrală independentă de drum $\Rightarrow \int_\gamma \omega = \int_{[AB]} \omega$

$$\begin{cases} x = 0 \\ y = t \end{cases}, t\in[-1,2]$$

$$\int_\gamma \omega = \int_{-1}^{2} t^2\,dt = \frac{t^3}{3}\Big|_{-1}^{2} = \frac{8}{3} + \frac{1}{3} = \frac{9}{3} = 3$$

7) $\iint_D (1 + x^2 + y^2)\,dx\,dy$



$D: x^2 + y^2 \le 2x,\ y \ge 0$

$x^2 + y^2 - 2x \le 0 \Leftrightarrow x^2 - 2x + 1 + y^2 \le 1 \Leftrightarrow$

$\Leftrightarrow (x-1)^2 + y^2 \le 1$

$D: (x-1)^2 + y^2 \le 1$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$ $\rho^2 \le 2\rho\cos\theta \Leftrightarrow \rho \le 2\cos\theta \Rightarrow \rho\in[0; 2\cos\theta]$

$\rho\sin\theta \ge 0 \Rightarrow \sin\theta \ge 0$

$\cos\theta \ge 0$ $\Rightarrow \theta\in[0; \frac{\pi}{2}]$

$$\iint_D (1+x^2+y^2)\,dx\,dy = \int_0^{\frac{\pi}{2}} \int_0^{2\cos\theta} (1+\rho^2)\rho\,d\rho\,d\theta =$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{\rho^2}{2} + \frac{\rho^4}{4}\right)\Big|_0^{2\cos\theta} d\theta = \int_0^{\frac{\pi}{2}} (2\cos^2\theta + 4\cos^4\theta)\,d\theta =$$

$$= \int_0^{\frac{\pi}{2}} 2\cdot\frac{1+\cos 2\theta}{2}\,d\theta + 4\int_0^{\frac{\pi}{2}} \cos^2\theta\cdot(1-\sin^2\theta)\,d\theta =$$

$$= \int_0^{\frac{\pi}{2}} 1+\cos 2\theta\,d\theta + \int_0^{\frac{\pi}{2}} 4\cdot\frac{1+\cos 2\theta}{2}\,d\theta - \int_0^{\frac{\pi}{2}} (2\sin\theta\cos\theta)^2\,d\theta =$$

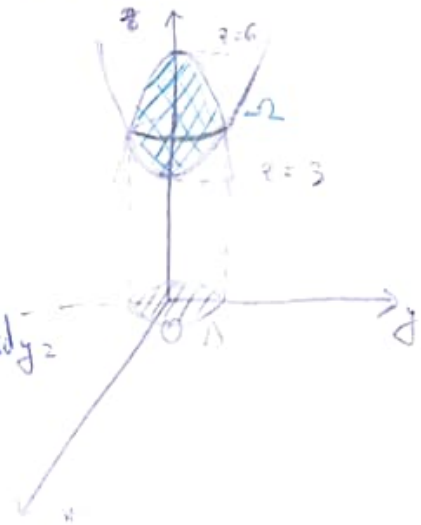
$$= \theta\Big|_0^{\frac{\pi}{2}} + \frac{1}{2}\sin 2\theta\Big|_0^{\frac{\pi}{2}} + 2\theta\Big|_0^{\frac{\pi}{2}} + \sin 2\theta\Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (\sin 2\theta)^2\,d\theta =$$

$$= \frac{\pi}{2} + \pi - \int_0^{\frac{\pi}{2}} \frac{1-\cos 4\theta}{2}\,d\theta = \frac{3\pi}{2} - \frac{1}{2}\theta\Big|_0^{\frac{\pi}{2}} + \frac{1}{8}\sin 4\theta\Big|_0^{\frac{\pi}{2}} =$$

$$= \frac{3\pi}{2} - \frac{\pi}{4} = \frac{5\pi}{4}$$

11) $Vol(\Omega) = ?$

Ω mărg. de $\begin{cases} z = 6 - x^2 - y^2 \\ z = 3 + 2x^2 + 2y^2 \end{cases}$



$$Vol(\Omega) = \iiint_\Omega dx\,dy\,dz = \iint_D \left(\int_{3+2x^2+2y^2}^{6-x^2-y^2} dz\right) dx\,dy =$$

$$= \iint_D (6-x^2-y^2-3-2x^2-2y^2)\,dx\,dy =$$

$$= \iint_D [3-3(x^2+y^2)]\,dx\,dy = 3\iint_D [1-(x^2+y^2)]\,dx\,dy$$

$\begin{cases} z = 6-x^2-y^2 \\ z = 3+2x^2+2y^2 \end{cases} \Rightarrow 3+2x^2+2y^2 = 6-x^2-y^2 \Leftrightarrow$

$\Leftrightarrow 3(x^2+y^2) = 3 \Rightarrow x^2+y^2 = 1$

$D: x^2+y^2 \le 1$, $\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho \in [0;1]$, $\theta \in [0;2\pi]$

$$Vol(\Omega) = 3\int_0^{2\pi}\int_0^1 (1-\rho^2)\rho\,d\rho\,d\theta = 3\int_0^{2\pi}\left(\frac{\rho^2}{2} - \frac{\rho^4}{4}\right)\Big|_0^1 d\theta =$$

$$= \frac{3}{4}\theta\Big|_0^{2\pi} = \frac{3\pi}{2}$$

(4) $\int_\gamma \bar{v}\, d\bar{r} = ?$

$\bar{v} = \underbrace{\frac{-y}{x^2+y^2}}_{P}\,\bar{i} + \underbrace{\frac{x}{x^2+y^2}}_{Q}\,\bar{j}$

a) $\int_{\mathcal{C}(O,R)} \bar{v}\, d\bar{r} = \int_{\mathcal{C}(O,R)} P\,dx + Q\,dy$ (*)

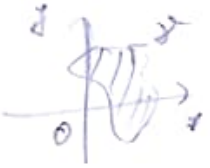
$\begin{cases} x = R\cos t \\ y = -R\sin t \end{cases}, t \in [0; 2\pi]$

$\int_{\mathcal{C}(O,R)} \bar{v}\, d\bar{r} = \int_0^{2\pi} \left[\frac{-R\sin t}{R^2} \cdot (-R\sin t) + \frac{R\cos t}{R^2} \cdot R\cos t \right] dt =$

$= \int_0^{2\pi} \frac{R^2\sin^2 t + R^2\cos^2 t}{R^2}\, dt = \int_0^{2\pi} \frac{R^2}{R^2}\, dt = t\Big|_0^{2\pi} = 2\pi$

b) γ curbă arbitrară închisă, $O \notin \gamma$

Cazul I) $O \in Ext(\gamma)$ $\Rightarrow$ se poate aplica



Green – Riemann

$\int_\gamma P\,dx + Q\,dy \overset{G-R}{=\!=} \iint_K \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy =$

$= \iint_K \frac{x^2+y^2-2x^2+x^2+y^2-2y^2}{(x^2+y^2)^2}\, dx\,dy = 0$

Cazul II) $O \in Int(\gamma)$



Fie $R > 0$ a.î. $D(O,R) \subset K$

$Fr\, A = \mathcal{C}(O,R) \cup \gamma$

$A = Int(\gamma) \setminus D(O,R)$

$\int_{Fr\,A} P\,dx + Q\,dy = 0$

$\int_{Fr\,A} P\,dx + Q\,dy = \int_\gamma P\,dx + Q\,dy - \int_{\mathcal{C}(O,R)} P\,dx + Q\,dy$ $\Big\}\Rightarrow$

$\Rightarrow \int_\gamma P\,dx + Q\,dy = \int_{\mathcal{C}(O,R)} P\,dx + Q\,dy = 2\pi$

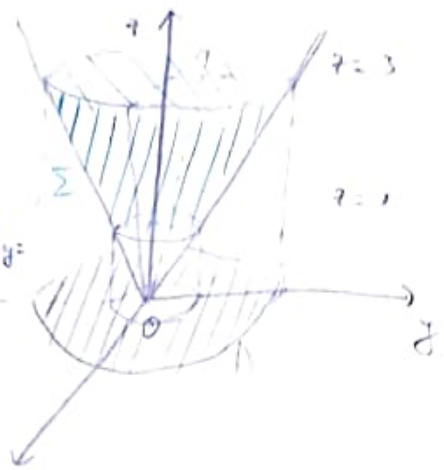
c) $\bar{v}$ câmp de gradienți $\Rightarrow \int_\gamma \bar{v}\, d\bar{r} = 0$, $(\forall)\ \gamma$ curbă închisă

5

16) Aria $(\Sigma) = ?$

$\Sigma : \rho^2 = 3x^2 + 3y^2, \ z \in [1,3]$

$z = \sqrt{3(x^2+y^2)} = f(x,y)$



$$\text{Aria}(\Sigma) = \iint_{\Sigma} d\sigma = \iint_{D} \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx dy =$$

$$= \iint_{D} \sqrt{1 + \left(\frac{3x}{\sqrt{3(x^2+y^2)}}\right)^2 + \left(\frac{3y}{\sqrt{3(x^2+y^2)}}\right)^2}\, dx dy =$$

$$= \iint_{D} \sqrt{\frac{3x^2 + 3y^2 + 3x^2 + 3y^2}{3(x^2+y^2)}}\, dx dy = \iint_{D} dx dy$$

$$\left.\begin{array}{l} z^2 = 3x^2 + 3y^2 \\ 1 \le z \le 3 \end{array}\right\} \Rightarrow 1 \le 3x^2 + 3y^2 \le 9 \Rightarrow \frac{1}{3} \le x^2 + y^2 \le 3$$

$D: \ \frac{1}{3} \le x^2 + y^2 \le 3$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \ \rho \in \left[\frac{1}{\sqrt{3}}; \sqrt{3}\right], \ \theta \in [0; 2\pi]$$

$$\text{Aria}(\Sigma) = 2\int_0^{2\pi} \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \rho \, d\rho \, d\theta = 2\int_0^{2\pi} \frac{\rho^2}{2}\Big|_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} d\theta = 2\int_0^{2\pi} \left(\frac{3}{2} - \frac{1}{6}\right) d\theta =$$

$$= \frac{8}{3}\theta\Big|_0^{2\pi} = \frac{16\pi}{3}$$

17) $\vec{v} = \underbrace{(x^2+y^2+z^2)}_{P}\vec{i} + \underbrace{(z^2 - 2yx)}_{Q}\vec{j} + \underbrace{(3z - x - y)}_{R}\vec{k}$

$\Sigma : \ x^2 + y^2 + z^2 = 4, \ z > 0$

$\Sigma_1 : \ z = 1$

$$\Phi_{\Sigma \cup \Sigma_1}(\vec{v}) \overset{G-O}{=\!=} \iiint_{\Omega} \text{div}(\vec{v})\, dx dy dz$$

$$\text{div}(\vec{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} =$$

$$= 2x - 2y + 3 = 3$$

$$\Phi_{\Sigma \cup \Sigma_1}(\vec{v}) = 3\iiint_{\Omega} dx dy dz = 3 \cdot \frac{4\pi \cdot 2^3}{3} \cdot \frac{1}{2} = 16\pi$$

$$\Phi_{\Sigma \cup \Sigma_1}(\vec{v}) = \Phi_{\Sigma}(\vec{v}) + \Phi_{\Sigma_1}(\vec{v})$$

$$\Rightarrow \Phi_{\Sigma}(\vec{v}) = 16\pi - \Phi_{\Sigma_1}(\vec{v})$$

$\bar N_1 = -\bar k$

$\Sigma_1 : z = 0$

$\Phi_{\Sigma_1}(\bar v) = \int_{\Sigma_1} \bar v \cdot \bar N_1 \, d\sigma = \iint_D (x+y)\,dx\,dy$

$z = 0 \Rightarrow x^2 + y^2 = 4$

$D: x^2 + y^2 \le 4$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \rho \in [0;2], \theta \in [0;2\pi]$

$\Phi_{\Sigma_1}(\bar v) = \int_0^{2\pi}\int_0^2 \rho(\cos\theta + \sin\theta)\cdot\rho\,d\rho\,d\theta =$

$= \int_0^{2\pi} \frac{1}{3}(\cos\theta + \sin\theta)\rho^3\Big|_0^2 d\theta =$

$= \frac{8}{3}\int_0^{2\pi}(\cos\theta + \sin\theta)\,d\theta = \frac{8}{3}(\sin\theta - \cos\theta)\Big|_0^{2\pi} = 0$

$\Phi_\Sigma(\bar v) = 16\bar a$

2°) $\bar v = \underbrace{(y-z)}_{P}\bar i + \underbrace{(z-x)}_{Q}\bar j + \underbrace{(x-y)}_{R}\bar k$

$\gamma: \begin{cases} x^2 + y^2 = a^2 \\ \frac{x}{a} + \frac{z}{h} = 1 \end{cases} \quad a, h > 0$

$C = \int_\gamma \bar v\,dr = \iint_D \operatorname{rot}(\bar v)\cdot(\bar n_x \times \bar n_y)\,dx\,dy$

$\operatorname{rot}\bar v = \begin{vmatrix} \bar i & \bar j & \bar k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y-z & z-x & x-y \end{vmatrix} = -2(\bar i + \bar j + \bar k)$

$\frac{z}{h} = 1 - \frac{x}{a} \Rightarrow z = h\left(1 - \frac{x}{a}\right)$

$\bar n = x\bar i + y\bar j + h\left(1 - \frac{x}{a}\right)\bar k$

$\bar n_x = \bar i - \frac{h}{a}\bar k$

$\bar n_y = y\bar j$

$\operatorname{rot}\bar v\,(\bar n_x \times \bar n_y) = \begin{vmatrix} -2 & -2 & -2 \\ 1 & 0 & -\frac{h}{a} \\ 0 & 1 & 0 \end{vmatrix} = 2\left(\frac{h}{a} - 1\right)$

$\Sigma: z = h\left(1 - \frac{x}{a}\right)$

$D: x^2 + y^2 \le a^2$

$$C = \iint_D 2\left(\frac{h}{a} - 1\right) dx\,dy = 2\left(\frac{h}{a} - 1\right) \cdot \pi \cdot a^2$$

22) $x^3 + 9x - 3 = 0$, $\varepsilon < 10^{-2}$

$f(x) = x^3 + 9x - 3$

$f(0) = -3$

$f(1) = 7$

$f(0) \cdot f(1) < 0 \Rightarrow f$ are o soluție reală în $(0,1)$ $\Big\} \Rightarrow$

$f'(x) = 3x^2 + 9 > 0, (\forall) x \in [0;1] \Rightarrow f$ crescătoare

$\Rightarrow f$ are o unică soluție reală în $(0,1)$

$x^3 + 9x - 3 = 0 \Leftrightarrow x^3 + 9x = 3 \Leftrightarrow x(x^2+9) = 3 \Rightarrow x = \frac{3}{x^2+9}$

Fie $T(x) = \frac{3}{x^2+9}$, $T: [0,1] \to [0,1]$

$[0,1]$ spațiu metric complet

Conform Teoremei Lagrange $\Rightarrow \exists c \in [0,1]$ a.î.

$T(x) - T(y) = T'(c)(x-y) \Rightarrow |T(x) - T(y)| = |T'(c)| \cdot |x-y| \leq$

$\leq \sup |T'(c)| \cdot |x-y|$ (1)

$$T'(x) = \frac{-3 \cdot 2x}{(x^2+9)^2} = \frac{-6x}{(x^2+9)^2}$$

$$g(x) = |T'(x)| = \frac{6x}{(x^2+9)^2}$$

$$g'(x) = \frac{6(x^2+9)^{x'} - 6x \cdot 2(x^2+9) \cdot 2x}{(x^2+9)^{*3}} = \frac{6x^2 + 54 - 24x^2}{(x^2+9)^3} =$$

$$= \frac{54 - 18x^2}{(x^2+9)^3} > 0 \ (\forall) x \in [0,1] \Rightarrow g \text{ crescătoare} \Rightarrow$$

$$\Rightarrow \sup g(x) = \sup T'(x) = g(1) = \frac{6}{100} = \frac{3}{50}$$

Din (1) $\Rightarrow$ T contracție de factor $K = \frac{3}{50}$ $\Big\} \Rightarrow$ Conform Teoremei

$[0,1]$ spațiu metric complet

Contracției că $\exists \xi \in [0,1]$ a.î. $T(\xi) = \xi$

$x_0 = 0$ și $x_n = T(x_{n-1})$, $(\forall) n \in \mathbb{N}$

$$|x_n - \xi| \leq \frac{K^n}{1-K} \cdot |x_1 - x_0| \leq 10^{-2} \Leftrightarrow$$

$x_1 = T(x_0) = \frac{1}{3}$

$\Leftrightarrow \dfrac{\left(\frac{3}{50}\right)^n}{1-\frac{3}{50}} \cdot \frac{1}{3} < \frac{1}{100} \Leftrightarrow$

$\Leftrightarrow \left(\frac{3}{50}\right)^n \cdot \frac{1}{3} \cdot \frac{50}{47} < \frac{1}{100} \Leftrightarrow$

$\Leftrightarrow \left(\frac{3}{50}\right)^{n-1} \cdot \frac{1}{47} < \frac{1}{100} \Leftrightarrow \left(\frac{50}{3}\right)^{n-1} \cdot 47 > 100$

$n=2 \Rightarrow \begin{cases} \simeq x_2 = T(x_1) = \dfrac{3}{\frac{1}{9}+9} = 3 \cdot \dfrac{9}{82} = \dfrac{27}{82} \end{cases}$

23) $x^3+4x-1=0$, $\varepsilon < 10^{-3}$

$f(x) = x^3+4x-1$

$f(0) = -1$

$f(1) = 4$

$f(0)\cdot f(1) < 0 \Rightarrow f$ are o soluție în $(0,1)$

$f'(x) = 3x^2+4 > 0$, $(\forall) x \in [0,1] \Rightarrow f$ crescătoare $\Big\} \Rightarrow$

$\Rightarrow f$ are o unică soluție în $(0,1)$

$x^3+4x-1=0 \Leftrightarrow x^3+4x=1 \Leftrightarrow x(x^2+4)=1 \Rightarrow x = \dfrac{1}{x^2+4}$

Fie $T(x) = \dfrac{1}{x^2+4}$, $T: [0,1] \to [0,1]$

Conform Teoremei Lagrange $\Rightarrow \exists c \in [0,1]$ a.î.

$T(x)-T(y) = T'(c)(x-y) \Rightarrow |T(x)-T(y)| = |T'(c)| \cdot |x-y| \leq$

$\leq \sup |T'(c)| |x-y|$ (1)

$T'(x) = \dfrac{-2x}{(x^2+4)^2}$

$g(x) = |T'(x)| = \dfrac{2x}{(x^2+4)^2}$

$g'(x) = \dfrac{2(x^2+4)^{2-1} - 2x \cdot 2(x^2+4)\cdot 2x}{(x^2+4)^{4}} = \dfrac{2x^2+8-8x^2}{(x^2+4)^3} = \dfrac{8-6x^2}{(x^2+4)^3} > 0$

$(\forall) x \in [0,1] \Rightarrow g$ crescătoare $\Rightarrow \sup g'(x) = \sup |T'(x)| = g(1) =$

$= \dfrac{2}{25}$

Dim (1) $\Rightarrow$ T contractie de factor $K = \frac{2}{25}$ $\Rightarrow$ Conform TH Contractiei

$[0,1]$ spatiu metric complet

Cu $\exists\, \xi \in [0,1]$ a.i. $T(\xi) = \xi$; $x_0 = 0$ si $x_n = T(x_{n-1})$, $n \in \mathbb{N}$

$$|x_n - \xi| \leq \frac{K^n}{1-K} \cdot |x_1 - x_0| < 10^{-3}$$

$$x_1 = T(x_0) = T(0) = \frac{1}{4}$$

$$\frac{\left(\frac{2}{25}\right)^n}{1-\frac{2}{25}} \cdot \frac{1}{4} < \frac{1}{1000} \Leftrightarrow \left(\frac{2}{25}\right)^n \cdot \frac{25}{23} \cdot \frac{1}{4} < \frac{1}{1000} \Leftrightarrow$$

$$\Leftrightarrow \left(\frac{2}{25}\right)^{n-1} \cdot \frac{1}{46} < \frac{1}{1000} \Leftrightarrow \left(\frac{25}{2}\right)^{n-1} \cdot 46 > 1000$$

$n = 3 \Rightarrow \xi \approx x_3 = T(x_2) = \frac{1}{\left(\frac{16}{65}\right)^2 + 4}$

$x_2 = T(x_1) = T\left(\frac{1}{4}\right) = \frac{1}{\frac{1}{16} + 4} = \frac{16}{65}$

Rezolvare examen - Nr 1 2022

1) $J = \int_0^\infty \frac{x^2}{1+x^4}\,dx$

~~$x^2 = y \Rightarrow x = \sqrt{y} \Rightarrow dx = \frac{1}{2\sqrt{y}}\,dy$~~

$\Rightarrow x^4 = y \Rightarrow x = \sqrt[4]{y} \Rightarrow x = y^{\frac{1}{4}} \Rightarrow dx = \frac{1}{4} y^{-\frac{3}{4}}\,dy$

$\Rightarrow J = \int_0^\infty \frac{y^{\frac{1}{2}}}{1+y} \cdot \frac{1}{4} y^{-\frac{3}{4}}\,dy = \frac{1}{4}\int_0^\infty \frac{y^{-\frac{1}{4}}}{(1+y)^1}\,dy$

$\beta(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = -\frac{1}{4} \Rightarrow p = \frac{3}{4}$

$p+q = 1 \Rightarrow q = 1 - \frac{3}{4} = \frac{1}{4}$

$\Rightarrow J = \frac{1}{4}\beta\left(\frac{3}{4}; \frac{1}{4}\right)$

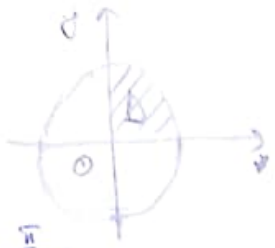
$\beta(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$\Rightarrow J = \frac{1}{4}\cdot\frac{\Gamma\left(\frac{3}{4}\right)\cdot\Gamma\left(\frac{1}{4}\right)}{\Gamma(1)} = \frac{1}{4}\cdot\frac{\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(1-\frac{1}{4}\right)}{0!} =$

$= \frac{1}{4}\cdot\frac{\pi}{\sin\left(\frac{\pi}{4}\right)} = \frac{1}{4}\cdot\frac{\pi}{\frac{\sqrt{2}}{2}} = \frac{1}{4}\cdot\pi\cdot\frac{2}{\sqrt{2}} = \frac{2\pi\sqrt{2}}{4\sqrt{2}} = \frac{\pi\sqrt{2}}{4}$

2) $\iint_D \sqrt{4-x^2-y^2}\,dx\,dy$; $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 1,\ x \ge 0,\ y \ge 0\}$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho \in [0,1]$, $\theta \in [0, 2\pi]$



$\begin{cases} \rho\cos\theta \ge 0 \\ \rho\sin\theta \ge 0 \end{cases} \Leftrightarrow \begin{cases} \cos\theta \ge 0 \\ \sin\theta \ge 0 \end{cases} \Rightarrow \theta \in [0; \frac{\pi}{2}]$

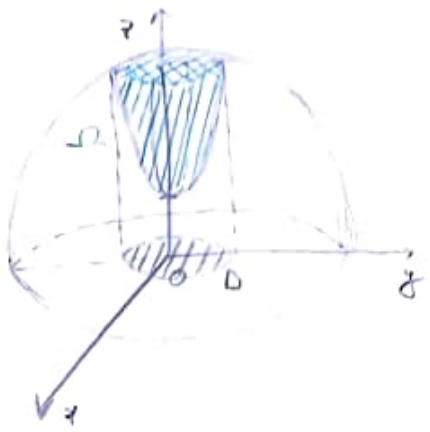
$\iint_D \sqrt{4-x^2-y^2}\,dx\,dy = \iint_D \sqrt{4-(x^2+y^2)}\,dx\,dy = \int_0^{\frac{\pi}{2}}\int_0^1 \sqrt{4-\rho^2}\cdot\rho\,d\rho\,d\theta =$

$= -\frac{1}{2}\int_0^{\frac{\pi}{2}} (4-\rho^2)^{\frac{3}{2}}\cdot\frac{2}{3}\Big|_0^1 d\theta = -\frac{1}{3}\int_0^{\frac{\pi}{2}} (3\sqrt{3} - 4\sqrt{4})\,d\theta =$

$= -\frac{1}{3}(3\sqrt{3}-8)\theta\Big|_0^{\frac{\pi}{2}} = \left(-\frac{3\sqrt{3}}{3}\sqrt{3} + \frac{8}{3}\right)\cdot\frac{\pi}{2} = -\frac{\pi\sqrt{3}}{2} + \frac{4\pi}{3}$

3) $S_1 : x^2+y^2+z^2 = 10$

$S_2 : z = 1+2x^2+2y^2$, $z \geq 0$



$$\iiint_\Omega dx\,dy\,dz = \iint_D \left(\int_{1+2x^2+2y^2}^{\sqrt{10-(x^2+y^2)}} dz \right) dx\,dy =$$

$$= \iint_D \left\{ \sqrt{10-(x^2+y^2)} - [1+2(x^2+y^2)] \right\} dx\,dy =$$

$$\left. \begin{array}{l} x^2+y^2+z^2 = 10 \Rightarrow x^2+y^2 = 10-z^2 \\ z = 1+2(x^2+y^2) \end{array} \right\} \Rightarrow$$

$$\Rightarrow z = 1+20-2z^2 \Leftrightarrow 2z^2+z-21=0$$

$$\Delta = 1+168 = 169$$

$$z_{1,2} = \frac{-1 \pm 13}{4} \Rightarrow z = 3 \Rightarrow x^2+y^2+9 = 10 \Rightarrow$$

$z \geq 0$

$$\Rightarrow x^2+y^2 = 1$$

$D: x^2+y^2 \leq 1$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \quad \rho \in [0;1],\ \theta \in [0;2\pi]$$

$$\Rightarrow \iiint_\Omega dx\,dy\,dz = \int_0^{2\pi} \int_0^1 \left[\sqrt{10-\rho^2} - (1+2\rho^2) \right] \rho\, d\rho\, d\theta =$$

$$= \int_0^{2\pi} \left(-\frac{1}{2} \int_0^1 \sqrt{10-\rho^2} \cdot (10-\rho^2)'\, d\rho - \int_0^1 (\rho+2\rho^3)\, d\rho \right) d\theta =$$

$$= \int_0^{2\pi} \left(-\frac{1}{2}\cdot\frac{2}{3}(10-\rho^2)^{\frac{3}{2}} - \frac{\rho^2}{2} - \frac{2\rho^4}{4} \right)\Big|_0^1 d\theta =$$

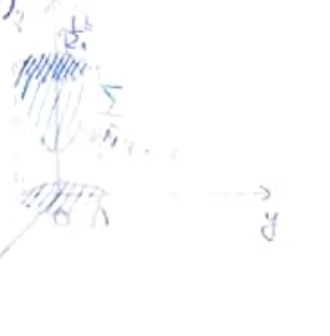
$$= \int_0^{2\pi} \left(-\frac{1}{3}\cdot 9\sqrt{9} - \frac{1}{2} - \frac{1}{2} + \frac{1}{3}\cdot 10\sqrt{10} \right) d\theta =$$

$$= \int_0^{2\pi} \left(\frac{10\sqrt{10}}{3} - 10 \right) d\theta = 10\left(\frac{\sqrt{10}}{3} - 1 \right)\theta\Big|_0^{2\pi} = \frac{20\pi\sqrt{10} - 60\pi}{3} =$$

$$= 2\pi \cdot \frac{10\sqrt{10}-30}{3}$$

4) $\bar{v} = (\underbrace{e^{yz}}_{P} + x^2 + xy,\ \underbrace{\cos xz - 2y - \frac{y^2}{2}}_{Q},\ \underbrace{1-z}_{R})$

$\Sigma : z = 2+x^2+y^2$, $z \in [2,3]$



$$\Phi_{\Sigma \cup \Sigma_1} \bar{v}\, d\bar{\sigma} \overset{G-O}{=\!=} \iiint_\Omega \operatorname{div}(\bar{v})\, dx\,dy\,dz$$

$$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 2x+y-2x-y-1 = -1 \Rightarrow$$

$$\Rightarrow \Phi_{\Sigma \cup \Sigma_1} \bar{v}\, d\bar{\sigma} = -\iiint_\Omega dx\,dy\,dz =$$

$$= -\iint_D \int_{2+x^2+y^2}^{3} dz\, dx\, dy = -\iint_D [3 - 2 - (x^2+y^2)]\, dx\, dy =$$

$$= \iint_D (x^2+y^2-1)\, dx\, dy$$

$$\left.\begin{array}{l} z = 2 + x^2 + y^2 \\ z = 3 \end{array}\right\} \Rightarrow x^2+y^2 = 1$$

$D: x^2+y^2 \le 1$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \quad \rho \in [0;1], \ \theta \in [0; 2\pi]$$

$$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \int_0^{2\pi}\int_0^1 (\rho^2-1)\cdot\rho\, d\rho\, d\theta = \int_0^{2\pi} \left(\frac{\rho^4}{4} - \frac{\rho^2}{2}\right)\Big|_0^1 d\theta =$$

$$= \int_0^{2\pi} -\frac{1}{4}\, d\theta = -\frac{1}{4}\theta\Big|_0^{2\pi} = -\frac{2\pi}{4} = -\frac{\pi}{2}$$

$$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \mathcal{F}_{\Sigma}(\bar{v}) + \mathcal{F}_{\Sigma_1}(\bar{v}) \Rightarrow \mathcal{F}_{\Sigma}(\bar{v}) = \mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) - \mathcal{F}_{\Sigma_1}(\bar{v}) =$$

$$= -\frac{\pi}{2} - \mathcal{F}_{\Sigma_1}(\bar{v})$$

$$\mathcal{F}_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v}\cdot\bar{N}_1\, d\sigma$$

$\bar{N}_1 = \bar{k}$

$\Sigma_1 : z = 3$

$$\mathcal{F}_{\Sigma_1}(\bar{v}) = \iint_D (1-3)\, dx\, dy = 2\iint_D dx\, dy = 2\pi \Rightarrow \mathcal{F}_{\Sigma}(\bar{v}) = -\frac{\pi}{2} - 2\pi = -\frac{5\pi}{2}$$

6) Aria $(\Sigma) = ?$, ~~$\Sigma: x^2+y^2+z^2 = R^2, z \ge 0$~~



$S: x^2+y^2+z^2 = R^2, z \ge 0$

$x^2+y^2 = Ry$, $R>0$ fixat

$x^2+y^2+z^2 = R^2 \Rightarrow z^2 = R^2 - x^2 - y^2 \Rightarrow$

$\Rightarrow z = \sqrt{R^2-x^2-y^2} = f(x,y)$

$x^2+y^2 = Ry \Rightarrow x^2+y^2-Ry = 0 \Rightarrow$

$\Rightarrow x^2 + y^2 - \frac{1}{2}\cdot 2\cdot Ry + \frac{R^2}{4} = \frac{R^2}{4} \Rightarrow x^2 + \left(y - \frac{R}{2}\right)^2 = \left(\frac{R}{2}\right)^2$

$D: x^2 + \left(y - \frac{R}{2}\right)^2 \le \left(\frac{R}{2}\right)^2$

$$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx\, dy = \iint_D \sqrt{1 + \frac{-2x}{\sqrt{R^2-(x^2+y^2)}} + \frac{-2y}{\sqrt{R^2-(x^2+y^2)}}}$$

$$= \iint_D \sqrt{1 + \ldots - 2\cdot\frac{x+y}{\sqrt{R^2-(x^2+y^2)}}}\, dx\, dy$$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho \in [0; \frac{R}{2}]$, $\theta \in [0; 2\pi]$

~~Aria (Σ) =~~ $\int_0^{2\pi}\int_0^{\frac{R}{2}}$

Aria $(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx\,dy =$

$= \iint_D \sqrt{1 + \left(\frac{-2x}{\sqrt{R^2-(x^2+y^2)}}\right)^2 + \left(\frac{-2y}{\sqrt{R^2-(x^2+y^2)}}\right)^2}\, dx\,dy =$

$= \iint_D \sqrt{1 + \frac{4(x^2+y^2)}{R^2-(x^2+y^2)}}\, dx\,dy =$

$= \iint_D \sqrt{\frac{R^2 + 3(x^2+y^2)}{R^2-(x^2+y^2)}}\, dx\,dy = \int_0^{2\pi}\int_0^{\frac{R}{2}} \sqrt{\frac{R^2+3\rho^2}{R^2-\rho^2}} \cdot \rho\, d\rho\, d\theta$

Aria $(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx\,dy =$

$= \iint_D \sqrt{1 + \left(\frac{-x}{\sqrt{R^2-(x^2+y^2)}}\right)^2 + \left(\frac{-y}{\sqrt{R^2-(x^2+y^2)}}\right)^2}\, dx\,dy =$

$= \iint_D \sqrt{\frac{R^2 - x^2 - y^2 + x^2 + y^2}{R^2-(x^2+y^2)}}\, dx\,dy = R \iint_D \frac{1}{\sqrt{R^2-(x^2+y^2)}}\, dx\,dy$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho \in [0, \frac{R}{2}]$, $\theta \in [0; 2\pi]$ ..... $\theta \in [0; \pi]$

Aria $(\Sigma) = R \int_0^{2\pi}\int_0^{\frac{R}{2}} \frac{1}{\sqrt{R^2-\rho^2}} \cdot \rho\, d\rho\, d\theta = -\frac{R}{2}\int_0^{2\pi} 2\left(R^2-\rho^2\right)^{\frac{1}{2}}\Big|_0^{\frac{R}{2}} d\theta =$

$= -\frac{R}{2}\cdot 2\int_0^{2\pi}\left(\sqrt{R^2-\frac{R^2}{4}} - \sqrt{R^2}\right)d\theta = -R\int_0^{2\pi}\left(\frac{R\sqrt{3}}{2} - R\right)d\theta =$

$= R^2\left(1 - \frac{\sqrt{3}}{2}\right)\theta\Big|_0^{2\pi} = 2\pi R^2\left(1 - \frac{\sqrt{3}}{2}\right)$ ~~$= \pi R^2(2-\sqrt{3})$~~

Rezolvare examen - Nr. 2 2022

1) $I = \int_0^\infty \frac{1}{1+x^4}\,dx$

$x^4 = y \Rightarrow x = \sqrt[4]{y} \Leftrightarrow x = y^{\frac{1}{4}} \Rightarrow dx = \frac{1}{4} y^{-\frac{3}{4}}\,dy$

$I = \frac{1}{4}\int_0^\infty \frac{y^{-\frac{3}{4}}}{1+y}\,dy =$

$\beta(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

~~$\frac{3}{4} = p$~~ $p-1 = -\frac{3}{4} \Rightarrow p = \frac{1}{4}$

$p+q = 1 \Rightarrow q = 1 - \frac{1}{4} = \frac{3}{4}$

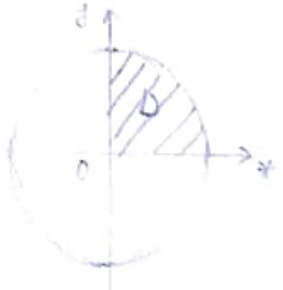
$\Rightarrow I = \frac{1}{4}\beta\left(\frac{1}{4}; \frac{3}{4}\right)$

$\beta(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$\Rightarrow I = \frac{1}{4}\cdot\frac{\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right)}{\Gamma(1)} = \frac{1}{4}\cdot\frac{\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(1-\frac{1}{4}\right)}{0!} = \frac{1}{4}\cdot\frac{\pi}{\sin\left(\frac{1}{4}\cdot\pi\right)} =$

$= \frac{1}{4}\cdot\frac{\pi}{\frac{\sqrt{2}}{2}} = \frac{1}{4}\cdot\frac{\pi\cdot 2}{\sqrt{2}} = \frac{2\pi}{4\sqrt{2}} = \frac{\pi}{2\sqrt{2}} = \frac{2\pi\cdot\sqrt{2}}{4}$

2) $\iint_D \sqrt{1+x^2+y^2}\,dx\,dy$; $D: \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 1, x \ge 0, y \ge 0\}$



$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \rho \in [0,1]$

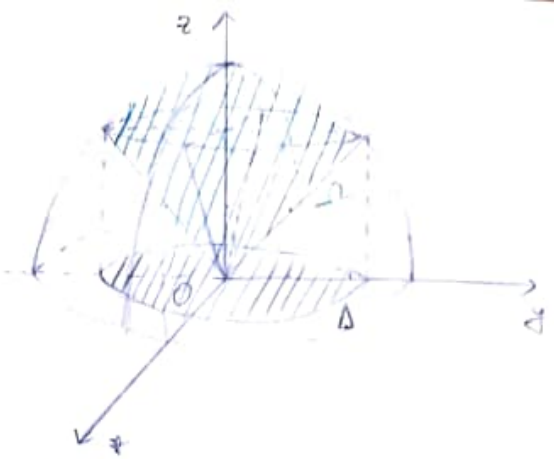
$\left.\begin{array}{l} \rho\cos\theta \ge 0 \\ \rho\sin\theta \ge 0 \\ \rho > 0 \end{array}\right\} \Rightarrow \theta \in \left[0, \frac{\pi}{4}\right]$

$\iint_D \sqrt{1+x^2+y^2}\,dx\,dy = \int_0^{\frac{\pi}{4}}\int_0^1 \sqrt{1+\rho^2}\cdot\rho\,d\rho\,d\theta =$

$= \frac{1}{2}\int_0^{\frac{\pi}{4}} \left(1+\rho^2\right)^{\frac{3}{2}}\cdot\frac{2}{3}\Big|_0^1 d\theta = \frac{1}{3}\int_0^{\frac{\pi}{4}} (2\sqrt{2}-1)\,d\theta =$

$= \frac{2\sqrt{2}}{3}\theta - \frac{\theta}{3}\Big|_0^{\frac{\pi}{4}} = \frac{2\pi\sqrt{2}-\pi}{12} = \pi\left(\frac{\sqrt{2}}{6} - \frac{1}{12}\right) = \frac{\pi}{6}\left(\sqrt{2} - \frac{1}{2}\right)$

3) $\begin{cases} S_1 : x^2+y^2+z^2=8 \\ S_2 : z^2 = x^2+y^2 \end{cases}$, $z \geq 0$



$$\iiint_\Omega dx\,dy\,dz = \iint_D \int_{\sqrt{x^2+y^2}}^{\sqrt{8-(x^2+y^2)}} dz\,dx\,dy =$$

$$= \iint_D \left(\sqrt{8-(x^2+y^2)} - \sqrt{x^2+y^2}\right) dx\,dy$$

$$\begin{cases} x^2+y^2+z^2=8 \\ z^2 = x^2+y^2 \end{cases} \Rightarrow 2z^2=8 \Rightarrow z^2=4 \Rightarrow z=2 \Rightarrow$$

$z \geq 0$

$$\Rightarrow x^2+y^2=2$$

$D: x^2+y^2 \leq 2$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases} \quad \rho \in [0;\sqrt{2}] \,,\ \theta \in [0;2\pi]$$

$$Vol(\Omega) = \int_0^{2\pi}\int_0^{\sqrt{2}} \left(\sqrt{8-\rho^2} - \sqrt{\rho^2}\right)\cdot \rho\, d\rho\, d\theta =$$

$$= \int_0^{2\pi} \left(-\frac{1}{2}\int_0^{\sqrt{2}} (8-\rho^2)\cdot\sqrt{8-\rho^2}\, d\rho - \int_0^{\sqrt{2}} \rho^2\, d\rho \right) d\theta =$$

$$= \int_0^{2\pi} \left(-\sqrt{8-\rho^2}\,\Big|_0^{\sqrt{2}} - \frac{\rho^3}{3}\Big|_0^{\sqrt{2}} \right) d\theta = \int_0^{2\pi} \left(-\sqrt{6}+2\sqrt{2} - \frac{2\sqrt{2}}{3}\right) d\theta =$$

$$= 2\pi\sqrt{2}\left(\frac{3}{2} - \sqrt{3} - \frac{2}{3}\right) = 2\sqrt{2}\pi \cdot \frac{4-3\sqrt{3}}{3}$$

4) $\Sigma : 2-z = x^2+y^2$, $z \in [1,2]$

$2-z = x^2+y^2 \Leftrightarrow z = 2-(x^2+y^2) = f(x,y)$

~~$\int_\Sigma d\sigma =$~~



$$Area(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}\, dx\,dy =$$

$$= \iint_D \sqrt{1+(-2x)^2+(-2y)^2}\, dx\,dy =$$

$$= \iint_D \sqrt{1+4x^2+4y^2}\, dx\,dy = \iint_D \sqrt{1+4(x^2+y^2)}\, dx\,dy$$

$$\text{com } \begin{cases} z = 2-(x^2+y^2) \\ 1 \leq z \leq 2 \end{cases} \Leftrightarrow 1 \leq 2-(x^2+y^2) \leq 2 \Leftrightarrow -1 \leq -(x^2+y^2) \leq 0 \Leftrightarrow$$

$$\Leftrightarrow 0 \leq x^2+y^2 \leq 1$$

$D: x^2+y^2 \leq 1$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases} , \rho \in [0;1] \,;\ \theta \in [0;2\pi]$$

3

$$\text{Aria}(\Sigma) = \int_0^{2\pi}\int_0^1 \sqrt{1+4\rho^2}\cdot\rho\, d\rho d\theta = \frac{1}{8}\int_0^{2\pi} 2\sqrt{1+4\rho^2}\,\Big|_0^1 d\theta =$$

$$= \frac{1}{4}\int_0^{2\pi}(\sqrt{5}-1)\,d\theta = \frac{\sqrt{5}-1}{4}\,\theta\Big|_0^{2\pi} = \frac{\pi(\sqrt{5}-1)}{2}$$

6) $q>0$, $\bar{r} = (x,y,z)$, $r = \|\bar{r}\|$

$$\bar{E} = \frac{q}{4\pi}\cdot\frac{\bar{r}}{r^3}$$

a) $S\ x^2+y^2+z^2 = R^2$, $R>0$

$$\mathcal{F}_S(\bar{E}) = \int_S \bar{E}\cdot\bar{N}\,d\sigma = \iint_D \left(\frac{q}{4\pi}\cdot\frac{\bar{r}}{r^3}\cdot\frac{\bar{r}}{R}\cdot R^2\sin\theta\right)d\theta d\rho =$$

$$= \iint_D \frac{q}{4\pi}\cdot 2\cdot 2\pi = q$$

b) $(K,\partial K)$ compact cu bord orientat a.î. $0\notin\partial K$

$\operatorname{div}(\bar{E}) = 0$

Cazul I: $0\notin K \overset{G-O}{\Rightarrow} \mathcal{F}_{\partial K}(\bar{E}) = \iiint_\Omega \operatorname{div}(\bar{E})\,dxdydz = 0$

Cazul II: $0\in K$ $\Rightarrow$ nu se poate aplica Gauss-Ostrogradski

Fie $S(0,R)\subset K$

Fie $K_1 = \operatorname{ext} S \cap K$

K_1 compact

$\partial K_1 = \partial K \cup S^-(0,R)$

$0\notin K_1$

$\mathcal{F}_{\partial K_1}(\bar{E}) = 0$

$$\mathcal{F}_{\partial K_1}(\bar{E}) = \mathcal{F}_{\partial K}(\bar{E}) + \mathcal{F}_{S^-(0,R)}(E) = \mathcal{F}_{\partial K}(\bar{E}) - \mathcal{F}_{S(0,R)}(\bar{E})$$

$\Rightarrow$

$$\mathcal{F}_{\partial K}(\bar{E}) = \mathcal{F}_{S(0,R)}(\bar{E}) \overset{\text{conform lui a)}}{=} q$$

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2) $\int_0^{\infty} \frac{\sqrt[6]{x}}{(1+x^2)^3}\,dx$

$x^2 = y \Rightarrow x = \sqrt{y} \Rightarrow dx = \frac{1}{2\sqrt{y}}\,dy$

$I = \int_0^{\infty} \frac{\sqrt[6]{y}}{(1+y)^3} \cdot \frac{1}{2\sqrt{y}}\,dy = \frac{1}{2}\int_0^{\infty} \frac{y^{\frac{1}{6}} \cdot y^{-\frac{1}{2}}}{(1+y)^3}\,dy =$

$= \frac{1}{2}\int_0^{\infty} \frac{y^{-\frac{1}{3}}}{(1+y)^3}\,dy$

$B(p,q) = \int_0^{\infty} \frac{y^{p-1}}{(1+y)^{p+q}}\,dy.$

$p-1 = -\frac{1}{3} \Rightarrow p = -\frac{1}{3} + \frac{3}{3} = \frac{2}{3}$

$p+q = 3 \Rightarrow q = \frac{3}{3} - \frac{2}{3} = \frac{7}{3}$

$\Rightarrow I = \frac{1}{2} B\left(\frac{2}{3}; \frac{7}{3}\right)$

$B(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)} \Rightarrow I = \frac{1}{2} \cdot \frac{\Gamma\left(\frac{2}{3}\right)\cdot\Gamma\left(\frac{7}{3}\right)}{\Gamma(3)} =$

$= \frac{1}{2} \cdot \frac{\Gamma\left(\frac{2}{3}\right)\cdot\Gamma\left(1+\frac{4}{3}\right)}{2!} = \frac{1}{4_1} \cdot \frac{4}{3} \cdot \Gamma\left(\frac{2}{3}\right)\cdot\Gamma\left(\frac{4}{3}\right) =$

$= \frac{1}{3} \cdot \Gamma\left(\frac{2}{3}\right)\cdot\Gamma\left(1+\frac{1}{3}\right) = \frac{1}{9}\Gamma\left(\frac{2}{3}\right)\cdot\Gamma\left(\frac{1}{3}\right) = \frac{1}{9}\Gamma\left(\frac{1}{3}\right)\cdot\Gamma\left(1-\frac{1}{3}\right) =$

$= \frac{1}{9} \cdot \frac{\pi}{\sin\left(\frac{\pi}{3}\right)} = \frac{1}{9} \cdot \pi \cdot \frac{2}{\sqrt{3}} = \frac{2\pi\sqrt{3}}{27}$

3) $\Sigma: 4 - z = x^2 + y^2,\ z \in [1,2]$

$4 - z = x^2 + y^2 \Rightarrow z = 4 - (x^2+y^2) = f(x,y)$



Aria $(\Sigma) = \int_{\Sigma} d\sigma = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{1 + (-2x)^2 + (-2y)^2}\,dx\,dy = \iint_D \sqrt{1 + 4(x^2+y^2)}\,dx\,dy$

$\left.\begin{array}{l} z = 4 - (x^2+y^2) \\ 1 \le z \le 2 \end{array}\right\} \Rightarrow 1 \le 4 - (x^2+y^2) \le 2 \,/-4 \Rightarrow -3 \le -(x^2+y^2) \le -2 \,/\cdot(-1)$

$\Rightarrow 2 \le x^2 + y^2 \le 3$

$D: 2 \le x^2 + y^2 \le 3$

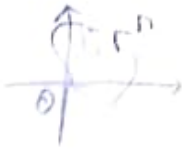
$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases} \quad \rho \in [\sqrt{2}; \sqrt{3}],\ \theta \in [0; 2\pi]$

Aria $(\Sigma) = \frac{1}{2}\int_0^{2\pi} \int_{\sqrt{2}}^{\sqrt{3}} \sqrt{1+4\rho^2} \cdot \rho\,d\rho\,d\theta = \frac{1}{2}\int_0^{2\pi} \frac{(1+4\rho^2)^{\frac{3}{2}}}{12} \Big|_{\sqrt{2}}^{\sqrt{3}} d\theta =$

$= \frac{1}{12}(13\sqrt{13} - 9\sqrt{3})\,\theta\Big|_0^{2\pi} = \frac{\pi}{6}(13\sqrt{13} - 27)$

5) $C = \int_\Gamma \underbrace{\frac{x-y}{x^2+y^2}}_{P}\,dx + \underbrace{\frac{x+y}{x^2+y^2}}_{Q}\,dy$

$O \notin \Gamma$

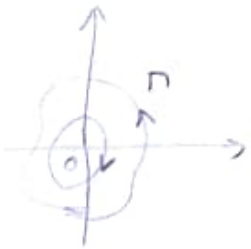


Caz 1) $O \in Ext(\Gamma) \Rightarrow$ se poate aplica Green - Riemann

$C = \iint_K \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,dx\,dy = \iint_K \frac{x^2+y^2-2y(x+y)+x^2+y^2+2xy+x^2+y^2+2y(x-y)}{(x^2+y^2)^2}\,dx\,dy =$

$= \iint_K \frac{2x^2+2y^2-2x^2-2xy+2xy-2y^2}{(x^2+y^2)^2}\,dx\,dy = 0$

Caz 2) $O \in Int(\Gamma)$



Fie $R > 0$ a.î. $D(0,R) \subset K$

$A = K \setminus D(0,R)$

$\partial A = \Gamma \cup \mathcal{C}(0,R)$

$\left.\begin{array}{l} \int_{\partial A} P\,dx + Q\,dy \overset{G-R}{=\!=} \iint_A \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,dx\,dy = 0 \\ \int_{\partial A} P\,dx + Q\,dy = \int_\Gamma P\,dx + Q\,dy + \int_{\mathcal{C}(0,R)} P\,dx + Q\,dy \end{array}\right\} \Rightarrow$

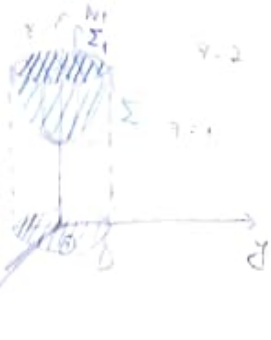
$\Rightarrow \int_\Gamma P\,dx + Q\,dy = \int_{\mathcal{C}(0,R)} P\,dx + Q\,dy$

$\begin{cases} x = R\cos t \\ y = R\sin t \end{cases}, t \in [0, 2\pi]$

$\int_{\mathcal{C}(0,R)} P\,dx + Q\,dy = \int_0^{2\pi} \frac{R(\cos t - \sin t)}{R^2} \cdot (-R\sin t) + \frac{R(\cos t + \sin t)}{R^2} \cdot R\cos t\,dt = \int_0^{2\pi} \frac{-R\cos t\sin t + R^2\sin^2 t + R^2\cos^2 t + R\cos t\sin t}{R^2}\,dt =$

$= \int_0^{2\pi} \frac{R^2}{R^2}\,dt = t\Big|_0^{2\pi} = 2\pi \Rightarrow \int_\Gamma P\,dx + Q\,dy = 2\pi$

4) $\vec{v} = (\underbrace{x+y^2}_{P}, \underbrace{z^2-y}_{Q}, \underbrace{x+z^2}_{R})$



$\Sigma: \begin{cases} z-1 = x^2+y^2 \\ z \in [1,2] \end{cases}$

$\mathcal{F}_{\Sigma \cup \Sigma_1}(\vec{v}) \overset{G-O}{=\!=} \iiint_\Omega div(\vec{v})\,dx\,dy\,dz$

3

$$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 1 - 1 + 2z = 2z$$

$$\Rightarrow \Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \iiint_{\Omega} 2z \, dx\,dz\,dy = \iint_D \left(\int_{x^2+y^2+1}^{2} 2z \, dz \right) dx\,dy =$$

$$= \iint_D z^2 \Big|_{x^2+y^2+1}^{2} dx\,dy = \iint_D \left[4 - (x^2+y^2+1)^2 \right] dx\,dy =$$

~~$= \iint_D (4 - x^4 - y^4 - 1 - 2x^2y^2 - 2x^2 - 2y^2)\, dx\,dy$~~

$$= 4 \iint_D dx\,dy - \iint_D (x^2+y^2+1)^2 \, dx\,dy$$

$$\left.\begin{array}{l} 1 \le z \le 2 \\ z = x^2+y^2+1 \end{array}\right\} \Rightarrow 1 \le x^2+y^2+1 \le 2 \;/-1 \Rightarrow 0 \le x^2+y^2 \le 1$$

$D: x^2+y^2 \le 1$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \quad \rho \in [0;1], \; \theta \in [0; 2\pi]$$

$$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = 4\pi - \int_0^{2\pi} \int_0^1 (\rho^2+1)^2 \cdot \rho \, d\rho\, d\theta = 4\pi - \frac{1}{2} \int_0^{2\pi} \frac{1}{3} \cdot (\rho^2+1)^3 \Big|_0^1 d\theta$$

$$= 4\pi - \frac{1}{6} \int_0^{2\pi} (2^3 - 1^3) d\theta = 4\pi - \frac{7}{6}\theta \Big|_0^{2\pi} = \frac{3}{4}\pi - \frac{7\pi}{3} = \frac{5\pi}{3}$$

$$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \Phi_{\Sigma}(\bar{v}) + \Phi_{\Sigma_1}(\bar{v}) \Rightarrow \Phi_{\Sigma}(\bar{v}) = \Phi_{\Sigma \cup \Sigma_1}(\bar{v}) - \Phi_{\Sigma_1}(\bar{v}) =$$

$$= \frac{5\pi}{3} - \Phi_{\Sigma_1}(\bar{v})$$

$N_1 = \bar{k}$

$\Sigma_1 : z = 2$

$$\Phi_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v} \cdot \bar{N}_1 \, d\sigma = \iint_D (2^2 + z) \, dx\,dy = 4 \iint_D dx\,dy + \iint_D x \, dx\,dy =$$

$$= 4\pi + \int_0^{2\pi} \int_0^1 \rho\cos\theta \cdot \rho \, d\rho\, d\theta = 4\pi + \int_0^{2\pi} \frac{\cos\theta}{3} \cdot \rho^3 \Big|_0^1 d\theta =$$

$$= 4\pi + \frac{1}{3} \int_0^{2\pi} \cos\theta \, d\theta = 4\pi + \frac{1}{3} \sin\theta \Big|_0^{2\pi} = 4\pi$$

$$\Rightarrow \Phi_{\Sigma}(\bar{v}) = \frac{5\pi}{3} - 4\pi = -\frac{7\pi}{3}$$

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2) $I = \int_0^{\infty} \frac{\sqrt{x}}{(1+x^4)^2}\,dx$

$x^2 = y \Rightarrow x = \sqrt{y} \Rightarrow dx = \frac{1}{2\sqrt{y}}\,dy$

$I = \frac{1}{2}\int_0^{\infty} \frac{y^{\frac{1}{2}} \cdot y^{-\frac{1}{2}}}{(1+y)^2}\,dy = \frac{1}{2}\int_0^{\infty} \frac{y^{-\frac{3}{8}}}{(1+y)^2}\,dy$

$B(p;q) = \int_0^{\infty} \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = -\frac{3}{8} \Rightarrow p = -\frac{3}{8} + \frac{8}{8} = \frac{5}{8}$

$p+q = 2 \Rightarrow q = 2 - \frac{5}{8} = \frac{11}{8}$

$\Rightarrow I = \frac{1}{2} B\left(\frac{5}{8}; \frac{11}{8}\right)$

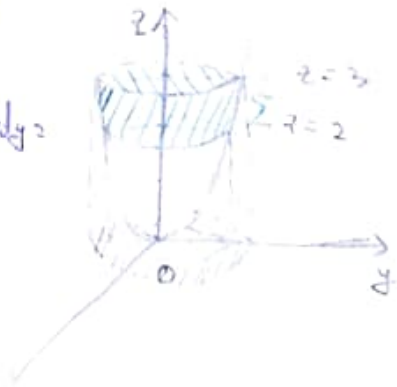
$B(p;q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$\Rightarrow I = \frac{1}{2} \cdot \frac{\Gamma\left(\frac{5}{8}\right)\cdot\Gamma\left(\frac{11}{8}\right)}{\Gamma(2)} = \frac{1}{2}\,\Gamma\left(\frac{5}{8}\right)\cdot\Gamma\left(1+\frac{3}{8}\right) =$

$= \frac{1}{2}\cdot\frac{3}{8}\cdot\Gamma\left(\frac{5}{8}\right)\cdot\Gamma\left(\frac{3}{8}\right) = \frac{3}{16}\,\Gamma\left(\frac{5}{8}\right)\cdot\Gamma\left(1-\frac{5}{8}\right) =$

$= \frac{3}{16}\cdot\frac{\pi}{\sin\left(\frac{5\pi}{8}\right)}$

3) $\Sigma: z = x^2 + y^2,\ z \in [2,3]$; $f(x,y,z) = x^2+y^2$



$\text{Aria}(\Sigma) = \int_{\Sigma} d\sigma = \iint_D \sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{1+(2x)^2+(2y)^2}\,dx\,dy =$

$= \iint_D \sqrt{1+4\rho^2}\,\rho\,d\rho\,d\theta$

$= \iint_D \sqrt{1+4(x^2+y^2)}\,dx\,dy$

$\left.\begin{array}{l} z = x^2+y^2 \\ 2 \le z \le 3 \end{array}\right\} \Rightarrow 2 \le x^2+y^2 \le 3$

$D:\ 2 \le x^2+y^2 \le 3$

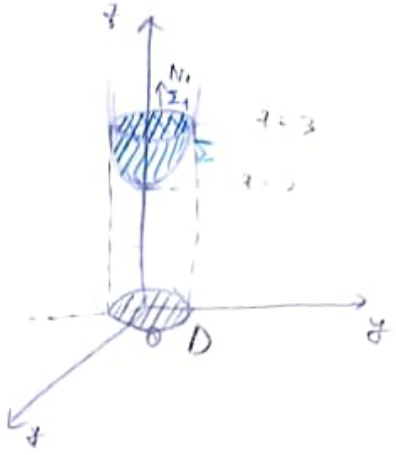
$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho \in [\sqrt{2};\sqrt{3}]$, $\theta \in [0, 2\pi]$

Aria $(\Sigma) = \int_0^{2\pi} \int_{\sqrt{2}}^{\sqrt{3}} \sqrt{1+4\rho^2} \cdot \rho \, d\rho d\theta = \frac{1}{8} \int_0^{2\pi} \frac{2}{3} \cdot (1+4\rho^2)^{\frac{3}{2}} \Big|_{\sqrt{2}}^{\sqrt{3}} d\theta =$

$= \frac{1}{12} (13\sqrt{13} - 27)\theta \Big|_0^{2\pi} = \frac{\pi}{6}(13\sqrt{13} - 27)$

4) $\bar{v} = (\underbrace{-x - y^2}_{P}, \underbrace{y - z^2}_{Q}, \underbrace{3z^2 - y}_{R})$

$\Sigma = \begin{cases} z - 2 = x^2 + y^2 \\ z \in [2,3] \end{cases}$



$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) \overset{G-O}{=} \iiint_\Omega \operatorname{div}(\bar{v}) \, dx dy dz$

$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} =$

$= -1 + 1 + 6z = 6z$

$\Rightarrow \Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \iiint_\Omega 6z \, dz \, dx dy =$

$= \iint_D \int_{x^2+y^2+2}^{3} 6z \, dz dx dy = \iint_D 6 \cdot \frac{z^2}{2} \Big|_{x^2+y^2+2}^{3} dx dy =$

$= 6 \iint_D [1 - (x^2+y^2)] \, dx dy$

$\begin{cases} 2 \le z \le 3 \\ z - 2 = x^2 + y^2 \end{cases} \Rightarrow 0 \le x^2 + y^2 \le 1$

$D: x^2 + y^2 \le 1$

$\begin{cases} x = \rho \cos\theta \\ y = \rho \sin\theta \end{cases}, \rho \in [0,1], \theta \in [0, 2\pi]$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = 6 \int_0^{2\pi} \int_0^1 (1 - \rho^2) \rho \, d\rho d\theta = 6 \int_0^{2\pi} \left(\frac{\rho^2}{2} - \frac{\rho^4}{4} \right) \Big|_0^1 d\theta =$

$= 6 \int_0^{2\pi} \frac{1}{4} \, d\theta = 3\theta \Big|_0^{2\pi} = 6\pi$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \Phi_\Sigma(\bar{v}) + \Phi_{\Sigma_1}(\bar{v}) \Rightarrow \Phi_\Sigma(\bar{v}) = \Phi_{\Sigma \cup \Sigma_1}(\bar{v}) - \Phi_{\Sigma_1}(\bar{v}) =$

$= 6\pi - \Phi_{\Sigma_1}(\bar{v})$

$\Phi_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v} \cdot \bar{N}_1 \, d\sigma$

$\bar{N}_1 = \bar{k}$

$\Sigma_1 : z = 3$

$\Phi_{\Sigma_1}(\bar{v}) = \iint_D (3 \cdot 3^2 - y) \, dx dy = 27 \iint_D dx dy - \iint_D y \, dx dy =$

3

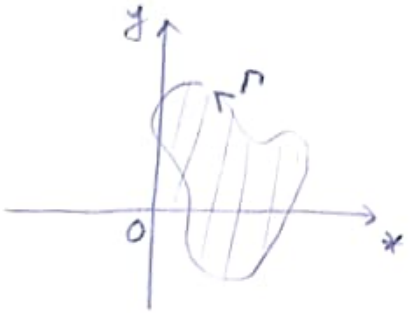
$= 27\bar{u} - \int_0^{2\bar{u}} \int_0^1 (\rho \sin\theta)\cdot\rho \, d\rho \, d\theta = 27\bar{u} - \int_0^{2\bar{u}} \left(\sin\theta \cdot \frac{\rho^3}{3}\right)\Big|_0^1 d\theta =$

$= 27\bar{u} - \int_0^{2\bar{u}} \frac{\sin\theta}{3} \, d\theta = 27\bar{u} + \frac{1}{3}\cos\theta \Big|_0^{2\bar{u}} = 27\bar{u} + \frac{1}{3} - \frac{1}{3} = 27\bar{u}$

$\Rightarrow T_2(\bar{v}) = 6\bar{u} - 27\bar{u} = -21\bar{u}$

5) $\int_\Gamma \underbrace{-\frac{y}{x^2+y^2}}_{P} dx + \underbrace{\frac{x}{x^2+y^2}}_{Q} dy$

Γ drum plan inchis arbitrar, $0 \notin \Gamma$



Cazul I) $0 \in Ext(\Gamma)$

$\frac{\partial Q}{\partial x} = \frac{x^2+y^2 - x\cdot 2x}{(x^2+y^2)^2}$

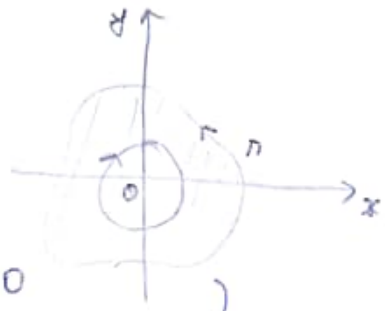
$\frac{\partial P}{\partial y} = \frac{-x^2-y^2+y\cdot 2y}{(x^2+y^2)^2}$

$\int_\Gamma P\,dx + Q\,dy \overset{G-R}{=} \iint_K \frac{x^2+y^2-2x^2+x^2+y^2-2y^2}{(x^2+y^2)^2} \, dx\,dy$

$= \iint_K \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \, dx\,dy = \iint_K \frac{x^2+y^2-2x^2+x^2+y^2-2y^2}{(x^2+y^2)^2} \, dx\,dy = 0$

Cazul II) $0 \in Int(\Gamma)$

Fie $R>0$ a.i. $D(0,R) \subset K$



$A = K \setminus D(0,R)$

$Fr A = \Gamma \cup C(0,R)$

$\int_{Fr A} P\,dx + Q\,dy \overset{G-R}{=} \iint_A \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \, dx\,dy = 0$

$\int_{Fr A} P\,dx + Q\,dy = \int_\Gamma P\,dx + Q\,dy - \int_{C(0,R)} P\,dx + Q\,dy \Big\} \Rightarrow$

$\Rightarrow \int_\Gamma P\,dx + Q\,dy = \int_{C(0,R)} P\,dx + Q\,dy$

$\begin{cases} x = R\cos t \\ y = R\sin t \end{cases}, t \in [0, 2\bar{u}]$

$\int_{C(0,R)} P\,dx + Q\,dy = \int_0^{2\bar{u}} \left(\frac{-R\sin t}{R^2} \cdot (-R\sin t) + \frac{R\cos t}{R^2} \cdot R\cos t \right) dt =$

$= \int_0^{2\bar{u}} \frac{R^2}{R^2} \, dt = \int_0^{2\bar{u}} dt = t\Big|_0^{2\bar{u}} = 2\bar{u} \Rightarrow \int_\Gamma P\,dx + Q\,dy = 2\bar{u}$

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2) $J = \int_0^\infty \frac{x^2}{1+x^4}\,dx$

$x^4 = y \Rightarrow x = \sqrt[4]{y} \Rightarrow dx = \frac{1}{4\sqrt[4]{y^3}}\,dy \; (=) \; dx = \frac{1}{4}\cdot y^{-\frac{3}{4}}\,dy$

$J = \frac{1}{4}\int_0^\infty \frac{y^{\frac{1}{2}}\cdot y^{-\frac{3}{4}}}{(1+y)^1}\,dy = \frac{1}{4}\int_0^\infty \frac{y^{-\frac{1}{4}}}{(1+y)^1}\,dy$

$\beta(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = -\frac{1}{4} \Rightarrow p = -\frac{1}{4}+\frac{4}{4} = \frac{3}{4}$

$p+q = 1 \Rightarrow q = \frac{4}{4}-\frac{3}{4} = \frac{1}{4}$

$\Rightarrow J = \frac{1}{4}\beta\left(\frac{3}{4};\frac{1}{4}\right)$

$\beta(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)} \Rightarrow J = \frac{1}{4}\cdot\frac{\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right)}{\Gamma(1)} =$

$= \frac{1}{4}\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(1-\frac{1}{4}\right) = \frac{1}{4}\cdot\frac{\pi}{\sin\frac{\pi}{4}} = \frac{\pi}{4}\cdot\frac{2}{\sqrt{2}} = \frac{\pi\sqrt{2}}{4}$

3) $x^2+y^2+z^2 = R^2$

$x^2+y^2 \le Ry,\ R>0, z\ge 0$

$x^2+y^2 \le Ry \Rightarrow x^2+y^2-2\cdot y\cdot\frac{R}{2} \le 0 \Rightarrow$

$\Rightarrow x^2+y^2-2y\cdot\frac{R}{2}+\frac{R^2}{4} \le \frac{R^2}{4} \Rightarrow$

$\Rightarrow x^2+\left(y-\frac{R}{2}\right)^2 \le \frac{R^2}{4}$

$x^2+y^2+z^2 = R^2 \Rightarrow z^2 = R^2-(x^2+y^2)$

$\Rightarrow z = \sqrt{R^2-x^2-y^2} = f(x,y)$

$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{1+\left(\frac{-x}{\sqrt{R^2-(x^2+y^2)}}\right)^2+\left(\frac{-y}{\sqrt{R^2-(x^2+y^2)}}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{\frac{R^2-x^2-y^2+x^2+y^2}{R^2-(x^2+y^2)}}\,dx\,dy = \iint_D \frac{R}{\sqrt{R^2-(x^2+y^2)}}\,dx\,dy =$

$= R\iint_D \frac{1}{\sqrt{R^2-(x^2+y^2)}}\,dx\,dy$

$D: x^2 + \left(y - \frac{R}{2}\right)^2 \le \frac{R^2}{4}$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$

$x^2+y^2 \le Ry \Leftrightarrow \rho^2 \le R\rho\sin\theta \Leftrightarrow \rho \le R\sin\theta \Rightarrow \rho \in [0, R\sin\theta]$

$\left.\begin{array}{l} R\sin\theta \ge \rho \\ \rho \ge 0 \\ R > 0 \end{array}\right\} \Rightarrow \sin\theta \ge 0 \Rightarrow \theta \in [0, \pi]$

$$\text{Aria}(\Sigma) = R\int_0^{\pi}\int_0^{R\sin\theta} \frac{1}{\sqrt{R^2-\rho^2}} \cdot \rho \, d\rho \, d\theta =$$

$$= -\frac{R}{2}\int_0^{\pi}\int_0^{R\sin\theta} (R^2-\rho^2)' \cdot (R^2-\rho^2)^{-\frac{1}{2}} \, d\rho \, d\theta =$$

$$= -R\int_0^{\pi} (R^2-\rho^2)^{\frac{1}{2}} \Big|_0^{R\sin\theta} d\theta = -R\int_0^{\pi} \left(\sqrt{R^2 - R^2\sin^2\theta} - \sqrt{R^2}\right) d\theta =$$

$$= -R\int_0^{\pi} (R\sqrt{1-\sin^2\theta} - R)\, d\theta = -R\int_0^{\pi} (R\cos\theta - R)\, d\theta =$$

$$= -R^2\int_0^{\pi} (\cos\theta - 1)\, d\theta = -R^2\left(\int_0^{\pi}\cos\theta \, d\theta - \int_0^{\pi} d\theta\right) =$$

$$= -R^2(\sin\theta - \theta)\Big|_0^{\pi} = \pi R^2$$

4) $\begin{cases} z^2 = x^2+y^2, & z \ge 0 \\ 2 - z = x^2+y^2 \end{cases}$



$$\text{Vol}(\Omega) = \iiint_{\Omega} dx\,dy\,dz =$$

$$= \iint_D \left(\int_{\sqrt{x^2+y^2}}^{2-x^2-y^2} dz\right) dx\,dy = \iint_D \left(2 - x^2 - y^2 - \sqrt{x^2+y^2}\right) dx\,dy =$$

$\left.\begin{array}{l} z^2 = x^2+y^2 \\ 2 - z = x^2+y^2 \end{array}\right\} \Rightarrow 2 - z = z^2 \Leftrightarrow z^2 + z - 2 = 0$

$\Delta = 1 + 8 = 9; \quad z_{1,2} = \frac{-1 \pm 3}{2}$

$z \ge 0$

$\Rightarrow z = 1 \Rightarrow x^2+y^2 = 1$

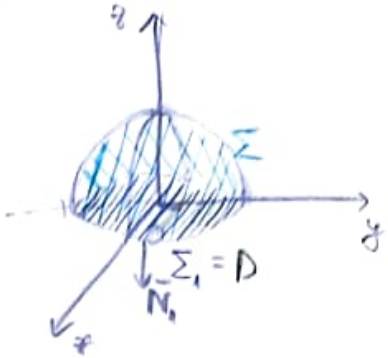
$D: x^2+y^2 \le 1$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases} \quad \rho \in [0,1], \ \theta \in [0, 2\pi]$

$$\text{Vol}(\Omega) = \int_0^{2\pi}\int_0^1 (2 - \rho^2 - \sqrt{\rho^2})\,\rho \, d\rho \, d\theta = \int_0^{2\pi} \left(\rho^2 - \frac{\rho^4}{4} - \frac{\rho^3}{3}\right)\Big|_0^1 d\theta =$$

3

$$= \int_0^{2\pi} \left(\frac{12}{1} - \frac{3}{4} - \frac{1}{3} \right) d\theta = \frac{5}{12}\theta \Big|_0^{2\pi} = \frac{5\pi}{6}$$



5) $\bar{v}(x,y,z) = (\underbrace{2x + z^4}_{P}, \underbrace{z^2 - 2y}_{Q}, \underbrace{2+3}_{R})$

$\Sigma = \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 4, \ z \geq 0\}$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) \overset{G-O}{=\!=} \iiint_\Omega \operatorname{div}(\bar{v}) \, dx\,dy\,dz$

$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 2 - 2 + 1 = 1$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \iiint_\Omega dx\,dy\,dz = \frac{4\pi \cdot 2^3}{6} = \frac{16\pi}{3}$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \Phi_\Sigma(\bar{v}) + \Phi_{\Sigma_1}(\bar{v}) \Rightarrow \Phi_\Sigma(\bar{v}) = \Phi_{\Sigma \cup \Sigma_1}(\bar{v}) - \Phi_{\Sigma_1}(\bar{v}) =$

$= \frac{16\pi}{3} - \Phi_{\Sigma_1}(\bar{v})$

$\bar{N_1} = -\bar{k}$

$\Sigma_1 : \ z = 0$

$\Phi_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v} \cdot N_1 \, d\bar{\sigma} = -\iint_D 3 \, dx\,dy = -3 \iint_D dx\,dy = -12\pi$

$z = 0 \Rightarrow x^2 + y^2 = 4 \Rightarrow D: x^2 + y^2 \leq 4$

$\Rightarrow \Phi_\Sigma(\bar{v}) = \frac{16\pi}{3} + 12\pi = \frac{52\pi}{3}$

Rezolvare examen Nr 2 2013

i) $J = \int_0^\infty \frac{\sqrt{x}}{(1+x^2)^2}\,dx$

$x^2 = y \Leftrightarrow x = \sqrt{y} \Rightarrow dx = \frac{1}{2\sqrt{y}}\,dy \Rightarrow dx = \frac{1}{2} y^{-\frac{1}{2}}\,dy$

$\Rightarrow J = \frac{1}{2} \int_0^\infty \frac{y^{\frac{1}{4}} \cdot y^{-\frac{1}{2}}}{(1+y)^2}\,dy = \frac{1}{2} \int_0^\infty \frac{y^{-\frac{1}{4}}}{(1+y)^2}\,dy$

$B(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = -\frac{1}{4} \Rightarrow p = -\frac{1}{4} + \frac{4}{4} = \frac{3}{4}$

$p+q = 2 \Rightarrow q = \frac{8}{4} - \frac{3}{4} = \frac{5}{4}$

$\Rightarrow J = \frac{1}{2} B\left(\frac{3}{4}; \frac{5}{4}\right)$

$B(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$\Rightarrow J = \frac{1}{2} \frac{\Gamma\left(\frac{3}{4}\right)\cdot\Gamma\left(\frac{5}{4}\right)}{\Gamma(2)} = \frac{1}{2} \cdot \Gamma\left(\frac{3}{4}\right)\cdot\Gamma\left(1+\frac{1}{4}\right) =$

$= \frac{1}{2} \cdot \frac{1}{4} \cdot \Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(1-\frac{1}{4}\right) = \frac{1}{8} \cdot \frac{\pi}{\operatorname{sen}\left(\frac{\pi}{4}\right)} = \frac{\pi}{8} \cdot \frac{4}{\sqrt{2}} = \frac{\pi}{4\sqrt{2}}$

3) $\int_\gamma \frac{x-y}{x^2+y^2}\,dx + \frac{x+y}{x^2+y^2}\,dy$

γ curbă netedă simplă $O \notin \gamma$

Cazul i) $O \in \operatorname{Int}\gamma \subset \operatorname{Ext}(\gamma^+) \Rightarrow$

$\Rightarrow$ se poate aplica Green-Riemann

$\int_\gamma P\,dx + Q\,dy \overset{G\text{-}R}{=\!=} \iint_K \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,dx\,dy$



$\frac{\partial Q}{\partial x} = \frac{x^2+y^2 - 2x(x+y)}{(x^2+y^2)^2}$

$\frac{\partial P}{\partial y} = \frac{-(x^2+y^2) - 2y(x-y)}{(x^2+y^2)^2}$

$\int_\gamma P\,dx + Q\,dy = \iint_K \frac{x^2+y^2-2x^2-2xy+x^2+y^2+2xy-2y^2}{(x^2+y^2)^2}\,dx\,dy =$

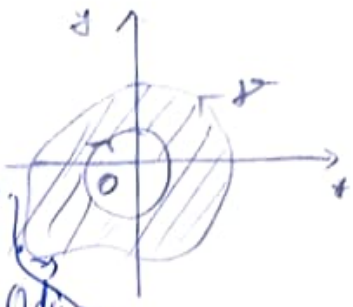
$= 0$

Cazul 2) $O \in Int(\gamma)$

$\exists R>0$ a.î. $D(O,R) \subset K$

$\gamma_A = \gamma \cup \mathcal{C}(O,R)$

$\int_{\gamma_A} P dx + Q dy = \iint_A \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dx dy = 0$



$\int_{\gamma_A} P dx + Q dy = \int_{\gamma} P dx + Q dy - \int_{\mathcal{C}(O,R)} P dx + Q dy$

$\Rightarrow \int_{\gamma} P dx + Q dy = \int_{\mathcal{C}(O,R)} P dx + Q dy$

$\begin{cases} x = R\cos t \\ y = R \sin t \end{cases}, t \in [0; 2\pi]$

$\int_{\mathcal{C}(O,R)} P dx + Q dy = \int_0^{2\pi} \left(\frac{R\cos t - R\sin t}{R^2} \cdot (-R\sin t) + \frac{R\cos t + R\sin t}{R^2} R\cos t \right) dt =$

$= \int_0^{2\pi} \frac{-R^2 \sin t \cos t + R^2 \sin^2 t + R^2 \cos^2 t + R^2 \sin t \cos t}{R^2} dt =$

$= \int_0^{2\pi} \frac{R^2}{R^2} dt = t\Big|_0^{2\pi} = 2\pi \Rightarrow \int_{\gamma} P dx + Q dy = 2\pi$

4) $\Sigma: 4 - z = x^2 + y^2, z \in [2,3]$

$2 \le z \le 3$

$z = 4 - (x^2+y^2) = f(x,y) \Rightarrow$

a) $2 \le 4 - (x^2+y^2) \le 3$ (·)

b) $-2 \le -(x^2+y^2) \le -1$ (·)

c) $1 \le x^2 + y^2 \le 2$

$D: 1 \le x^2 + y^2 \le 2$



$Aria(\Sigma) = \int_\Sigma d\sigma =$

$= \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx dy = \iint_D \sqrt{1 + (-2x)^2 + (-2y)^2}\, dx dy =$

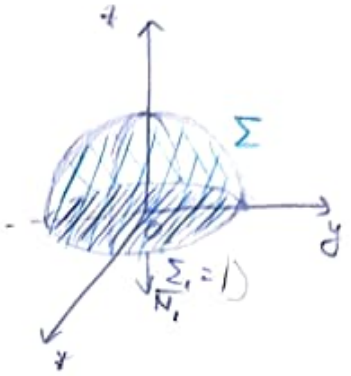
$= \iint_D \sqrt{1 + 4(x^2+y^2)}\, dx dy$

$\begin{cases} x = \rho\cos\theta \\ y = \rho \sin\theta \end{cases} \quad \rho \in [1; \sqrt{2}] \; ; \; \theta \in [0; 2\pi]$

$Aria(\Sigma) = \int_0^{2\pi} \int_1^{\sqrt{2}} \sqrt{1+4\rho^2} \cdot \rho \, d\rho \, d\theta = \frac{1}{8} \int_0^{2\pi} \frac{2}{3} \cdot (1+4\rho^2)^{\frac{3}{2}} \Big|_1^{\sqrt{2}} d\theta =$

$$= \frac{1}{12}\int_0^{2\pi}(27-5\sqrt{5})\,d\theta = \frac{1}{12}(27-5\sqrt{5})\theta\Big|_0^{2\pi} = \frac{\pi}{6}(27-5\sqrt{5})$$

5) $\bar{v}(x,y,z) = (\underbrace{3x+y^4}_{P}, \underbrace{x^3-3y}_{Q}, \underbrace{1-z}_{R})$



$\Sigma = \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 9,\ z \geq 0\}$

$$\Phi_{\Sigma\cup\Sigma_1}(\bar{v}) \overset{G.O}{=\!=} \iiint_{\Omega} \operatorname{div}(\bar{v})\,dx\,dy\,dz$$

$$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} =$$

$$= 3 - 3 - 1 = -1$$

$$\Phi_{\Sigma\cup\Sigma_1}(\bar{v}) = -\iiint_{\Omega} dx\,dy\,dz = -\frac{4\cdot\pi\cdot 3^3}{3\cdot 2} = -2\cdot\pi\cdot 3^2 = -18\pi$$

$$\Phi_{\Sigma}(\bar{v}) = \Phi_{\Sigma\cup\Sigma_1}(\bar{v}) - \Phi_{\Sigma_1}(\bar{v}) = -18\pi - \Phi_{\Sigma_1}(\bar{v})$$

$\bar{N}_1 = -\bar{k}$

$\Sigma_1 : z = 0$

$$\Phi_{\Sigma_1}(\bar{v}) = \iint_{\Sigma_1} \bar{v}\cdot\bar{N}_1\,d\sigma = -\iint_D 1\,dx\,dy = -\iint_D dx\,dy$$

$\left.\begin{array}{l} x^2+y^2+z^2 = 9 \\ z = 0 \end{array}\right\} \Rightarrow x^2+y^2 = 9$

$D: x^2+y^2 \leq 3^2$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases},\ \rho\in[0;3]\ ;\ \theta\in[0;2\pi]$$

$\Phi_{\Sigma_1}(\bar{v}) = -9\pi$

$\Rightarrow \Phi_{\Sigma}(\bar{v}) = -18\pi + 9\pi = -9\pi$

Rezolvare examen 2024 Nr. 1

1) $I = \int_0^\infty \frac{x^2}{1+x^4}\,dx$

$x^4 = y \Leftrightarrow x = \sqrt[4]{y} \Rightarrow dx = \frac{1}{4\sqrt[4]{y^3}}\,dy$

$I = \int_0^\infty \frac{y^{\frac{1}{2}}}{1+y} \cdot \frac{1}{4y^{\frac{3}{4}}}\,dy = \frac{1}{4}\int_0^\infty \frac{y^{-\frac{1}{4}}}{(1+y)^1}\,dy$

$\beta(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = -\frac{1}{4} \Rightarrow p = -\frac{1}{4} + \frac{4}{4} = \frac{3}{4}$

$p+q = 1 \Rightarrow q = \frac{4}{4} - \frac{3}{4} = \frac{1}{4}$

$I = \frac{1}{4} \cdot \beta\left(\frac{3}{4}; \frac{1}{4}\right)$

$\beta(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$I = \frac{1}{4} \cdot \frac{\Gamma\left(\frac{3}{4}\right)\cdot\Gamma\left(\frac{1}{4}\right)}{\Gamma(1)} = \frac{1}{4}\cdot\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(1-\frac{1}{4}\right) =$

$= \frac{1}{4}\cdot\frac{\pi}{\sin\frac{\pi}{4}} = \frac{1}{4}\cdot\pi\cdot\frac{2}{\sqrt{2}} = \frac{\pi}{2\sqrt{2}}$

2) $x^2+y^2+z^2 = R^2$, $R>0$

$x^2 + \left(y-\frac{R}{2}\right)^2 = \frac{R^2}{4}$ cu $z \geq 0$

Σ: $x^2+y^2+z^2 = R^2 \Rightarrow$

$\Rightarrow z = \sqrt{R^2-x^2-y^2} = f(x,y)$

$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{1+\left(\frac{-x}{\sqrt{R^2-x^2-y^2}}\right)^2+\left(\frac{-y}{\sqrt{R^2-x^2-y^2}}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{1+\frac{x^2+y^2}{R^2-x^2-y^2}}\,dx\,dy = \iint_D \sqrt{\frac{R^2-x^2-y^2+x^2+y^2}{R^2-x^2-y^2}}\,dx\,dy =$

$$= \iint_D \frac{R}{\sqrt{R^2 - x^2 - y^2}}\,dx\,dy = R \iint_D \frac{1}{\sqrt{R^2 - x^2 - y^2}}\,dx\,dy$$

$$D: x^2 + \left(y - \frac{R}{2}\right)^2 \le \frac{R^2}{4}$$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$$

$$x^2 + y^2 - Ry + \frac{R^2}{4} - \frac{R^2}{4} \le 0 \Rightarrow x^2 + y^2 \le Ry$$

$$\Rightarrow \rho^2 \le R\rho\sin\theta \Rightarrow \rho \le R\sin\theta \Rightarrow \rho \in [0, R\sin\theta]$$

$$\left.\begin{array}{l} R\sin\theta \ge \rho \\ R > 0,\ \rho \ge 0 \end{array}\right\} \Rightarrow \sin\theta \ge 0 \Rightarrow \theta \in [0; \pi]$$

$$\text{Aria}(\Sigma) = R \iint_{0}^{\pi}\int_{0}^{R\sin\theta} \frac{1}{\sqrt{R^2 - \rho^2}}\,\rho\,d\rho\,d\theta = \cancel{R}$$

$$= -\frac{R}{2}\int_0^{\pi} \left(R^2 - \rho^2\right)^{\frac{1}{2}} \cdot 2 \Big|_0^{R\sin\theta} d\theta =$$

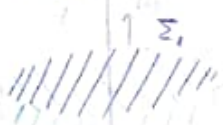
$$= -R\int_0^{\pi}\left(\sqrt{R^2 - R^2\sin^2\theta} - \sqrt{R^2}\right)d\theta =$$

$$= -R\int_0^{\pi}\left(R\sqrt{1 - \sin^2\theta} - R\right)d\theta = -R^2\int_0^{\pi}(\cos\theta - 1)\,d\theta =$$

$$= -R^2(\sin\theta - \theta)\Big|_0^{\pi} = +R^2\theta\Big|_0^{\pi} = \pi R^2$$

3) $\bar{v} = \left(\frac{\rho}{1 + 2xy}, \frac{Q}{z^2 - y^2}, \frac{R}{1}\right)$

$$\Sigma: \begin{cases} z = x^2 + y^2 \\ z \in [0, 1] \end{cases}$$



$$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) \overset{G-O}{=\!=} \iiint_{\Omega} \cancel{dx\,dy\,dz}\ \text{div}(\bar{v})\,dx\,dy\,dz$$

$$\text{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} =$$

$$= 2y - 2y + 0 = 0 \Rightarrow$$

$$\Rightarrow \Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = 0$$

$$\left.\begin{array}{l} \Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \Phi_{\Sigma}(\bar{v}) + \Phi_{\Sigma_1}(\bar{v}) \end{array}\right\} \Rightarrow \Phi_{\Sigma}(\bar{v}) = -\Phi_{\Sigma_1}(\bar{v})$$

$$\Phi_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v} \cdot \bar{N}_1\,d\sigma$$

3

$\overline{N_1} = \overline{k}$

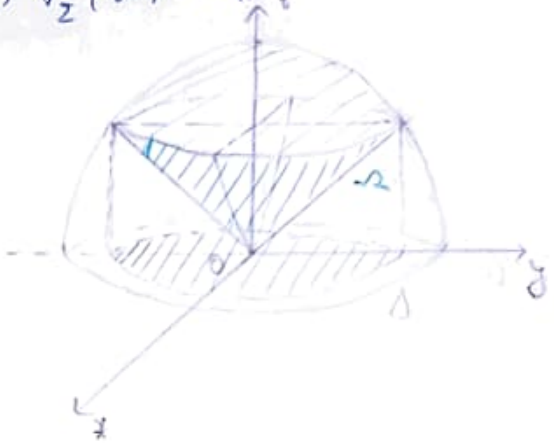
$\left.\begin{array}{l} z = x^2+y^2 \\ 0 \le z \le 1 \end{array}\right\} \Rightarrow \quad 0 \le x^2+y^2 \le 1$

$D: \; x^2+y^2 \le 1$

$\Sigma_1: \; z = 1$

$\Phi_{\Sigma_1}(\overline{v}) = \iint\limits_D 1 \, dx dy = \overline{\pi} \cdot 1^2 = \overline{\pi} \Rightarrow \Phi_{\Sigma}(\overline{v}) = -\overline{\pi}$

4) $\Omega: \begin{cases} x^2+y^2+z^2 = 2 \\ z^2 = x^2+y^2 \end{cases} \quad z \ge 0$



$Vol(\Omega) = \iiint_{\Omega} dx dy dz =$

$= \iint\limits_D \left(\int\limits_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} dz \right) dx dy =$

$= \iint\limits_D \left(\sqrt{2-x^2-y^2} - \sqrt{x^2+y^2} \right) dx dy$

$\begin{cases} x^2+y^2+z^2 = 2 \\ z^2 = x^2+y^2 \end{cases} \Rightarrow 2z^2 = 2 \Rightarrow z^2 = 1 \left.\begin{array}{l} \\ z \ge 0 \end{array}\right\} \Rightarrow z = 1 \Rightarrow$

$\Rightarrow x^2+y^2 = 1$

$D: \; x^2+y^2 \le 1$

$\begin{cases} x = \rho \cos\theta \\ y = \rho \sin\theta \end{cases} \quad \rho \in [0;1], \; \theta \in [0; 2\pi]$

$Vol(\Omega) = \int\limits_0^{2\pi} \int\limits_0^1 \left(\sqrt{2-\rho^2} - \sqrt{\rho^2} \right) \rho \, d\rho d\theta =$

$= \int\limits_0^{2\pi} \int\limits_0^1 \left(\rho\sqrt{2-\rho^2} - \rho^2 \right) d\rho d\theta = \int\limits_0^{2\pi} \left(-\frac{1}{2} \int\limits_0^1 -2\rho\sqrt{2-\rho^2} \, d\rho - \frac{\rho^3}{3} \Big|_0^1 \right) d\theta =$

$= \int\limits_0^{2\pi} \left[\left((2-\rho^2)^{\frac{3}{2}} \cdot \frac{2}{3} \right) \Big|_0^1 - \frac{1}{3} \right] d\theta = \int\limits_0^{2\pi} \left(\frac{1}{3}(1-2\sqrt{2}) - \frac{1}{3} \right) d\theta =$

$= \frac{1}{3}(4+2\sqrt{2}-1)\theta \Big|_0^{2\pi} = \frac{1}{3}(1-4\sqrt{2}) \cdot 2\pi - \frac{2\pi}{3}(1-4\sqrt{2})$

$= \frac{1}{3}(2\sqrt{2}-2)\theta \Big|_0^{2\pi} = \frac{4\pi}{3}(4\sqrt{2}-1)$

5) $\varepsilon < 10^{-3}$ $\quad x^3 + 5x - 2 = 0$

$f(x) = x^3 + 5x - 2$

$f(0) = -2$

$f(1) = 4$

$f(0) \cdot f(1) = -6 < 0$ $\Big\}$ $\Rightarrow$ $f(x) = 0$ are cel puțin o soluție reală în f continuă $(0,1)$

$f'(x) = 3x^2 + 5 \Rightarrow$ ~~f'~~ f crescătoare $\Big\}$ $\Rightarrow f$ are ~~o~~ unică soluție în $(0,1)$

$x^3 + 5x - 2 = 0 \Leftrightarrow x^3 + 5x = 2 \Leftrightarrow x(x^2+5) = 2 \Rightarrow$

$\Rightarrow x = \dfrac{2}{x^2+5}$

Fie $T(x) = \dfrac{2}{x^2+5}$, $T : [0,1] \to [0,1]$

$[0;1]$ - spațiu metric complet

Conform teoremei Lagrange $\Rightarrow \exists\, c \in [0,1]$ a.î. $T(x) - T(y) =$

$= T'(c)(x-y) \Rightarrow |T(x) - T(y)| = |T'(c)| \cdot |x-y| \le \sup |T'(c)| \cdot |x-y|$ (1)

$T'(x) = \dfrac{-3 \cdot 2x}{(x^2+9)^2} = -\dfrac{6x}{(x^2+5)^2}$

$T'(x) = \dfrac{-2 \cdot 2x}{(x^2+5)^2} = \dfrac{-4x}{(x^2+5)^2} = g(x)$ $\quad g(x) = |T'(x)| = \dfrac{4x}{(x^2+5)^2}$

$g'(x) = \dfrac{-4(x^2+5)^{2-1} - (-4x) \cdot 2(x^2+5)^{2-1} \cdot 2x}{(x^2+5)^{2\cdot 2}} =$

$= \dfrac{-4x^2 - 20 + 16x^2}{(x^2+5)^3} = \dfrac{12x^2 + 20}{(x^2+5)^3} > 0$, $(\forall)\, x \in [0,1] \Rightarrow$

$\Rightarrow g$ crescătoare $\Rightarrow \sup g(x) = \sup |T'(x)| = g(1) = \dfrac{4}{6} = \dfrac{1}{9}$

Din (1) $\Rightarrow$ T contracție de factor $K = \dfrac{1}{9}$ $\Big\}$ $\Rightarrow$ Conform Th. Contracției

$[0,1]$ spațiu metric complet

$\Rightarrow \exists!\, \xi \in [0,1]$ a.î. $T(\xi) = \xi$

$x_0 = 0$, $x_{n+1} = T(x_n)$, $(\forall)\, n \in \mathbb{N}$

5

$$|x_m - \xi| \leq \frac{K^m}{1-K} \cdot |x_1 - x_0| < 10^{-3}$$

$$x_1 = T(x_0) = T(0) = \frac{2}{5}$$

$$\frac{\left(\frac{1}{9}\right)^m}{1 - \frac{1}{9}} \cdot \left|\frac{2}{5} - 0\right| < 10^{-3} \Rightarrow \left(\frac{1}{9}\right)^m \cdot \frac{9}{4} \cdot \frac{2}{5} < 10^{-3} \Rightarrow$$

$$\Rightarrow \frac{(-1)^m}{9^{m-1}} \cdot \frac{1}{25} < \frac{1}{1000}$$

$$\Rightarrow \frac{1}{9^{m-1}} \cdot \frac{1}{20} < \frac{1}{1000} \Rightarrow 9^{m-1} \cdot 20 > 1000 \quad m = 3 \Rightarrow$$

$$\Rightarrow \xi \simeq x_3 = T(x_2)$$

$$x_2 = T(x_1) = \frac{2}{\frac{4}{25} + \frac{125}{25}} = \frac{2}{\frac{129}{25}} = \frac{50}{129}$$

$$\xi = T\left(\frac{50}{129}\right) = \frac{2}{\left(\frac{50}{129}\right)^2 + 5} = \dots$$

Rezolvare examen 2022 Nr. 1

1) $I = \int_0^\infty \frac{x^2}{1+x^4}\,dx$

$x^4 = y \Leftrightarrow x = \sqrt[4]{y} \Rightarrow dx = \frac{1}{4\sqrt[4]{y^3}}\,dy$

$I = \frac{1}{4}\int_0^\infty \frac{y^{\frac{1}{2}} \cdot y^{-\frac{3}{4}}}{1+y}\,dy = \frac{1}{4}\int_0^\infty \frac{y^{-\frac{1}{4}}}{(1+y)^1}\,dy$

$\beta(p,q) = \int_0^\infty \frac{x^{p-1}}{(1+x)^{p+q}}\,dy$

$p-1 = -\frac{1}{4} \Rightarrow p = -\frac{1}{4} + \frac{4}{4} = \frac{3}{4}$

$p+q = 1 \Rightarrow q = \frac{4}{4} - \frac{3}{4} = \frac{1}{4}$

$\Rightarrow I = \frac{1}{4}\,\beta\left(\frac{3}{4}; \frac{1}{4}\right)$

$\beta(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$I = \frac{1}{4}\,\frac{\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right)}{\Gamma(1)} = \frac{1}{4}\cdot\sqrt{\pi}\cdot\sqrt{4} = \frac{\pi}{4}$

$= \frac{1}{4}\,\frac{\pi}{\sin\frac{\pi}{4}} = \frac{\pi}{4}\cdot\frac{2}{\sqrt{2}} = \frac{\pi}{2\sqrt{2}}$

2) $\iint_D \sqrt{4-x^2-y^2}\,dx\,dy$, $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 1,\ x \ge 0,\ y \ge 0\}$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho =$



$\left.\begin{array}{l} \rho\cos\theta \ge 0 \\ \rho\sin\theta \ge 0 \\ \rho \ge 0 \end{array}\right\} \Rightarrow \begin{cases} \cos\theta \ge 0 \\ \sin\theta \ge 0 \end{cases} \Rightarrow \theta \in \left[0, \frac{\pi}{2}\right]$

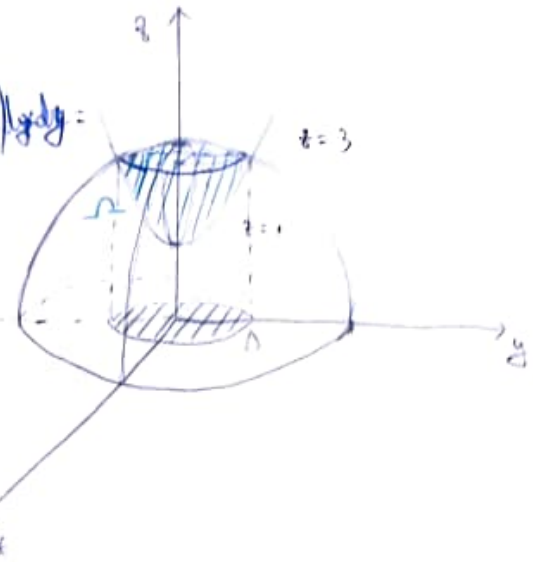
$\rho \in [0,1]$

$\iint_D \sqrt{4-x^2-y^2}\,dx\,dy = \int_0^{\frac{\pi}{2}} \int_0^1 \sqrt{4-\rho^2}\,\rho\,d\rho\,d\theta =$

$= -\frac{1}{2}\int_0^{\frac{\pi}{2}} (4-\rho^2)^{\frac{3}{2}} \cdot \frac{2}{3}\,\Big|_0^1 d\theta = -\frac{1}{3}\int_0^{\frac{\pi}{2}} (3\sqrt{3}-8)\,d\theta =$

$= -\frac{1}{3}(3\sqrt{3}-8)\theta\,\Big|_0^{\frac{\pi}{2}} = -\frac{\pi}{6}(3\sqrt{3}-8) = \frac{4\pi}{3} - \frac{\pi\sqrt{3}}{2}$

3) $\Omega \begin{cases} x^2+y^2+z^2 = 10 \\ z = 1+2x^2+2y^2 \end{cases}, z \geq 0$



$Vol(\Omega) = \iiint_\Omega dx\,dy\,dz = \iint_D \left(\int_{1+2x^2+2y^2}^{\sqrt{10-x^2-y^2}} dz \right) dx\,dy =$

$= \iint_D \sqrt{10-x^2-y^2} - 1 - 2(x^2+y^2)\,dx\,dy$

$\begin{cases} x^2+y^2+z^2 = 10 \\ z = 1+2x^2+2y^2 \end{cases}$ (1)

(1) $\Rightarrow z = 1 + 2(10-z^2) \Leftrightarrow$

(1) $\Rightarrow z = 1 + 20 - 2z^2$ (1) $\Rightarrow 2z^2 + z - 21 = 0$

$\Delta = 1 + 168 = 169 \Rightarrow z_{1,2} = \frac{-1 \pm 13}{4}$ $\Big\} \Rightarrow z = 3$

$z \geq 0$

$x^2+y^2 = 10-3^2 \Rightarrow x^2+y^2 = 1$

$D: x^2+y^2 \leq 1$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \rho \in [0;1], \theta \in [0; 2\pi]$

$Vol(\Omega) = \int_0^{2\pi} \int_0^1 \left(\sqrt{10-\rho^2} - 1 - 2\rho^2 \right) \rho\, d\rho\, d\theta =$

$= \int_0^{2\pi} \left(\left[-\frac{1}{2} \left(10-\rho^2\right)^{\frac{3}{2}} \cdot \frac{2}{3} \right]_0^1 - \frac{\rho^2}{2}\Big|_0^1 - 2\frac{\rho^4}{4}\Big|_0^1 \right) d\theta =$

$= \int_0^{2\pi} \left(-\frac{1}{3}\left(1-10\sqrt{10}\right) + \frac{1}{3} \cdot 10\sqrt{10} - \frac{1}{2} - \frac{1}{2} \right) d\theta = \left[-\frac{1}{3}\left(1+10\sqrt{10}\right) + 1 \right]\theta\Big|_0^{2\pi} =$

$= -\frac{2\pi}{3}\left(1+10\sqrt{10}\right) - 2\pi$

$= \left(-10 + \frac{10\sqrt{10}}{3} \right)\theta\Big|_0^{2\pi} = -20\pi\left(1 - \frac{\sqrt{10}}{3}\right)$

4) $\bar{v} = \underbrace{\left(e^{y^2+z^2+xy}\right.}_{P}, \underbrace{\cos xz - 2xy - \frac{x^2}{2}z}_{Q}, \underbrace{\left.1 \cdot z\right)}_{R}$

$\Sigma = S: z = 2+x^2+y^2, z \in [2;3]$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) \overset{G-O}{=\!=} \iiint_\Omega div(\bar{v})\,dx\,dy\,dz$

$div(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} =$

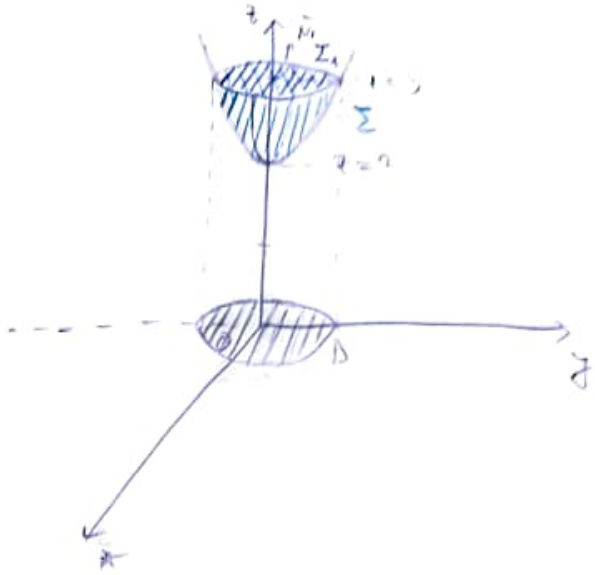
$= 2x + y - 2x - y - 1 = -1 \Rightarrow$

$\Rightarrow \Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = -\iiint_{\Omega} dx\,dy\,dz =$

$= -\iint_D \left(\int_{2+x^2+y^2}^{3} dz \right) dx\,dy =$

$= -\iint_D \left[1 - (x^2+y^2) \right] dx\,dy$

$\begin{cases} z = 2 + x^2 + y^2 \\ 1 \le z \le 3 \end{cases} \Rightarrow$



$\Rightarrow 1 \le 2 + x^2 + y^2 \le 3 \Rightarrow 0 \le x^2 + y^2 \le 1$

$D: x^2 + y^2 \le 1 \; ; \begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \rho \in [0;1], \theta \in [0; 2\pi]$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = -\int_0^{2\pi} \int_0^1 (1-\rho^2)\rho \, d\rho\, d\theta =$

$= -\int_0^{2\pi} \left(\frac{\rho^2}{2} - \frac{\rho^4}{4} \right) \Big|_0^1 d\theta =$

$= -\int_0^{2\pi} \left(\frac{1}{2} - \frac{1}{4} \right) d\theta = -\frac{1}{4}\theta \Big|_0^{2\pi} = -\frac{2\pi}{4} = -\frac{\pi}{2}$ $\Rightarrow$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \Phi_{\Sigma}(\bar{v}) + \Phi_{\Sigma_1}(\bar{v})$

$\Rightarrow \Phi_{\Sigma}(\bar{v}) = -\frac{\pi}{2} - \Phi_{\Sigma_1}(\bar{v})$

$\Phi_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v} \cdot \bar{N}_1 \, d\sigma$

$\Sigma_1 : z = 3$

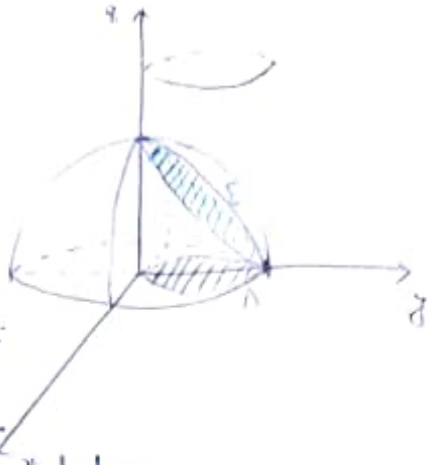
$\bar{N}_1 = \bar{k}$

$\Phi_{\Sigma_1}(\bar{v}) = \iint_D -2 \, dx\,dy = -2 \iint_D dx\,dy = -2 \cdot \bar{u} \cdot 1^2 = -2\bar{u}$

$\Phi_{\Sigma}(\bar{v}) = -\frac{\bar{u}}{2} + 2\bar{u} = \frac{3\bar{u}}{2}$

$\square$

6) $\Sigma: \begin{cases} x^2+y^2+z^2=R^2 \\ z^2+y^2=Ry \end{cases}, z\geq 0$



$R>0$

$z^2 = R^2 - x^2 - y^2 \Leftrightarrow z = \sqrt{R^2-x^2-y^2} = f(x,y)$.

$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}\,dxdy =$

$= \iint_D \sqrt{1+\left(\frac{-x}{\sqrt{R^2-x^2-y^2}}\right)^2+\left(\frac{-y}{\sqrt{R^2-x^2-y^2}}\right)^2}\,dxdy =$

$= \iint_D \sqrt{\frac{R^2-x^2-y^2+x^2+y^2}{R^2-x^2-y^2}}\,dxdy = R\iint_D \frac{1}{\sqrt{R^2-x^2-y^2}}\,dxdy$

$x^2+y^2-Ry+\frac{R^2}{4} = \frac{R^2}{4} \Leftrightarrow x^2+\left(y-\frac{R}{2}\right)^2 = \frac{R^2}{4}$

$D: x^2+\left(y-\frac{R}{2}\right)^2 \leq \frac{R^2}{4}$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho^2 \leq R\rho\sin\theta \Leftrightarrow \rho \leq R\sin\theta \Rightarrow$

$\Rightarrow \rho \in [0; R\sin\theta]$

$R\sin\theta \geq \rho$

$R>0,\ \rho\geq 0 \Big\} \Rightarrow \sin\theta \geq 0 \Rightarrow \theta\in[0;\pi]$

$\text{Aria}(\Sigma) = R\int_0^\pi \int_0^{R\sin\theta} \frac{1}{\sqrt{R^2-\rho^2}}\,\rho\,d\rho\,d\theta =$

$= -\frac{R}{2}\cdot 2\int_0^\pi \left(R^2-\rho^2\right)^{\frac{1}{2}}\Big|_0^{R\sin\theta} d\theta =$

$= -R\int_0^\pi \left(\sqrt{R^2(1-\sin^2\theta)} - \sqrt{R^2}\right)d\theta =$

$= -R\int_0^\pi \left(R\cos\theta - R\right)d\theta = -R^2\int_0^\pi (\cos\theta - 1)\,d\theta =$

$= -\left(-R^2\sin\theta + R^2\theta\right)\Big|_0^\pi = R^2\theta\Big|_0^\pi = \pi R^2$

Rezolvare examen 2022 Nr 2

1) $I = \int_0^\infty \frac{1}{1+x^4}\,dx$

$x^4 = y \Rightarrow x = \sqrt[4]{y} \Rightarrow dx = \frac{1}{4\sqrt[4]{y^3}}\,dy$

$I = \frac{1}{4}\int_0^\infty \frac{y^{-\frac{3}{4}}}{(1+y)^1}\,dy$

$\beta(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = -\frac{3}{4} \Rightarrow p = -\frac{3}{4}+\frac{4}{4} = \frac{1}{4}$

$p+q = 1 \Rightarrow q = \frac{4}{4}-\frac{1}{4} = \frac{3}{4}$

$I = \frac{1}{4}\beta\left(\frac{1}{4};\frac{3}{4}\right)$

$\beta(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$I = \frac{1}{4}\cdot\frac{\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right)}{\Gamma(1)} = \frac{1}{4}\cdot\frac{\pi}{\sin\frac{\pi}{4}} = \frac{\pi}{4}\cdot\frac{2}{\sqrt{2}} = \frac{\pi}{2\sqrt{2}}$

2) $\iint_D \sqrt{1+x^2+y^2}\,dx\,dy$

$D = \{(x,y)\in\mathbb{R}^2 \mid x^2+y^2\le 1,\ x\ge 0,\ y\ge 0\}$



$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \rho\in[0;1]$; ~~$\theta \in$~~

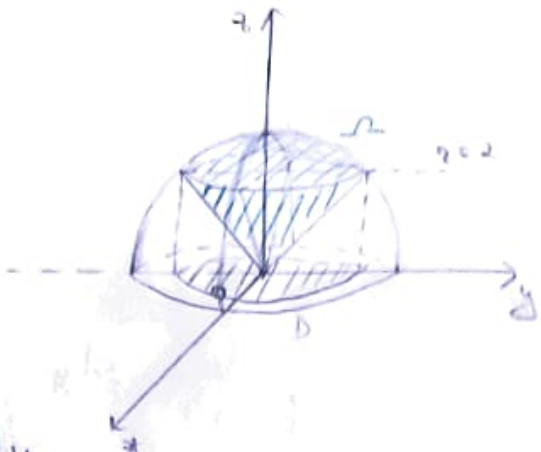
$\left.\begin{array}{l}\rho\sin\theta\ge 0\\ \rho\cos\theta\ge 0\\ \rho\ge 0\end{array}\right\} \Rightarrow \sin\theta\ge 0,\ \cos\theta\ge 0 \Rightarrow \theta\in\left[0;\frac{\pi}{2}\right]$

$\iint_D \sqrt{1+x^2+y^2}\,dx\,dy = \int_0^{\frac{\pi}{2}}\int_0^1 \sqrt{1+\rho^2}\cdot\rho\,d\rho\,d\theta =$

$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \left(1+\rho^2\right)^{\frac{3}{2}}\cdot\frac{2}{3}\Big|_0^1 d\theta = \frac{1}{2}\int_0^{\frac{\pi}{2}}(2\sqrt{2}-1)\,d\theta =$

$= \frac{1}{2}\cdot\frac{2}{3}(2\sqrt{2}-1)\,\theta\Big|_0^{\frac{\pi}{2}} = \frac{\pi}{6}(2\sqrt{2}-1)$

3) $\Omega: \begin{cases} x^2+y^2+z^2=8 \\ z^2=x^2+y^2 \end{cases}, z\geq 0$



$Vol(\Omega) = \iiint_{\Omega} dx\,dy\,dz =$

$= \iint_D \left(\int_{\sqrt{x^2+y^2}}^{\sqrt{8-x^2-y^2}} dz \right) dx\,dy =$

$= \iint_D \left(\sqrt{8-x^2-y^2} - \sqrt{x^2+y^2} \right) dx\,dy$

$\left.\begin{aligned} z^2 &= x^2+y^2 \\ x^2+y^2+z^2 &= 8 \end{aligned}\right\} \Rightarrow 2(x^2+y^2)=8 \Rightarrow x^2+y^2=4$

$D: x^2+y^2 \leq 4$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \rho\in[0;2];\ \theta\in[0;2\pi]$

$Vol(\Omega) = \int_0^{2\pi}\int_0^2 \left(\sqrt{8-\rho^2} - \sqrt{\rho^2}\right)\rho\,d\rho\,d\theta =$

$= \int_0^{2\pi} \left[-\frac{1}{2}\int_0^2 \sqrt{8-\rho^2}\cdot(8-\rho^2)'\,d\rho - \int_0^2 \rho^2\,d\rho \right] d\theta =$

$= \int_0^{2\pi} \left[-\frac{1}{2}\cdot\frac{2}{3}\cdot(8-\rho^2)^{\frac{3}{2}}\Big|_0^2 - \frac{\rho^3}{3}\Big|_0^2 \right] d\theta =$

$= \int_0^{2\pi} \left(-\frac{1}{3}\cdot 8 + \frac{1}{3}\cdot 16\sqrt{2} - \frac{8}{3} \right) d\theta =$

$= \left(-\frac{16}{3} + \frac{16}{3}\sqrt{2} \right)\theta\Big|_0^{2\pi} = \frac{32\pi}{3}\left(\sqrt{2}-1\right)$

4) $\Sigma: 2-z=x^2+y^2, z\in[1,2]$

$z = 2-x^2-y^2 = f(x,y)$

~~$2-z=x$~~

$\left.\begin{aligned} z &= 2-x^2-y^2 \\ 1 &\leq z \leq 2 \end{aligned}\right\} \Rightarrow 1 \leq 2-x^2-y^2 \leq 2 \Rightarrow$

$\Rightarrow -1 \leq -x^2-y^2 \leq 0 \Rightarrow$

$\Rightarrow 0 \leq x^2+y^2 \leq 1$

$D: x^2+y^2 \leq 1$



$$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}\,dx\,dy =$$

$$= \iint_D \sqrt{1+(-2x)^2+(-2y)^2}\,dx\,dy = \iint_D \sqrt{1+4(x^2+y^2)}\,dx\,dy$$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \rho\in[0,1],\ \theta\in[0,2\pi]$$

$$\text{Aria}(\Sigma) = \int_0^{2\pi}\int_0^1 \sqrt{1+4\rho^2}\cdot\rho\,d\rho\,d\theta = \frac{1}{8}\int_0^{2\pi} (1+4\rho^2)^{\frac{3}{2}}\cdot\frac{2}{3}\Big|_0^1 \,d\theta =$$

$$= \frac{1}{12}\int_0^{2\pi}(5\sqrt{5}-1)\,d\theta = \frac{1}{12}(5\sqrt{5}-1)\theta\Big|_0^{2\pi} =$$

$$= \frac{\pi}{6}(5\sqrt{5}-1)$$

b) $\bar{E} = \frac{q}{4\pi}\cdot\frac{\bar{r}}{r^3}$

a) $S: x^2+y^2+z^2=R^2,\ R>0$

$$\mathcal{F}_{S(O,R)}(\bar{E}) = \int_S \bar{E}\cdot\bar{N}\,d\sigma = \iint_D \frac{q}{4\pi}\cdot\frac{\bar{r}}{r^3}\cdot\frac{\bar{r}}{R}\cdot R^2\sin\theta\,d\theta\,d\rho =$$

$$= \iint_D \frac{q}{4\pi}\cdot 2\cdot 2\pi = q$$

b) Cazul I) $O\notin K \Rightarrow$ se poate aplica Gauss-Ostrogradski $\Rightarrow$

$$\mathcal{F}_{\partial K}(\bar{E}) \overset{G-O}{=\!=} \iiint_\Omega \operatorname{div}(\bar{E})\,dx\,dy\,dz \Big\} \Rightarrow \mathcal{F}_{\partial K}(\bar{E}) = 0$$

$$\operatorname{div}(\bar{E}) = 0$$

Cazul II) $O\in K$

$K_1 = \text{Ext}\,S(O,R)\cap K$

K_1 compact, $O\notin K_1$

$\partial K_1 = \partial K \cup S^-(O,R)$.



$$\mathcal{F}_{\partial K_1}(\bar{E}) \overset{G-O}{=\!=} 0$$

$$\mathcal{F}_{\partial K_1}(\bar{E}) = \mathcal{F}_{\partial K}(\bar{E}) - \mathcal{F}_{S(O,R)}(\bar{E}) \Big\} \Rightarrow \mathcal{F}_{\partial K}(\bar{E}) = \mathcal{F}_{S(O,R)}(\bar{E}) = q$$

Rezolvare examen 2023 I Nr.1

2) $I = \int_0^\infty \frac{\sqrt{x}}{(1+x^2)^3}\,dx$

$x^2 = y \Rightarrow x = \sqrt{y} \Rightarrow dx = \frac{1}{2\sqrt{y}}\,dy$

$I = \frac{1}{2}\int_0^\infty \frac{y^{\frac{1}{6}} \cdot y^{-\frac{1}{2}}}{(1+y)^3}\,dy = \frac{1}{2}\int_0^\infty \frac{y^{-\frac{1}{3}}}{(1+y)^3}\,dy$

$\beta(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = -\frac{1}{3} \Rightarrow p = -\frac{1}{3} + \frac{3}{3} = \frac{2}{3}$

$p+q = 3 \Rightarrow q = \frac{9}{3} - \frac{2}{3} = \frac{7}{3}$

$I = \frac{1}{2}\beta\left(\frac{2}{3}; \frac{7}{3}\right)$

$\beta(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

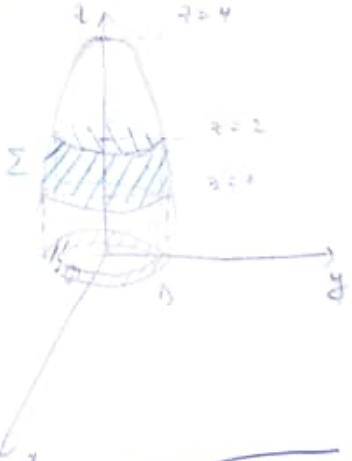
$I = \frac{1}{2}\cdot\frac{\Gamma\left(\frac{2}{3}\right)\cdot\Gamma\left(\frac{7}{3}\right)}{\Gamma(3)} = \frac{1}{2\cdot 2}\cdot\Gamma\left(\frac{2}{3}\right)\cdot\Gamma\left(1+\frac{4}{3}\right) =$

$= \frac{1}{4}\cdot\frac{4}{3}\cdot\Gamma\left(\frac{4}{3}\right)\cdot\Gamma\left(\frac{2}{3}\right) = \frac{1}{3}\Gamma\left(1+\frac{1}{3}\right)\cdot\Gamma\left(\frac{2}{3}\right) =$

$= \frac{1}{9}\Gamma\left(\frac{1}{3}\right)\cdot\Gamma\left(1-\frac{1}{3}\right) = \frac{1}{9}\cdot\frac{\pi}{\sin\frac{\pi}{3}} = \frac{1}{9}\cdot\frac{2}{\sqrt{3}} = \frac{2\pi}{9\sqrt{3}}$

3) $\Sigma: \begin{cases} 4 - z = x^2 + y^2 \\ z \in [1,2] \end{cases}$

$z = 4 - x^2 - y^2 = f(x,y)$



$\left.\begin{array}{l} 1 \le z \le 2 \\ z = 4 - x^2 - y^2 \end{array}\right\} \Rightarrow 1 \le 4 - x^2 - y^2 \le 2 \Leftrightarrow$

$\Leftrightarrow -3 \le -x^2 - y^2 \le -2 \Leftrightarrow$

$\Leftrightarrow 2 \le x^2 + y^2 \le 3$

$D: 2 \le x^2 + y^2 \le 3$

$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\,dxdy = \iint_D \sqrt{1 + (-2x)^2 + (-2y)^2}\,dxdy =$

$= \iint_D \sqrt{1 + 4(x^2+y^2)}\,dxdy$

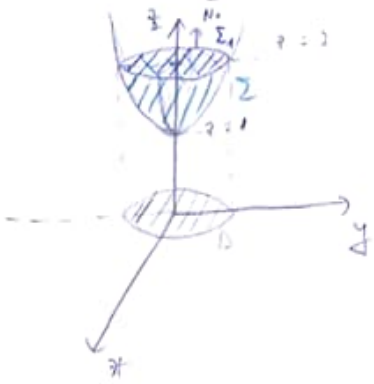
$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho \in [\sqrt{2};\sqrt{3}]$; $\theta \in [0;2\pi]$

$$\text{Aria}(\Sigma) = \int_0^{2\pi}\int_{\sqrt{2}}^{\sqrt{3}} \sqrt{1+4\rho^2}\cdot\rho\, d\rho\, d\theta =$$

$$= \frac{1}{8}\int_0^{2\pi} (1+4\rho^2)^{\frac{3}{2}}\cdot\frac{2}{3}\Big|_{\sqrt{2}}^{\sqrt{3}} d\theta = \frac{1}{12}\int_0^{2\pi}(13\sqrt{13}-27)\,d\theta =$$

$$= \frac{1}{12}(13\sqrt{13}-27)\theta\Big|_0^{2\pi} = \frac{\pi}{6}(13\sqrt{13}-24)$$ ✓

4) $\bar{v} = (\underbrace{x+y^2}_{P}, \underbrace{z^2-y}_{Q}, \underbrace{x+z^2}_{R})$



$\Sigma = S: \begin{cases} z-1 = x^2+y^2 \\ z \in [1,2] \end{cases}$

$$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) \overset{G-O}{=\!=} \iiint_\Omega \operatorname{div}(\bar{v})\,dx\,dy\,dz$$

$$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} =$$

$$= 1 - 1 + 2z = 2z$$

$$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \cancel{\iiint_\Omega 2z\,dx} \iint_D \left(\int_{1+x^2+y^2}^{2} 2z\,dz\right)dx\,dy =$$

$$= \iint_D (4-1-x^2-y^2)\,dx\,dy = \iint_D [3-(x^2+y^2)]\,dx\,dy$$

$\left.\begin{array}{l} z = 1+x^2+y^2 \\ 1 \le z \le 2 \end{array}\right\} \Rightarrow 1 \le 1+x^2+y^2 \le 2 \Leftrightarrow 0 \le x^2+y^2 \le 1$

$D: x^2+y^2 \le 1$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, ~~$x \in [0$~~ $\rho \in [0;1]$; $\theta \in [0;2\pi]$

$$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \int_0^{2\pi}\int_0^1 (3-\rho^2)\rho\,d\rho\,d\theta = \int_0^{2\pi}\int_0^1 (3\rho-\rho^3)\,d\rho\,d\theta =$$

$$= \int_0^{2\pi}\left(\frac{3\rho^2}{2} - \frac{\rho^4}{4}\right)\Big|_0^1 d\theta = \left(\frac{3}{2}-\frac{1}{4}\right)\theta\Big|_0^{2\pi} =$$

$$= \pi\left(3-\frac{1}{2}\right) = \frac{5\pi}{2}$$

$\left.\begin{array}{l} \mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \mathcal{F}_{\Sigma}(\bar{v}) + \mathcal{F}_{\Sigma_1}(\bar{v}) \end{array}\right\} \Rightarrow \mathcal{F}_{\Sigma}(\bar{v}) = \frac{5\pi}{2} - \mathcal{F}_{\Sigma_1}(\bar{v})$

3 $\Phi_{\Sigma_1}(\vec{v}) = \iint_{\Sigma_1} \vec{v} \cdot \vec{N}_1 \, d\sigma$

$\vec{N}_1 = \vec{k}$

$\Sigma_1 : z = 2$

$\Phi_{\Sigma_1}(\vec{v}) = \iint_D (z+4)\,dx\,dy = \int_0^{2\pi} \int_0^1 (\rho\cos\theta + 4)\,\rho\,d\rho\,d\theta =$

$= \int_0^{2\pi} \left(\frac{\cos\theta}{3}\rho^3 + 2\rho^2 \right)\Big|_0^1 d\theta = \int_0^{2\pi} \left(\frac{\cos\theta}{3} + 2 \right) d\theta =$

$= \frac{1}{3}\int_0^{2\pi} \cos\theta\, d\theta + 2\int_0^{2\pi} d\theta = \left(\frac{1}{3}\sin\theta + 2\theta \right)\Big|_0^{2\pi} =$

$= 4\pi$

$\Rightarrow \Phi_\Sigma(\vec{v}) = \frac{5\pi}{2} - 4\pi = \frac{5\pi}{2} - \frac{8\pi}{2} = -\frac{3\pi}{2}$

5) $\int_\gamma \underbrace{\frac{x-y}{x^2+y^2}}_{P} dx + \underbrace{\frac{x+y}{x^2+y^2}}_{Q} dy$

Cazul i) $0 \in Ext(\gamma) \Rightarrow$ se poate aplica Green–Riemann



$\int_\gamma P\,dx + Q\,dy \overset{G-R}{=} \iint_K \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy =$

$= \iint_K \frac{x^2+y^2 - 2x(x+y) + x^2+y^2 + 2y(x-y)}{(x^2+y^2)^2}\,dx\,dy =$

$= \iint_K \frac{2x^2+2y^2 - 2x^2 - 2y^2 - 2xy + 2xy}{(x^2+y^2)^2}\,dx = 0$



Cazul ii) $0 \in Int(\gamma)$

Fie $R > 0$ a.i. $D(0,R) \subset K$

Fie $A = \gamma \cup \mathcal{C}(0,R)$

$A = K \setminus D(0,R)$

$\int_{Fr A} P\,dx + Q\,dy \overset{G-R}{=\!=} \iint_A \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,dx\,dy = 0$

$\int_{Fr A} P\,dx + Q\,dy = \int_\gamma P\,dx + Q\,dy + \int_{\mathcal{C}(0,R)} P\,dx + Q\,dy$ $\Big\} \Rightarrow$

$\Rightarrow \int_\gamma P\,dx + Q\,dy = \int_{\mathcal{C}(0,R)} P\,dx + Q\,dy$

$$\begin{cases} x = R\cos t \\ y = R\sin t \end{cases}, \quad t \in [0, 2\pi]$$

$$\int_{\mathcal{C}(O,R)} P\,dx + Q\,dy = \int_0^{2\pi} \left[\frac{R\cos t - R\sin t}{R^2} \cdot (-R\sin t) + \frac{R\cos t + R\sin t}{R^2} \cdot R\cos t \right] dt$$

$$= \int_0^{2\pi} \frac{R^2(\sin^2 t + \cos^2 t)}{R^2}\,dt = \int_0^{2\pi} dt = t\Big|_0^{2\pi} = 2\pi \Rightarrow$$

$$\Rightarrow \int_\gamma P\,dx + Q\,dy = 2\pi$$

Rezolvare examen 2021 I Nr. 2

2) $J = \int_0^\infty \frac{\sqrt[4]{x}}{(1+x^2)^2}\,dx$

$x^2 = y \Rightarrow x = \sqrt{y} \Rightarrow dx = \frac{1}{2\sqrt{y}}\,dy$

$J = \frac{1}{2}\int_0^\infty \frac{y^{\frac{1}{8}} \cdot y^{-\frac{1}{2}}}{(1+y)^2}\,dy = \frac{1}{2}\int_0^\infty \frac{y^{-\frac{3}{8}}}{(1+y)^2}\,dy$ *

$\beta(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = -\frac{3}{8} \Rightarrow p = -\frac{3}{8} + \frac{8}{8} = \frac{5}{8}$

$p+q = 2 \Rightarrow q = \frac{16}{8} - \frac{5}{8} = \frac{11}{8}$

$J = \frac{1}{2}\beta\left(\frac{5}{8}; \frac{11}{8}\right)$

$\beta(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$J = \frac{1}{2}\cdot\frac{\Gamma\left(\frac{5}{8}\right)\cdot\Gamma\left(\frac{11}{8}\right)}{\Gamma(2)} = \frac{1}{2}\Gamma\left(\frac{5}{8}\right)\cdot\Gamma\left(1+\frac{3}{8}\right) =$

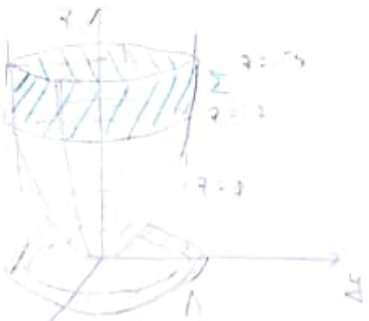
$= \frac{3}{16}\Gamma\left(\frac{3}{8}\right)\cdot\Gamma\left(\frac{5}{8}\right) = \frac{3}{16}\cdot\frac{\pi}{\sin\frac{3\pi}{8}} = \frac{3\pi}{16}\cdot\frac{1}{\sin\frac{3\pi}{8}}$

3) $\Sigma: z = x^2+y^2,\ z\in[2,3]$

$z = x^2+y^2 = f(x,y)$

$\left.\begin{array}{l} z = x^2+y^2 \\ 2 \le z \le 3 \end{array}\right\} \Rightarrow 2 \le x^2+y^2 \le 3$

$D: 2 \le x^2+y^2 \le 3$



$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{1+4(x^2+y^2)}\,dx\,dy$ *

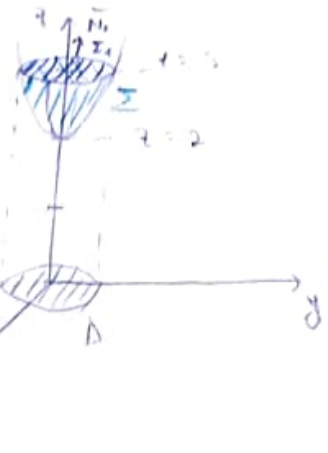
$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases},\ \rho\in[\sqrt{2};\sqrt{3}],\ \theta\in[0,2\pi]$

$\text{Aria}(\Sigma) = \int_0^{2\pi}\int_{\sqrt{2}}^{\sqrt{3}} \sqrt{1+4\rho^2}\,\rho\,d\rho\,d\theta = \frac{1}{8}\int_0^{2\pi}(1+4\rho^2)^{\frac{3}{2}}\cdot\frac{2}{3}\Big|_{\sqrt{2}}^{\sqrt{3}}\,d\theta =$

$= \frac{1}{12}\int_0^{2\pi}(13\sqrt{13}-27)\,d\theta = \frac{1}{12}(13\sqrt{13}-27)\theta\Big|_0^{2\pi} =$

$= \frac{\pi}{6}(13\sqrt{13}-27)$

4) $\bar{v} = (\overbrace{-x-y^2}^{P}, \overbrace{y-z^2}^{Q}, \overbrace{3z^2-x}^{R})$



$\Sigma: \begin{cases} z-2 = x^2+y^2 \\ z \in [2,3] \end{cases}$

$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) \overset{G-O}{=\!=} \iiint_\Omega \operatorname{div}(\bar{v})\,dx\,dy\,dz$

$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = -1+1+6z$

$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \iint_D \int_{2+x^2+y^2}^{3} 6z\,dz\,dx\,dy = 3\iint_D z^2\Big|_{2+x^2+y^2}^{3} dx\,dy =$

$= 3\iint_D [9-(2+x^2+y^2)^2]\,dx\,dy$

$\left.\begin{array}{l} z = 2+x^2+y^2 \\ 2 \le z \le 3 \end{array}\right\} \Rightarrow 0 \le x^2+y^2 \le 1$

$D: x^2+y^2 \le 1$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \rho \in [0;1], \theta \in [0;2\pi]$

$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = 3\int_0^{2\pi}\int_0^1 [9-(2+\rho^2)^2]\,\rho\,d\rho\,d\theta =$

$= 3\int_0^{2\pi}\left[\frac{9\rho^2}{2}\Big|_0^1 - \frac{1}{2}\int_0^1 2\rho(2+\rho^2)^2\,d\rho\right]d\theta =$

$= 3\int_0^{2\pi}\left[\frac{9}{2} - \frac{1}{2}\cdot\frac{1}{3}\cdot(2+\rho^2)^3\Big|_0^1\right]d\theta = 3\int_0^{2\pi}\left(\frac{9}{2}-\frac{9}{2}+\frac{4}{3}\right)d\theta =$

$= 3\cdot\frac{4}{3}\theta\Big|_0^{2\pi} = 8\pi$

$\left.\begin{array}{l} \mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \mathcal{F}_\Sigma(\bar{v}) + \mathcal{F}_{\Sigma_1}(\bar{v}) \end{array}\right\} \Rightarrow \mathcal{F}_\Sigma(\bar{v}) = 8\pi - \mathcal{F}_{\Sigma_1}(\bar{v})$

$\mathcal{F}_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v}\cdot\bar{N}_1\,d\sigma$

$\bar{N}_1 = \bar{k}$

$\Sigma_1: z = 3$

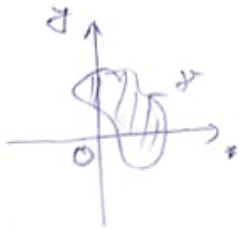
$$\mathcal{F}_{z_1}(\bar{v}) = \iint_{\Delta} (27 - y)\,dx\,dy = \int_0^{2\bar{u}} \int_0^1 (27 - \rho\sin\theta)\,\rho\,d\rho\,d\theta =$$

$$= \int_0^{2\bar{u}} \left(\frac{27}{2}\rho^2 - \frac{\sin\theta}{3}\rho^3\right)\Big|_0^1 d\theta = \int_0^{2\bar{u}} \left(\frac{27}{2} - \frac{1}{3}\sin\theta\right) d\theta =$$

$$= \frac{27\theta}{2}\Big|_0^{2\bar{u}} + \frac{1}{3}\cos\theta\Big|_0^{2\bar{u}} = 27\bar{u}$$

$$\mathcal{F}_z(\bar{v}) = 8\bar{v} - 27\bar{u} = -19\bar{u}$$

5) $\int_{\gamma} \underbrace{\frac{-y}{x^2+y^2}}_{P}\,dx + \underbrace{\frac{x}{x^2+y^2}}_{Q}\,dy$



Cazul I) $0 \in \mathrm{Ext}(\gamma)$ => se poate aplica Green - Rieman

$$\int_{\gamma} P\,dx + Q\,dy \overset{G-R}{=\!=} \iint_K \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy =$$

$$= \iint_K \frac{x^2+y^2 - 2x^2 + x^2 + y^2 - 2y^2}{(x^2+y^2)^2}\,dx\,dy = 0$$



Cazul II) $0 \in \mathrm{Int}(\gamma)$

Fie $R > 0$ a.î. $D(0,R) \subset K$

$\mathrm{Fr}\,A = \gamma \cup \mathcal{C}(0,R)$

$A = K \setminus D(0,R)$

$$\left.\begin{aligned} &\int_{\mathrm{Fr}A} P\,dx + Q\,dy \overset{G-R}{=\!=} \iint_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = 0 \\ &\int_{\mathrm{Fr}A} P\,dx + Q\,dy = \int_{\gamma} P\,dx + Q\,dy - \int_{\mathcal{C}(0,R)} P\,dx + Q\,dy \end{aligned}\right\} \Rightarrow$$

$$\Rightarrow \int_{\gamma} P\,dx + Q\,dy = \int_{\mathcal{C}(0,R)} P\,dx + Q\,dy$$

$$\begin{cases} x = R\cos t \\ y = R\sin t \end{cases}, \quad t \in [0; 2\bar{u}]$$

$$\int_{\mathcal{C}(0,R)} P\,dx + Q\,dy = \int_0^{2\bar{u}} \left[\frac{-R\sin t}{R^2}(-R\sin t) + \frac{R\cos t}{R^2} R\cos t\right] dt =$$

$$= \int_0^{2\pi} \frac{R^2(\sin^2 t + \cos^2 t)}{R^2}\, dt = \int_0^{2\pi} dt = t\Big|_0^{2\pi} = 2\pi \Rightarrow$$

$$\Rightarrow \int_{\partial\Gamma} P\,dx + Q\,dy = 2\pi$$

Rezolvare examen 2023 ≈ Nr 1

2) $I = \int_0^\infty \frac{x^2}{1+x^4}\,dx$

$x^4 = y \Rightarrow x = \sqrt[4]{y} \Rightarrow dx = \frac{1}{4\sqrt[4]{y^3}}\,dy$

$I = \frac{1}{4}\int_0^\infty \frac{y^{\frac{1}{2}-\frac{3}{4}}}{1+y}\,dy = \frac{1}{4}\int_0^\infty \frac{y^{-\frac{1}{4}}}{(1+y)^1}\,dy$

$\beta(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = -\frac{1}{4} \Rightarrow p = -\frac{1}{4}+\frac{4}{4} = \frac{3}{4}$

$p+q = 1 \Rightarrow q = \frac{4}{4}-\frac{3}{4} = \frac{1}{4}$

$I = \frac{1}{4}\,\beta\left(\frac{3}{4};\frac{1}{4}\right)$

$\beta(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$I = \frac{1}{4}\,\frac{\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right)}{\Gamma(1)} = \frac{1}{4}\cdot\frac{\pi}{\sin\frac{\pi}{4}} = \frac{1}{4}\cdot\frac{2}{\sqrt{2}} = \frac{\pi}{2\sqrt{2}}$

3) $\begin{cases} x^2+y^2+z^2 = R^2, & z\geq 0 \\ x^2+y^2 = Ry, & R>0 \end{cases}$

$x^2+y^2-2\cdot\frac{1}{2}\cdot R\cdot y+\frac{R^2}{4} = \frac{R^2}{4} \Leftrightarrow$

$\Leftrightarrow x^2+\left(y-\frac{R}{2}\right)^2 = \frac{R^2}{4}$

$D: x^2+\left(y-\frac{R}{2}\right)^2 \leq \frac{R^2}{4}$

$x^2+y^2+z^2 = R^2 \Rightarrow z = \sqrt{R^2-x^2-y^2} = f(x,y)$

$\text{Aria}(\Sigma) = \iint_\Sigma d\sigma = \iint_D \sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{1+\left(\frac{-x}{\sqrt{R^2-x^2-y^2}}\right)^2+\left(\frac{-y}{\sqrt{R^2-x^2-y^2}}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{\frac{R^2-x^2-y^2+x^2+y^2}{R^2-x^2-y^2}}\,dx\,dy = R\iint_D \frac{1}{\sqrt{R^2-x^2-y^2}}\,dx\,dy$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$ $\quad \rho^2 \leq R\rho\sin\theta \Leftrightarrow \rho \leq R\sin\theta \Rightarrow \rho \in [0; R\sin\theta]$

$\left.\begin{array}{l}\rho \le R\sin\theta \\ \rho \ge 0 \\ R>0\end{array}\right\} \Rightarrow \sin\theta \ge 0 \Rightarrow \theta \in [0,\pi]$

$$\text{Aria}(\Sigma) = R\int_0^{\pi}\int_0^{R\sin\theta} \frac{1}{\sqrt{R^2-\rho^2}}\,\rho\,d\rho\,d\theta =$$

$$= -\frac{R}{2}\cdot 2\int_0^{\pi} \sqrt{R^2-\rho^2}\Big|_0^{R\sin\theta} d\theta = -R\int_0^{\pi}\left(\sqrt{R^2-R^2\sin^2\theta} - \sqrt{R^2}\right)d\theta =$$

~~$$= -R\int_0^{\pi}(R\cos\theta - R)\,d\theta = -R^2\int_0^{\pi}(\cos\theta - 1)\,d\theta =$$~~

~~$$= (-R^2\sin\theta + R^2\theta)\Big|_0^{\pi} = \pi R^2$$~~

$$= -R\int_0^{\pi}(R\cdot|\cos\theta| - R)\,d\theta = -R^2\int_0^{\pi}(|\cos\theta| - 1)\,d\theta =$$

$$= -R^2\left(\int_0^{\frac{\pi}{2}}\cos\theta\,d\theta - \int_{\frac{\pi}{2}}^{\pi}\cos\theta\,d\theta - \int_0^{\pi}d\theta\right) =$$

$$= -R^2\left(\sin\theta\Big|_0^{\frac{\pi}{2}} - \sin\theta\Big|_{\frac{\pi}{2}}^{\pi} - \theta\Big|_0^{\pi}\right) =$$

$$= -R^2(1-0-0+1-\pi+0) = -R^2(2-\pi)$$

4) $\Omega : \begin{cases} z^2 = x^2+y^2, & z\ge 0 \\ 2-z = x^2+y^2 \end{cases}$



$$\text{Vol}(\Omega) = \iiint_{\Omega} dx\,dy\,dz =$$

$$= \iint_D\left(\int_{\sqrt{x^2+y^2}}^{2-x^2-y^2} dz\right)dx\,dy =$$

$$= \iint_D \left[2-(x^2+y^2)-\sqrt{x^2+y^2}\right]dx\,dy$$

$\left.\begin{array}{l} z^2 = x^2+y^2 \\ 2-z = x^2+y^2 \end{array}\right\} \Rightarrow 2-z = z^2 \Leftrightarrow z^2+z-2=0$

$\Delta = 1+8 = 9 \;;\; z_{1,2} = \dfrac{-1\pm 3}{2}$, $z\ge 0$ $\Rightarrow z = 1$

$D : x^2+y^2 \le 1$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$, $\rho\in[0,1]$, $\theta\in[0,2\pi]$

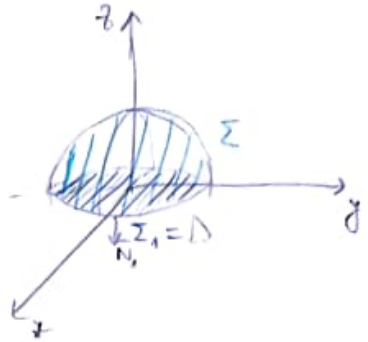
$$Vol(\Omega) = \int_0^{2\pi}\int_0^1 \left(2-\rho^2-\sqrt{\rho^2}\right)\rho\, d\rho\, d\theta =$$

$$= \int_0^{2\pi}\int_0^1 (2\rho - \rho^3 - \rho^2)\, d\rho\, d\theta =$$

$$= \int_0^{2\pi} \left(\rho^2 - \frac{1}{4}\rho^4 - \frac{1}{3}\rho^3\right)\Big|_0^1 d\theta = \int_0^{2\pi}\left(1-\frac{1}{4}-\frac{1}{3}\right)d\theta =$$

$$= \frac{5}{12}\theta\Big|_0^{2\pi} = \frac{10\pi}{12} = \frac{5\pi}{6}$$

5) $\bar{v}(x,y,z) = (\overbrace{2x+z^4}^{P}, \overbrace{z^2-2y}^{Q}, \overbrace{z+3}^{R})$



$\Sigma = S = \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 4, z \geq 0\}$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \overset{G-O}{=\!=\!=} \iiint_\Omega dx\,dy\,dz$

$\overset{G-O}{=\!=\!=} \iiint_\Omega div(\bar{v})\, dx\,dy\,dz$

$div(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 2 - 2 + 1 = 1$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \iiint_\Omega dx\,dy\,dz = \frac{4\pi \cdot 2^3}{3}\cdot\frac{1}{2} = \frac{16\pi}{3}$ $\Big\}$ (1)

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \Phi_\Sigma(\bar{v}) + \Phi_{\Sigma_1}(\bar{v})$

$\Rightarrow \Phi_\Sigma(\bar{v}) = \frac{16\pi}{3} - \Phi_{\Sigma_1}(\bar{v})$

$\bar{N}_1 = -\bar{k}$

$\Sigma_1 : z = 0$

$\Delta : x^2 + y^2 \leq 2$

$\Phi_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v}\cdot\bar{N}_1\, d\sigma = -\iint_\Delta 3\, dx\,dy = -3\iint_\Delta dx\,dy = -3\pi\cdot 4 = -12\pi$

$\Phi_\Sigma(\bar{v}) = \frac{16\pi}{3} + 12\pi = \frac{16\pi}{3} + \frac{36\pi}{3} = \frac{52\pi}{3}$

Rezolvare examen 2023 – Nr 2

2) $I = \int_0^\infty \frac{\sqrt{x}}{(1+x^2)^2}\,dx$

$x^2 = y \Rightarrow x = \sqrt{y} \Rightarrow dx = \frac{1}{2\sqrt{y}}\,dy$

$I = \frac{1}{2}\int_0^\infty \frac{y^{\frac{1}{4}} \cdot y^{-\frac{1}{2}}}{(1+y)^2}\,dy = \frac{1}{2}\int_0^\infty \frac{y^{\frac{1}{4}}}{(1+y)^2}\,dy$

$B(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = \frac{1}{4} \Rightarrow p = \frac{1}{4} + \frac{4}{4} = \frac{5}{4}$

$p+q = 2 \Rightarrow q = \frac{8}{4} - \frac{5}{4} = \frac{3}{4}$

$I = \frac{1}{2} B\left(\frac{5}{4}; \frac{3}{4}\right)$

$B(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$I = \frac{1}{2}\,\frac{\Gamma\left(\frac{5}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right)}{\Gamma(2)} = \frac{1}{2}\,\Gamma\left(1+\frac{1}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right) =$

$= \frac{1}{2}\cdot\frac{1}{8}\,\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(1-\frac{1}{4}\right) = \frac{1}{16}\cdot\frac{\pi}{\sin\frac{\pi}{4}} =$

$= \frac{\pi}{16}\cdot\frac{2}{\sqrt{2}} = \frac{\pi}{8\sqrt{2}}$

3) $\int_C \underbrace{\frac{x-y}{x^2+y^2}}_{P}\,dx + \underbrace{\frac{x+y}{x^2+y^2}}_{Q}\,dy$

Fie $\gamma \subset C$

Cazul 1) $O \in Ext(\gamma) \Rightarrow$ se poate aplica Green-Riemann

$\int_\gamma P\,dx + Q\,dy = \iint_K \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)dx\,dy =$

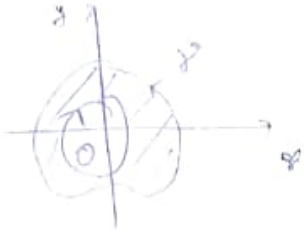
$= \iint_K \frac{x^2+y^2-2x(x+y)+x^2+y^2+2y(x-y)}{(x^2+y^2)^2}\,dx\,dy = 0$

Cazul 2) $O \in Int(\gamma)$

Fie $R>0$ a.î. $D(O,R) \subset K$

Fie $A = \gamma \cup \mathcal{C}(O,R)$

$A = K \setminus D(O,R)$



$$\int_{\text{Fr } A} P\,dx + Q\,dy \overset{G-R}{=} \iint_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = 0$$

$$\int_{\text{Fr } A} P\,dx + Q\,dy = \int_{\gamma} P\,dx + Q\,dy - \int_{\gamma(0,R)} P\,dx + Q\,dy \Rightarrow$$

$$\Rightarrow \int_{\gamma} P\,dx + Q\,dy = \int_{\gamma(0,R)} P\,dx + Q\,dy$$

$$\begin{cases} x = R\cos t \\ y = R\sin t \end{cases}, \; t \in [0; 2\pi]$$

$$\int_{\gamma(0,R)} P\,dx + Q\,dy = \int_0^{2\pi} \left(\frac{R\cos t - R\sin t}{R^2} (-R\sin t) + \frac{R\cos t + R\sin t}{R^2} R\cos t \right) dt$$

$$= \int_0^{2\pi} \frac{R^2}{R^2}\,dt = t\Big|_0^{2\pi} = 2\pi$$

$$\int_{\gamma} P\,dx + Q\,dy = 2\pi$$

4) $\Sigma: 4 - z = x^2 + y^2, \; z \in [2, 3]$

$z = 4 - x^2 - y^2 = f(x, y)$

$2 \le z \le 3 \Leftrightarrow 2 \le 4 - x^2 - y^2 \le 3 \Leftrightarrow$

$\Leftrightarrow -2 \le -x^2 - y^2 \le -1 \Leftrightarrow$

$\Leftrightarrow 1 \le x^2 + y^2 \le 2$

$D: \; 1 \le x^2 + y^2 \le 2$



$$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\,dx\,dy =$$

$$= \iint_D \sqrt{1 + (-2x)^2 + (-2y)^2}\,dx\,dy = \iint_D \sqrt{1 + 4(x^2 + y^2)}\,dx\,dy$$

$$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}, \; \rho \in [1; \sqrt{2}] \; ; \; \theta \in [0, 2\pi]$$

$$\text{Aria}(\Sigma) = \int_0^{2\pi} \int_1^{\sqrt{2}} \sqrt{1 + 4\rho^2} \cdot \rho\,d\rho\,d\theta =$$

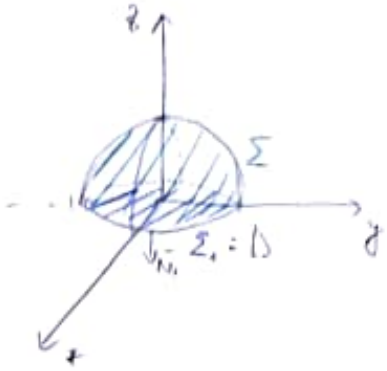
$$= \frac{1}{8} \cdot \frac{2}{3} \int_0^{2\pi} (1 + 4\rho^2)^{\frac{3}{2}} \Big|_1^{\sqrt{2}} d\theta = \frac{1}{12} \int_0^{2\pi} (27 - 5\sqrt{5})\,d\theta =$$

$$= \frac{1}{12}(27 - 5\sqrt{5})\theta\Big|_0^{2\pi} = \frac{\pi}{6}(27 - 5\sqrt{5})$$

5) $\bar{v}(x,y,z) = (\overbrace{3x+y^4}^{P}, \overbrace{x^3-3y}^{Q}, \overbrace{1-z}^{R})$

$\Sigma = S = \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 9,\ z \geq 0\}$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) \overset{G-O}{=\!=} \iiint_\Omega \operatorname{div}(\bar{v})\, dx\,dy\,dz$



$\operatorname{div}(\bar{v}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} =$

$= 3 - 3 - 1 = -1$

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = -\iiint_\Omega dx\,dy\,dz = -\frac{\frac{2}{3}\pi \cdot 3^{x_2}}{x_1} \cdot \frac{1}{x_1} = -2\pi \cdot 9 = -18\pi$ }

$\Phi_{\Sigma \cup \Sigma_1}(\bar{v}) = \Phi_{\Sigma}(\bar{v}) + \Phi_{\Sigma_1}(\bar{v})$

$\Rightarrow \Phi_{\Sigma}(\bar{v}) = -18\pi - \Phi_{\Sigma_1}(\bar{v})$

$\Sigma_1 \div z = 0$

$\bar{N}_1 = -\bar{k}$

$D: x^2 + y^2 \leq 9$

$\Phi_{\Sigma_1}(\bar{v}) = \int_{\Sigma_1} \bar{v} \cdot \bar{N}_1\, d\sigma = -\iint_D dx\,dy = -\pi \cdot 3^2 = -9\pi$

$\Phi_{\Sigma}(\bar{v}) = -18\pi + 9\pi = -9\pi$

Rezolvare examen 2024 Nr. 1

1) $J = \int_0^\infty \frac{x^2}{1+x^4}\,dx$

$x^4 = y \Rightarrow x = \sqrt[4]{y} \Rightarrow dx = \frac{1}{4\sqrt[4]{y^3}}\,dy$

$J = \frac{1}{4}\int_0^\infty \frac{y^{\frac{1}{2}} \cdot y^{-\frac{3}{4}}}{1+y}\,dy = \frac{1}{4}\int_0^\infty \frac{y^{-\frac{1}{4}}}{(1+y)^1}\,dy$

$B(p,q) = \int_0^\infty \frac{y^{p-1}}{(1+y)^{p+q}}\,dy$

$p-1 = -\frac{1}{4} \Rightarrow p = -\frac{1}{4}+\frac{4}{4} = \frac{3}{4}$

$p+q = 1 \Rightarrow q = \frac{4}{4}-\frac{3}{4} = \frac{1}{4}$

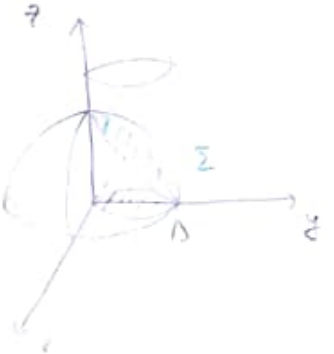
$J = \frac{1}{4} B\left(\frac{3}{4};\frac{1}{4}\right)$

$B(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$

$J = \frac{1}{4}\cdot\frac{\Gamma\left(\frac{3}{4}\right)\cdot\Gamma\left(\frac{1}{4}\right)}{\Gamma(1)} = \frac{1}{4}\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(1-\frac{1}{4}\right) = \frac{1}{4}\cdot\frac{\pi}{\sin\frac{\pi}{4}} =$

$= \frac{\pi}{4}\cdot\frac{2}{\sqrt{2}} = \frac{\pi}{2\sqrt{2}}$

2) $\Sigma: \begin{cases} x^2+y^2+z^2 = R^2, \ R>0 \\ x^2+\left(y-\frac{R}{2}\right)^2 = \frac{R^2}{4}, \ z\geq 0 \end{cases}$



$z = \sqrt{R^2-x^2-y^2} = f(x,y)$

$D: x^2+\left(y-\frac{R}{2}\right)^2 \leq \frac{R^2}{4}$

$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{1+\left(\frac{-x}{\sqrt{R^2-x^2-y^2}}\right)^2+\left(\frac{-y}{\sqrt{R^2-x^2-y^2}}\right)^2}\,dx\,dy =$

$= \iint_D \sqrt{\frac{R^2-x^2-y^2+x^2+y^2}{\sqrt{R^2-x^2-y^2}}}\,dx\,dy = R\iint_D \frac{1}{\sqrt{R^2-x^2-y^2}}\,dx\,dy$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$

$\rho^2 \le R \cdot \rho \cdot \sin\theta \Rightarrow \rho \le R\sin\theta \Rightarrow \rho \in [0; R\sin\theta]$

$\left.\begin{array}{l} R\sin\theta \ge \rho \\ R > 0 \\ \rho \ge 0 \end{array}\right\} \Rightarrow \sin\theta \ge 0 \Rightarrow \theta \in [0; \pi]$

$$\text{Aria}(\Sigma) = R\int_0^{\pi} \int_0^{R\sin\theta} \frac{1}{\sqrt{R^2-\rho^2}} \cdot \rho \, d\rho \, d\theta =$$

$$= -\frac{R}{2} \cdot 2 \int_0^{\pi} \sqrt{R^2-\rho^2} \Big|_0^{R\sin\theta} d\theta =$$

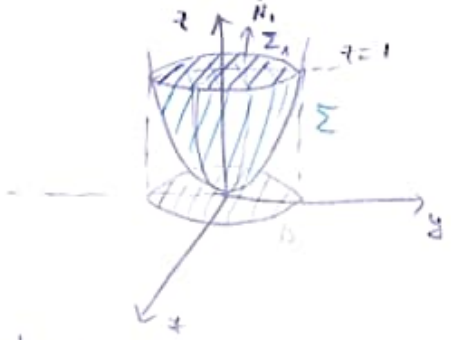
$$= -R \int_0^{\pi} \left(R\sqrt{1-\sin^2\theta} - R \right) d\theta =$$

$$= -R^2 \int_0^{\pi} (|\cos\theta| - 1) \, d\theta = -R^2 \left(\int_0^{\frac{\pi}{2}} \cos\theta \, d\theta - \int_{\frac{\pi}{2}}^{\pi} \cos\theta \, d\theta - \int_0^{\pi} d\theta \right) =$$

$$= -R^2 \left(\sin\theta \Big|_0^{\frac{\pi}{2}} - \sin\theta \Big|_{\frac{\pi}{2}}^{\pi} - \theta \Big|_0^{\pi} \right) =$$

$$= -R^2 (1 - 0 - 0 + 1 - \pi) = -R^2 (2 - \pi)$$

3) $\vec{v} = (\underbrace{1+2xy}_{P}, \underbrace{z^2 - y^2}_{Q}, \underbrace{1}_{R})$



$\Sigma: \begin{cases} z = x^2 + y^2 \\ z \in [0,1] \end{cases}$

$0 \le x^2 + y^2 \le 1$

$D: x^2 + y^2 \le 1$

$$\Phi_{\Sigma \cup \Sigma_1}(\vec{v}) \overset{G-O}{=\!=} \iiint_{\Omega} \text{div}(\vec{v}) \, dx\,dy\,dz$$

$\text{div}(\vec{v}) = 2y - 2y + 0 = 0$

$\left.\begin{array}{l} \Phi_{\Sigma \cup \Sigma_1}(\vec{v}) = 0 \end{array}\right\} \Rightarrow$

$\Phi_{\Sigma \cup \Sigma_1}(\vec{v}) = \Phi_{\Sigma}(\vec{v}) + \Phi_{\Sigma_1}(\vec{v})$

$\Rightarrow \Phi_{\Sigma}(\vec{v}) = 0 - \Phi_{\Sigma_1}(\vec{v}) = -\Phi_{\Sigma_1}(\vec{v})$

$\Sigma_1: z = 1$

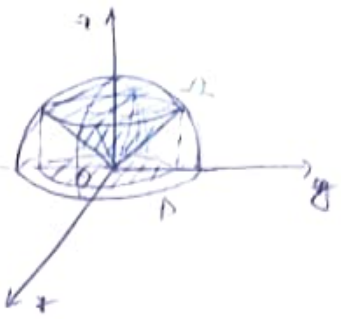
$\vec{N}_1 = \vec{k}$

$$\Phi_{\Sigma_1}(\vec{v}) = \iint_{\Sigma_1} \vec{v} \cdot \vec{N}_1 \, d\sigma = \iint_D dx\,dy = \pi \cdot 1^2 = \pi$$

$\Rightarrow \Phi_{\Sigma}(\vec{v}) = -\pi$

3

4) $\Omega \begin{cases} x^2+y^2+z^2 = 2 \\ z^2 = x^2+y^2 \end{cases}, z \geq 0$



$$\text{Vol}(\Omega) = \iiint_{\Omega} dx\,dy\,dz = \iint_{\Delta}\left(\int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} dz\right)dx\,dy =$$

$$= \iint_{\Delta}\left(\sqrt{2-x^2-y^2} - \sqrt{x^2+y^2}\right)dx\,dy$$

$2z^2 = 2 \Rightarrow z^2 = 1$, $z \geq 0$ $\Rightarrow z = 1$

$\Delta: x^2+y^2 \leq 1$

$\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$ $\rho \in [0,1]$, $\theta \in [0, 2\pi]$

$$\text{Vol}(\Omega) = \int_0^{2\pi}\int_0^1 \left(\sqrt{2-\rho^2} - \sqrt{\rho^2}\right)\rho\, d\rho\, d\theta =$$

$$= \int_0^{2\pi}\left[-\frac{1}{2}(2-\rho^2)^{\frac{3}{2}} \cdot \frac{2}{3}\Big|_0^1 - \frac{\rho^3}{3}\Big|_0^1\right]d\theta =$$

$$= \int_0^{2\pi}\left(-\frac{1}{3}\cdot 1 + \frac{1}{3}\cdot 2\sqrt{2} - \frac{1}{3}\right)d\theta =$$

$$= \frac{2}{3}(\sqrt{2}-1)\theta\Big|_0^{2\pi} = \frac{4\pi}{3}(\sqrt{2}-1)$$

5) $x^3+5x-2=0$, $\varepsilon < 10^{-3}$

$f(x) = x^3+5x-2$

$f(0) = -2$

$f(1) = 4$

$f(0)\cdot f(1) = -8 < 0 \Rightarrow f$ are o soluție în $(0,1)$ $\Rightarrow$

$f'(x) = 3x^2+5 > 0$, $(\forall)\, x \in [0,1] \Rightarrow f$ crescătoare

$\Rightarrow f$ are o unică soluție reală în $(0,1)$

$x^3+5x-2=0 \Leftrightarrow x^3+5x = 2 \Leftrightarrow x(x^2+5) = 2 \Rightarrow x = \frac{2}{x^2+5}$

Fie $T(x) = \frac{2}{x^2+5}$, $T: [0;1] \to [0;1]$

~~$T'(x) =$~~ Conform Teoremei Lagrange $\Rightarrow \exists\, c \in [0,1]$ a.î.

$T(x) - T(y) = T'(c)(x-y) \Leftrightarrow |T(x)-T(y)| = |T'(c)|\cdot|x-y| \leq$

$\leq \sup|T'(c)|\cdot|x-y|$ (1)

$T'(x) = \frac{-2 \cdot 2x}{(x^2+5)^2} = \frac{-4x}{(x^2+5)^2}$

$g(x) = |T'(x)| = \frac{4x}{(x^2+5)^2}$

$g'(x) = \frac{4(x^2+5)^{x'} - 4x \cdot 2(x^2+5)^{1} \cdot 2x}{(x^2+5)^{4}} = \frac{4x^2+20-16x^2}{(x^2+5)^3} =$

$= \frac{20-12x^2}{(x^2+5)^3} \; 4 > 0 \; , (\forall) x \in [0,1] \Rightarrow g$ crescătoare $\Rightarrow$

$\Rightarrow \sup_{x} g(x) = \sup_{x} |T'(x)| = g(1) = \frac{4}{36} = \frac{1}{9}$

Din (1) $\Rightarrow$ T contracție de factor $K = \frac{1}{9}$
$[0,1]$ - spațiu metric complet
$\Rightarrow$ Conform Th. Contracției

$\Rightarrow \exists! \; \xi \in [0,1]$ a.î. $T(\xi) = \xi$

$x_0 = 0, \; x_n = T(x_{n-1}) \; , (\forall) n \in \mathbb{N} \; ; \; x_1 = T(x_0) = \frac{2}{5}$

$|x_n - \xi| \leq \frac{K^n}{1-K} |x_1 - x_0| < 10^{-3}$

$\frac{\left(\frac{1}{9}\right)^n}{1-\frac{1}{9}} \cdot \frac{2}{5} < 10^{-3} \Leftrightarrow \left(\frac{1}{9}\right)^n \cdot \frac{9}{8} \cdot \frac{2}{5} < \frac{1}{1000} \Leftrightarrow$

$\Leftrightarrow \frac{1}{9^{n-1}} \cdot \frac{1}{20} < \frac{1}{1000} \Leftrightarrow 9^{n-1} \cdot 20 > 1000$

$n = 3 \Rightarrow \xi \simeq x_3 = T(x_2)$

$x_2 = T(x_1) = T\left(\frac{2}{5}\right) = \frac{2}{\frac{129}{25}} = \frac{50}{129}$

$\xi \simeq \frac{2}{\left(\frac{50}{129}\right)^2 + 5} = \dots$